

ECE/CS 757 Advanced Computer Architecture II

Spring 2025

Coherence (Part 3)

Instructor: Prof. Joshua San Miguel

Some slides adopted from Greg Steffan, Natalie Enright Jerger and Mikko Lipasti

Last Lecture

- Exclusive state (MESI)
- Owned state (MOESI)
- Forward state (MESIF)

This Lecture

- Coherence misses
- Directory-based coherence – MSI
- Transient states

Snooping-Based Coherence – Requirements

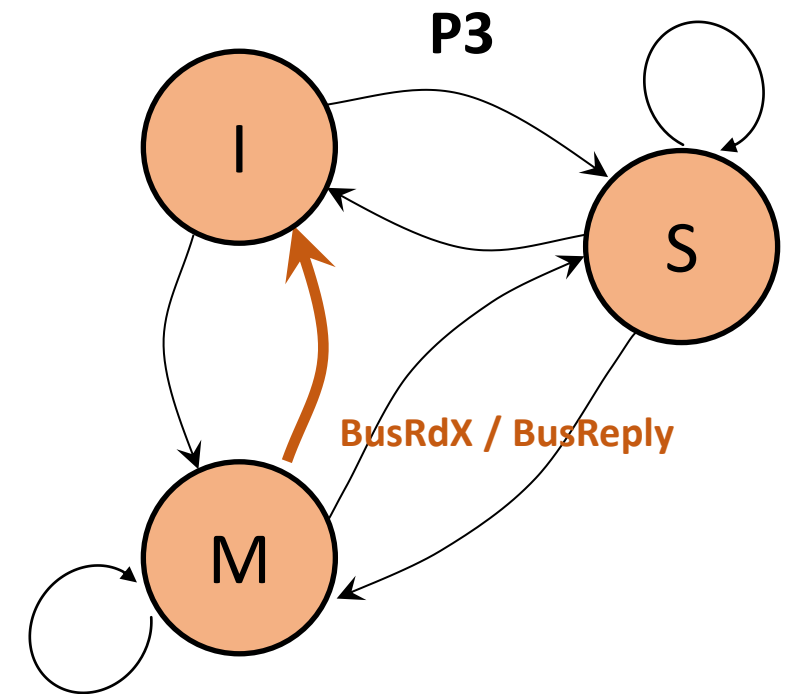
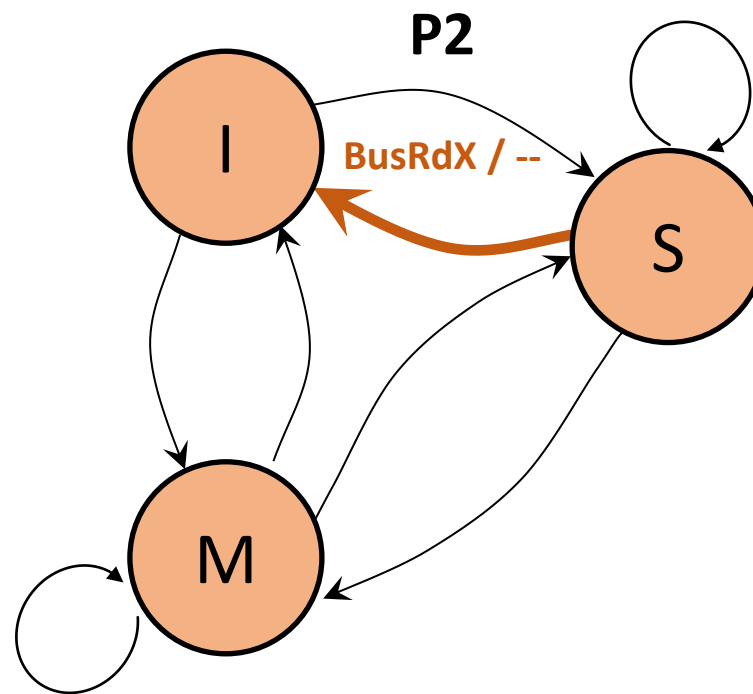
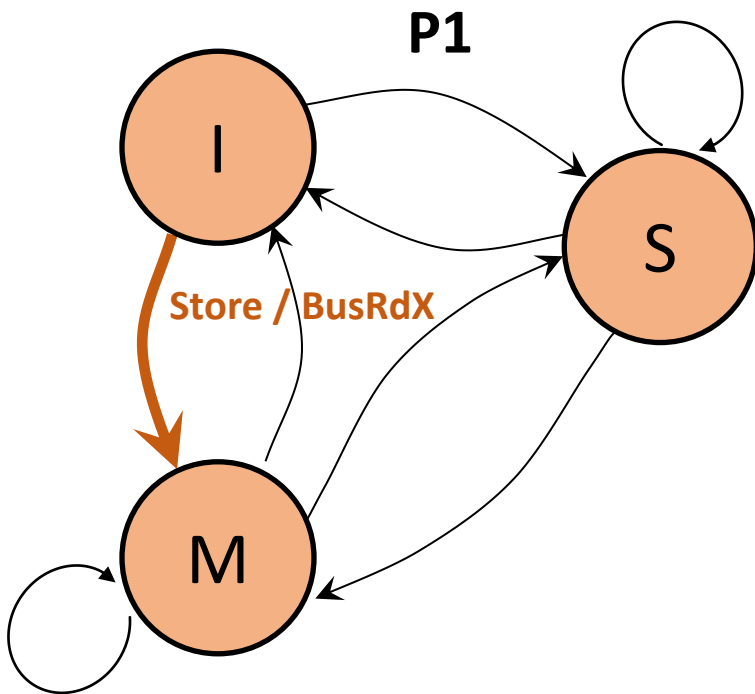
Atomicity:

Bus transactions (i.e., request-response sequences) must not be interleaved.

Snooping-Based Coherence – Requirements

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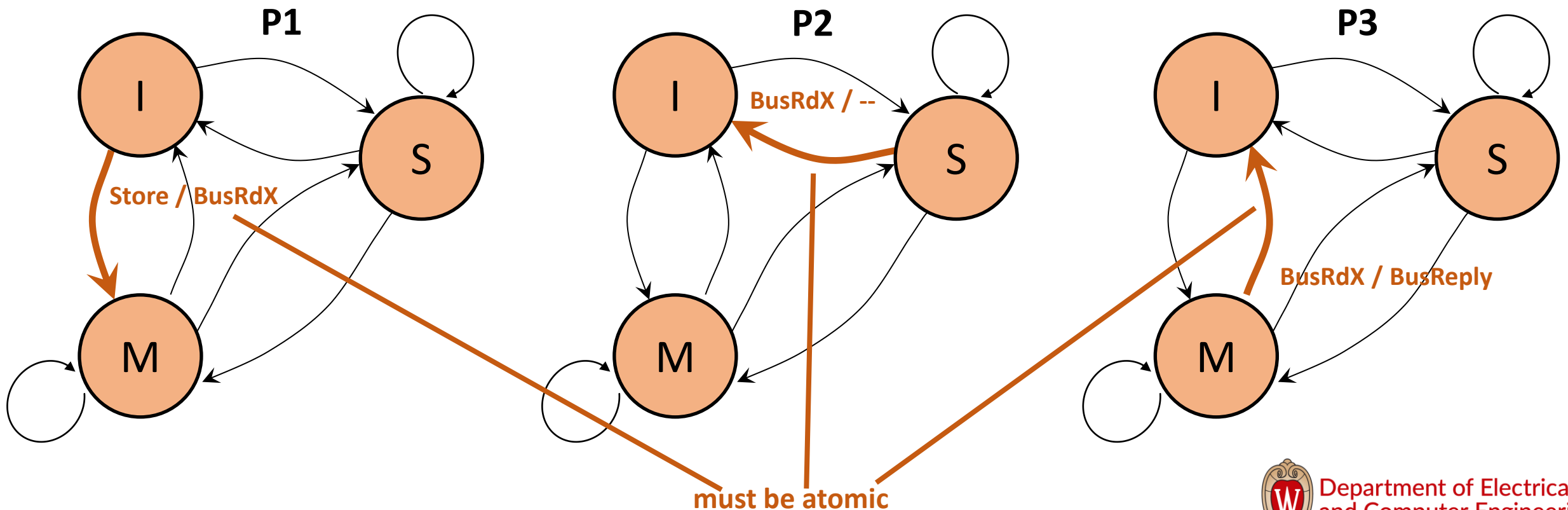
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Response messages (e.g., BusReply, BusWB) must be prioritized over request messages (e.g., Load, Store, Evict, BusRd, BusRdX, BusWr, BusInv).

Starvation avoidance:

Bus requests must be fairly arbitrated among caches.

Cache Misses

Cold (Compulsory) Miss:

First ever access to a given block of memory.

- If infinite cache, every miss is a cold miss.

Capacity Miss:

Working set exceeds cache capacity.

- If fully-associative finite cache, every miss is a cold miss or a capacity miss.

Conflict Miss:

Placement restrictions cause useful blocks to be evicted.

- Conflict misses only occur when a block can only be placed on a subset of the cache.

Coherence Miss:

Invalidations cause useful blocks to be evicted.

- Coherence misses only occur with invalidation-based protocols.



Coherence Misses

True Coherence Miss:

Compulsory miss with invalidation-based protocols due to true sharing in application.

- If infinite cache with minimum block size, every miss is a cold miss or a true coherence miss.

False Coherence Miss:

Unnecessary miss with invalidation-based protocols due to false sharing in application.

- Maximum false coherence misses occurs on infinite cache with maximum block size.

Coherence Decoupling

J. Huh, J. Chang, D. Burger and G. S. Sohi, *Coherence Decoupling: Making Use of Incoherence*, ASPLOS, 2014.

1. After invalidation, keep stale data in cache
2. On next load, use stale value speculatively
3. Verify/rollback based on actual value from BusReply

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+ Can eliminate false coherence misses

+ Can eliminate true coherence misses of **silent stores**

stores that do not change
the current data value



Snooping-Based Coherence – Scalability Issues

Bus Serialization:

Forcing transactions to serialize hurts performance

- Most transactions can run in parallel (e.g., requests for different blocks)

Snooping/Broadcasting:

Forcing caches to always snoop/broadcast hurts energy efficiency

- Most snoops lead to no transition nor action (e.g., blocks are usually in Invalid state)

Cache Coherence

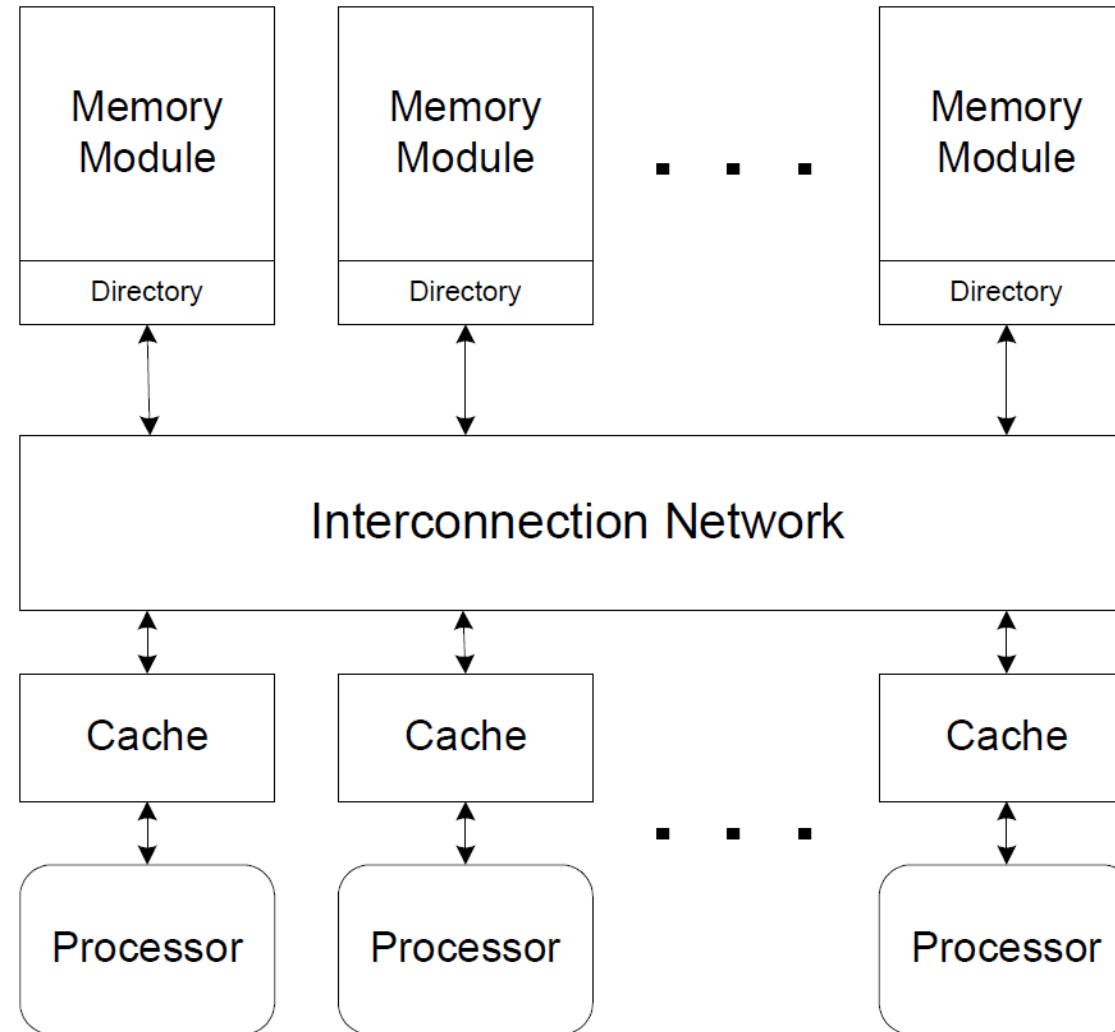
Snooping-based coherence

Caches *snoop* (i.e., observe) all read/write requests (and their responses) to memory and other caches via broadcasting.

Directory-based coherence

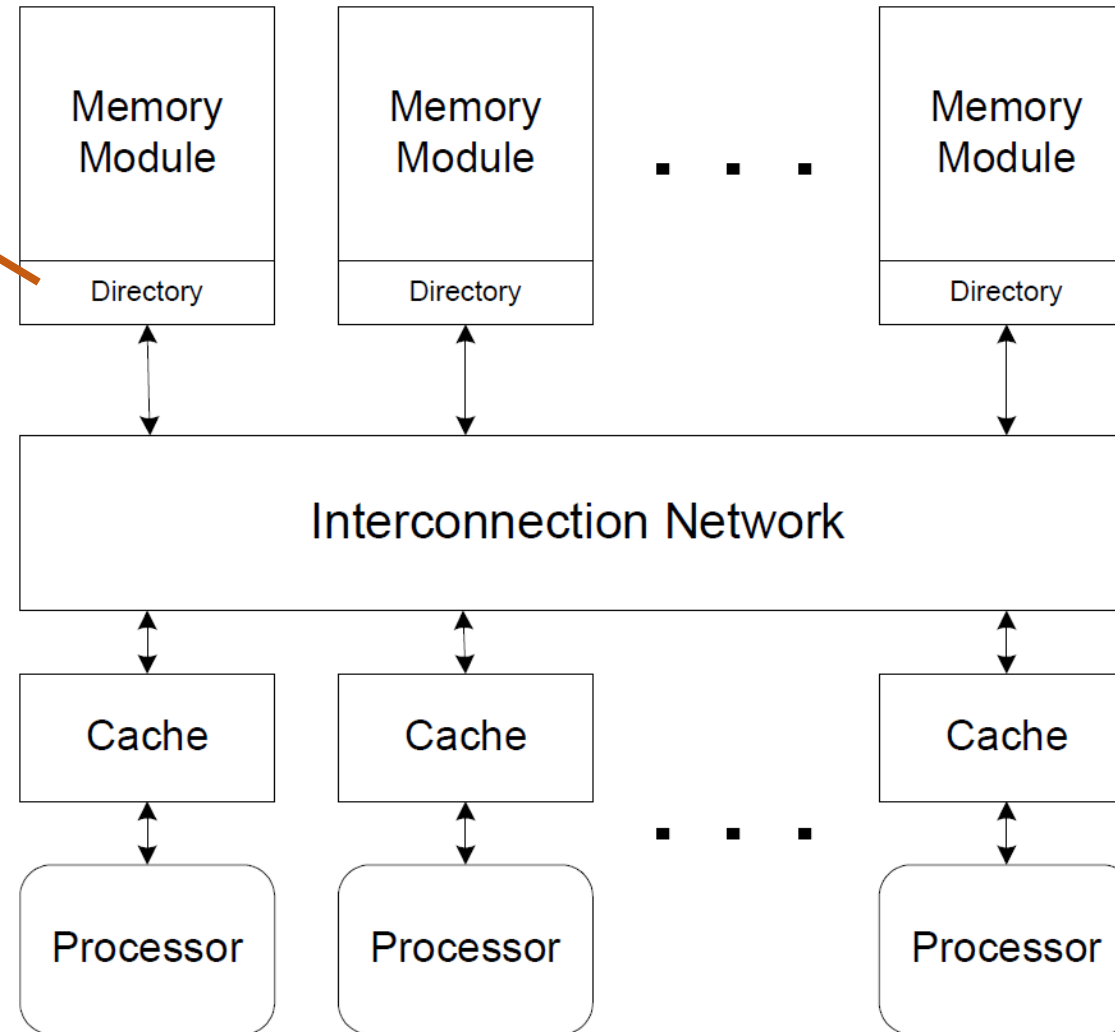
Read/write requests (and their responses) are point-to-point; a separate *directory* structure must be referenced first to see who has copies of the data.

Directory-Based Coherence



Directory-Based Coherence

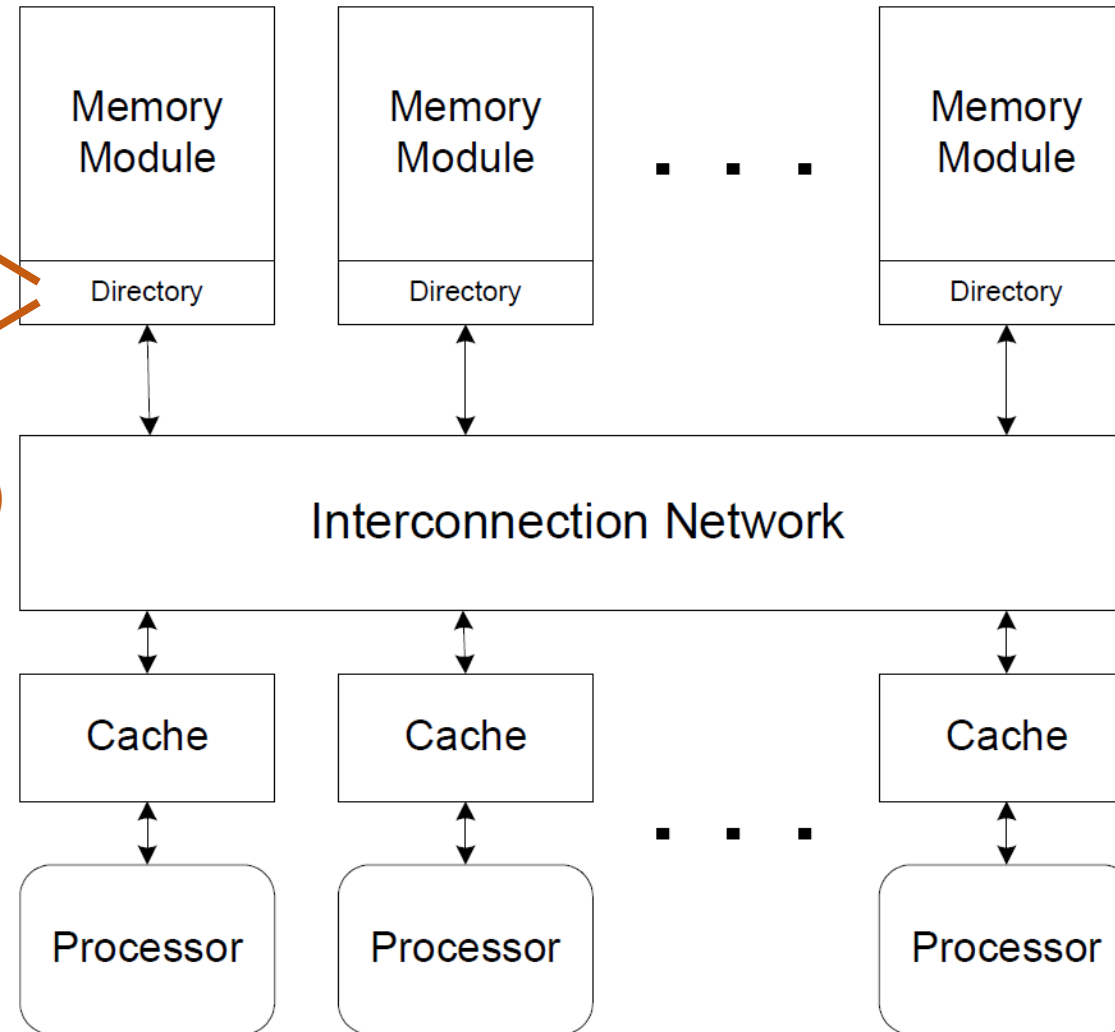
write serialization
(each address exists in only
one *home* directory bank)



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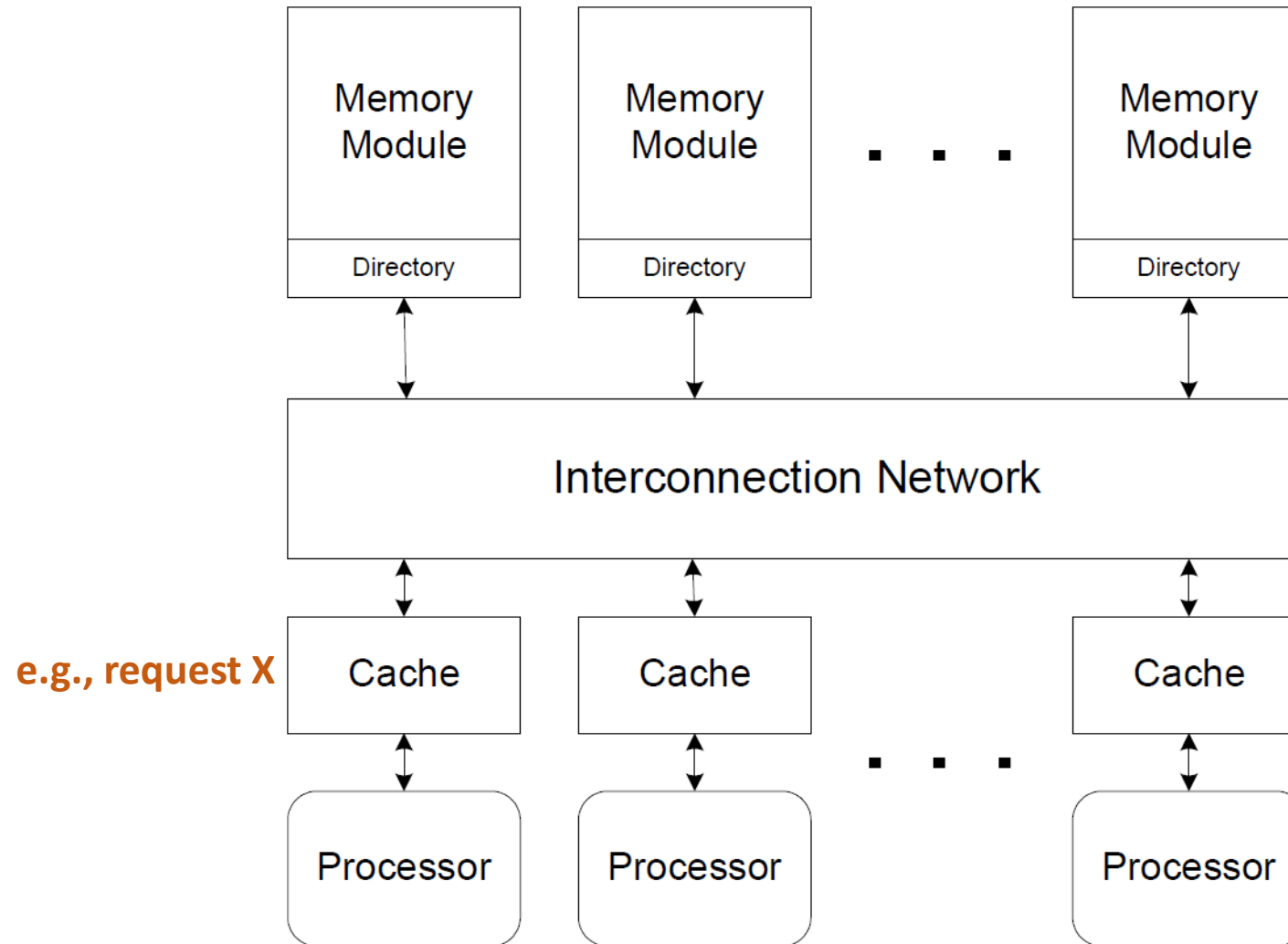
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write propagation
(track which caches are sharers)

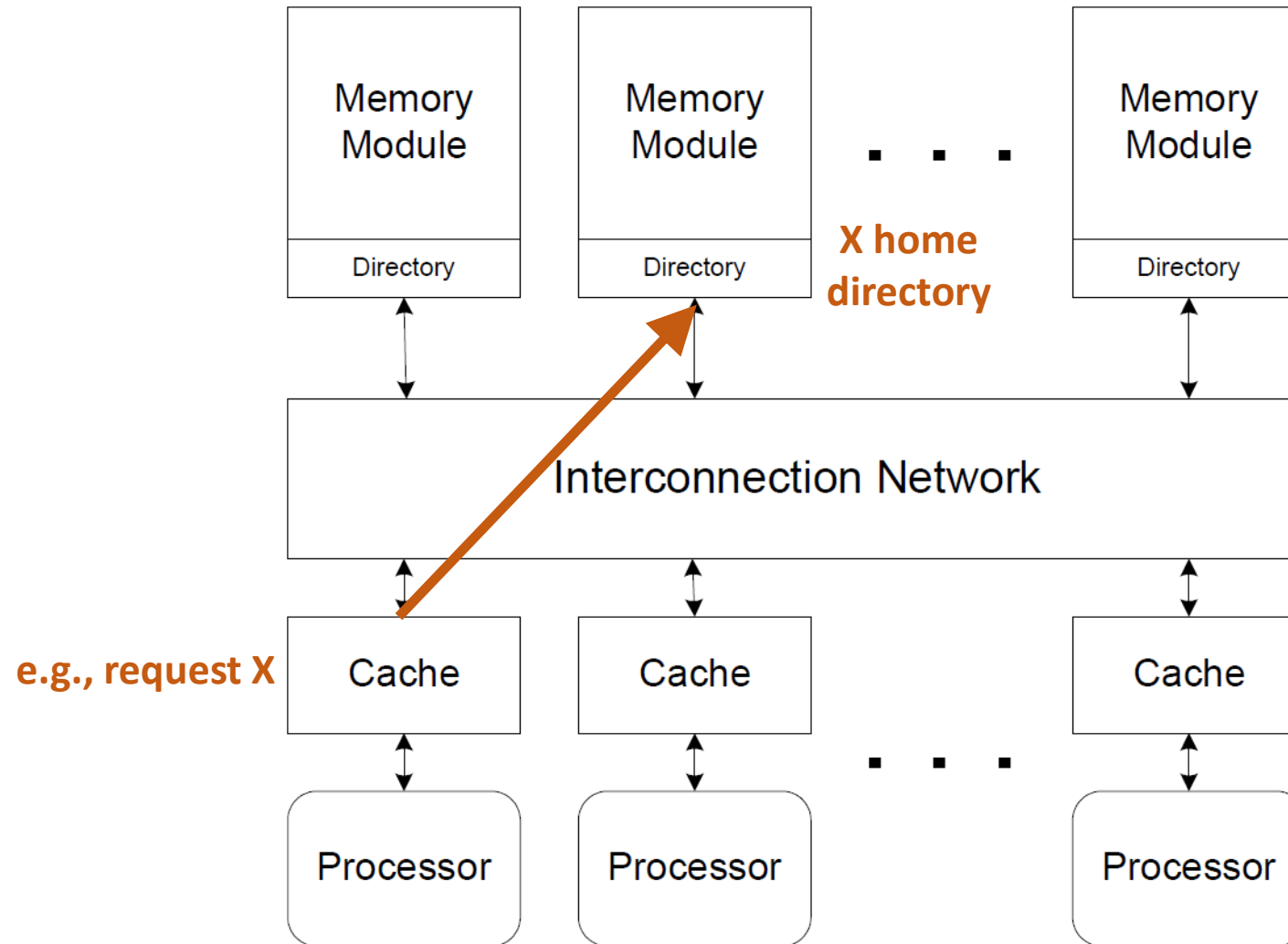


state	owner	sharers
owned	8	0000011100001100000000

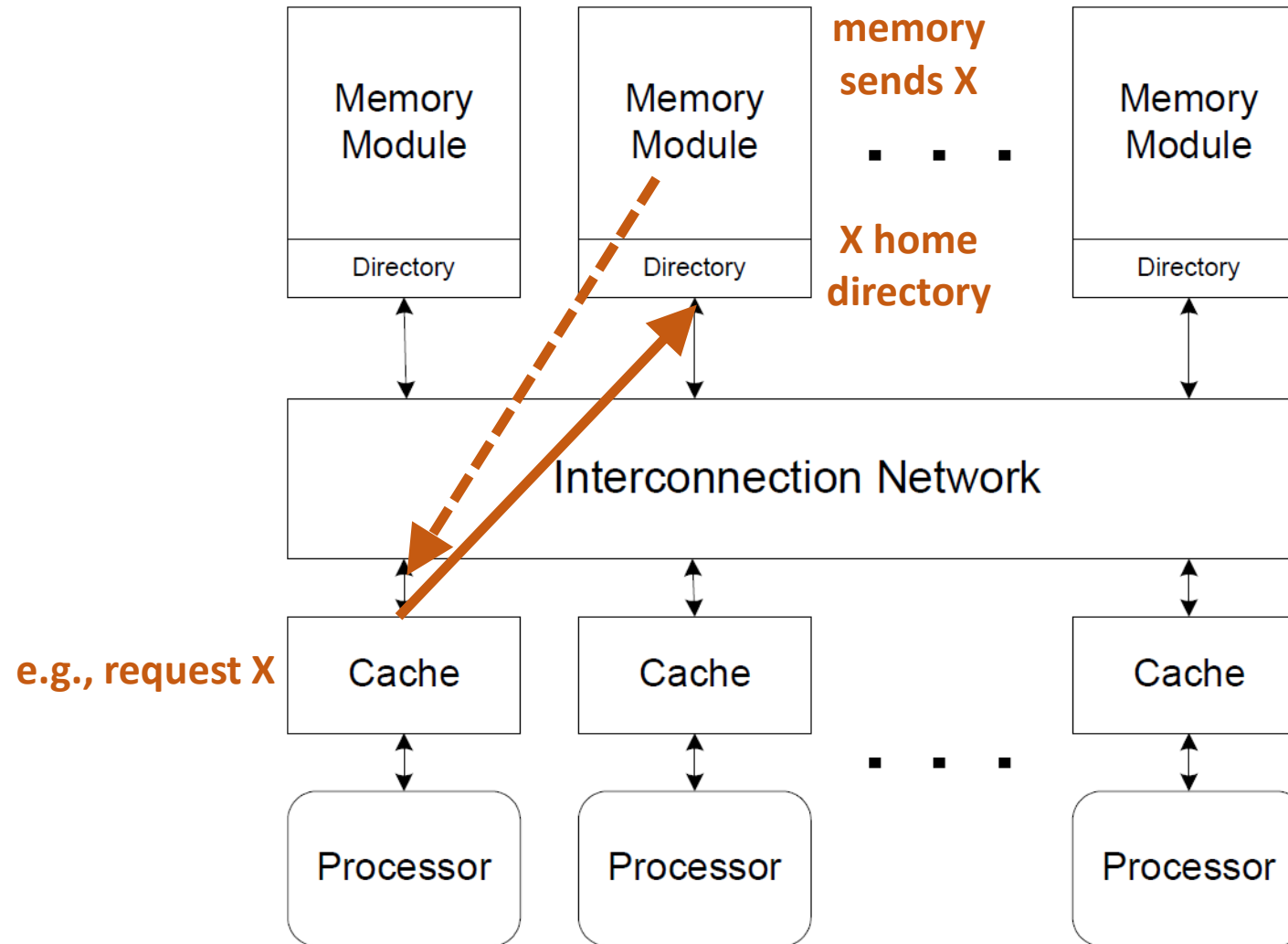
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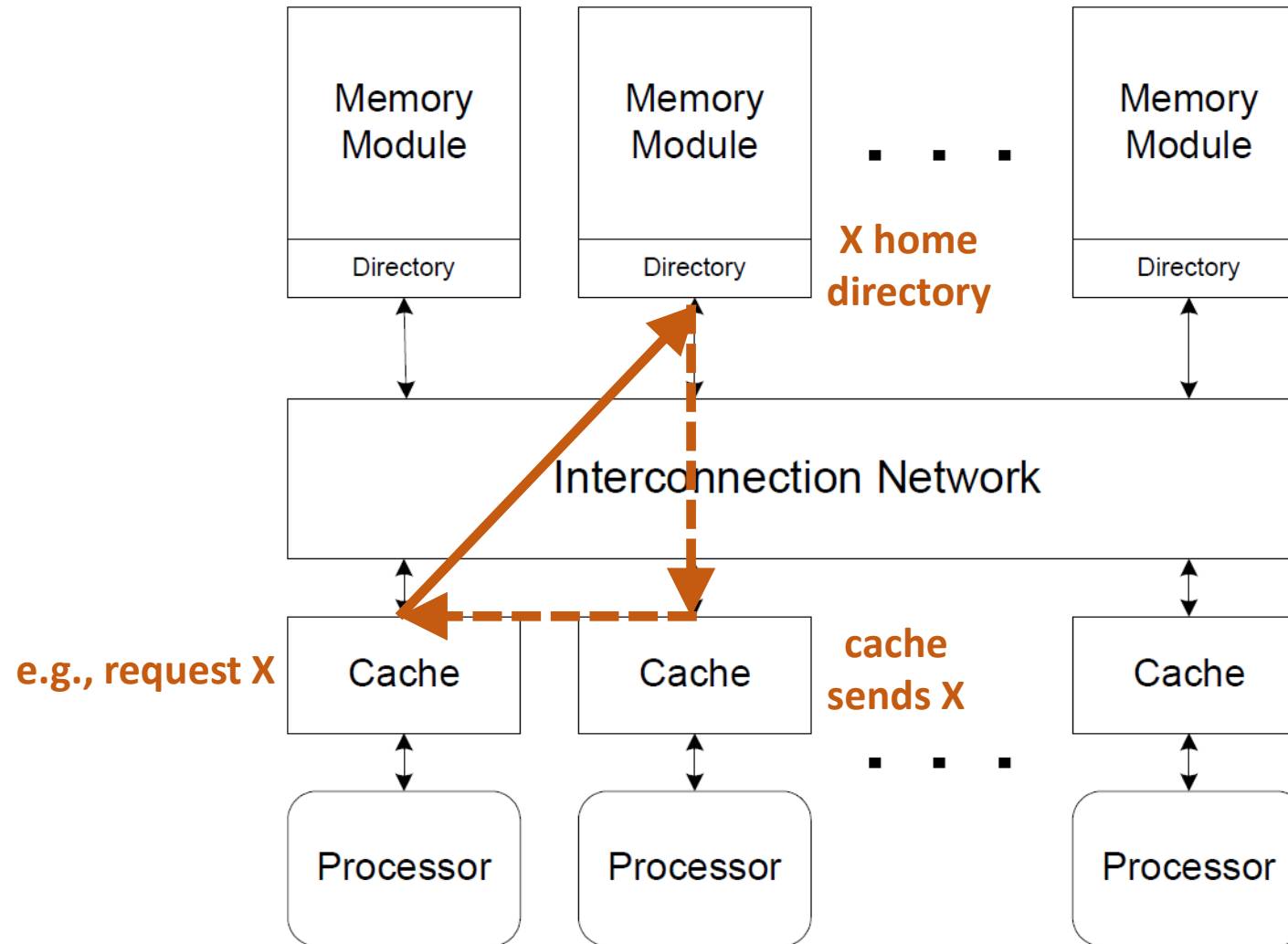
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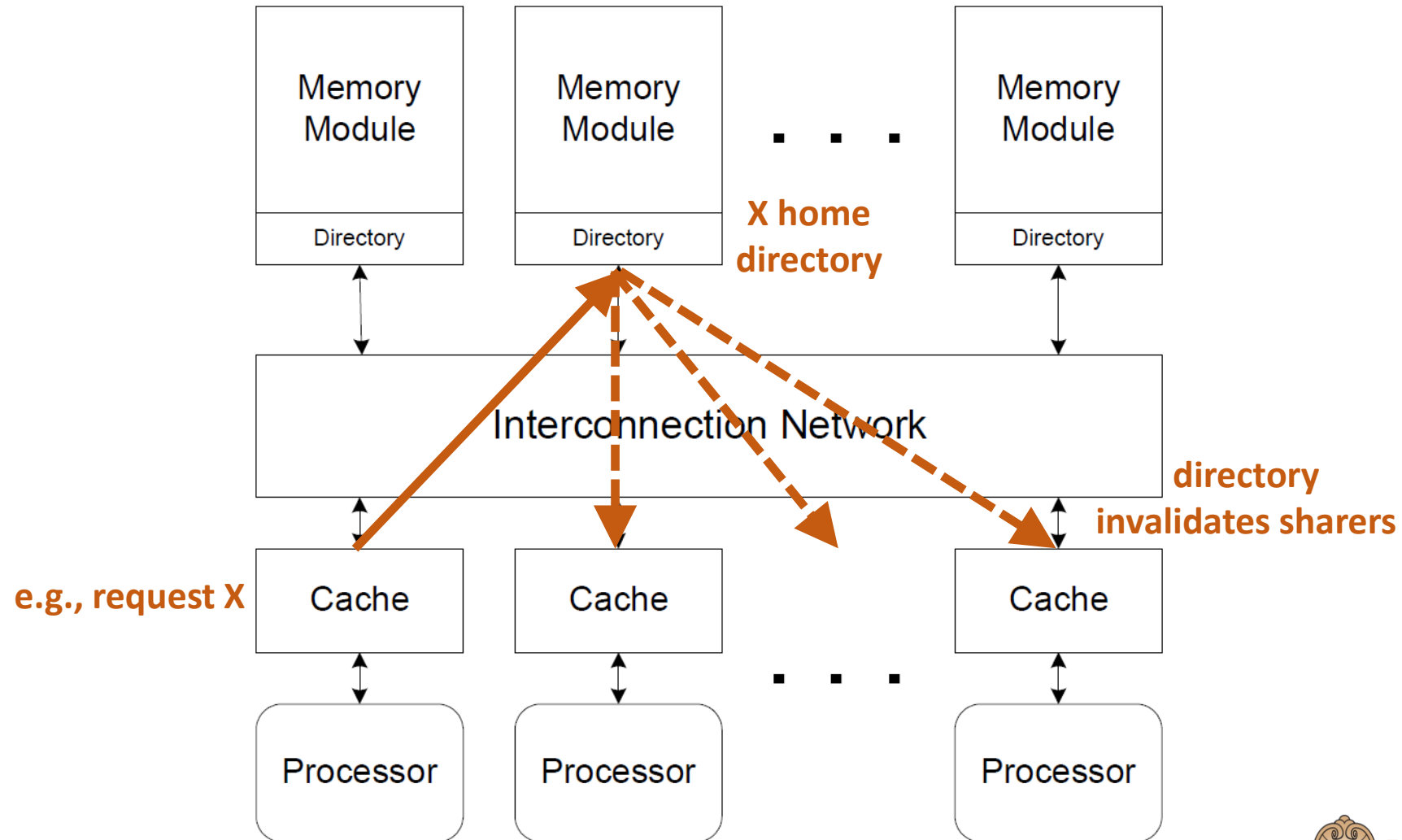
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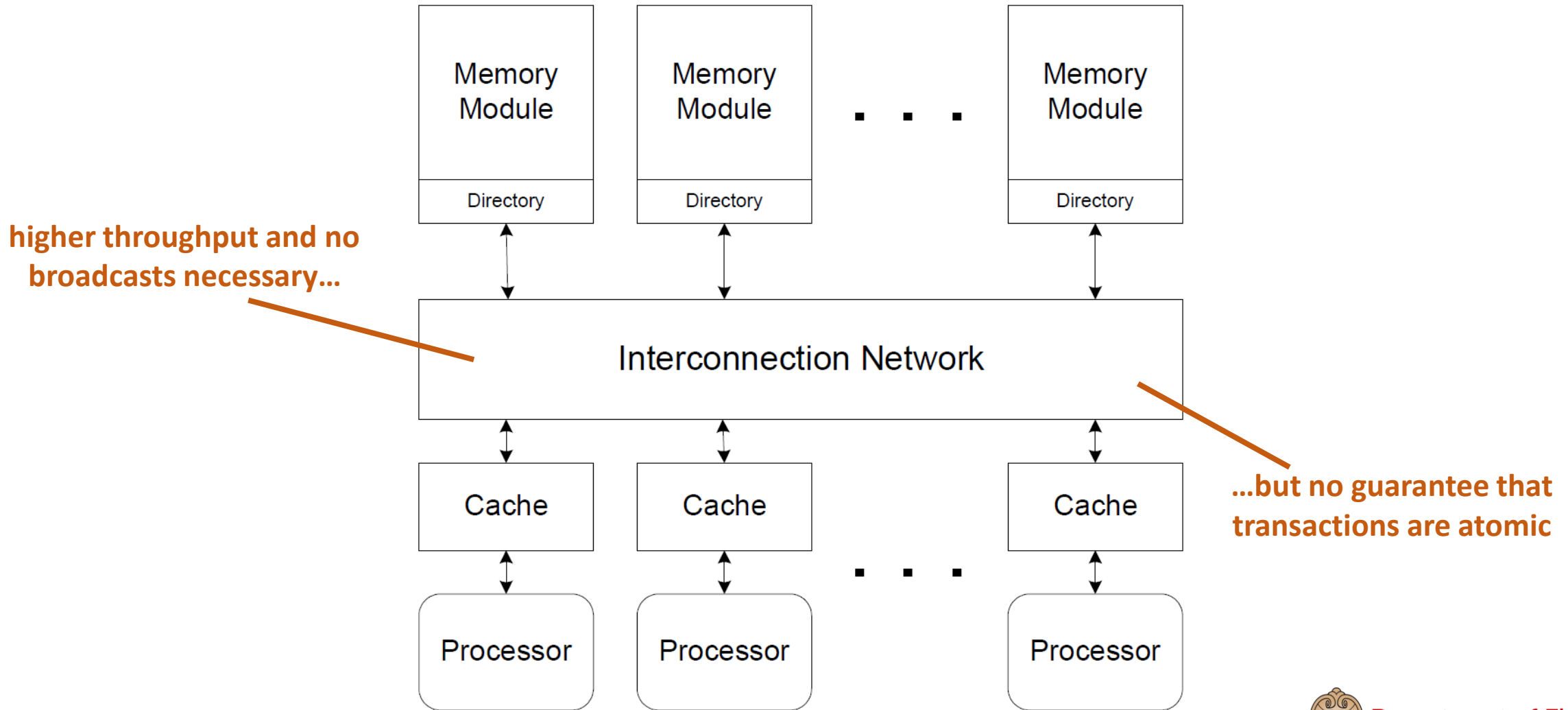
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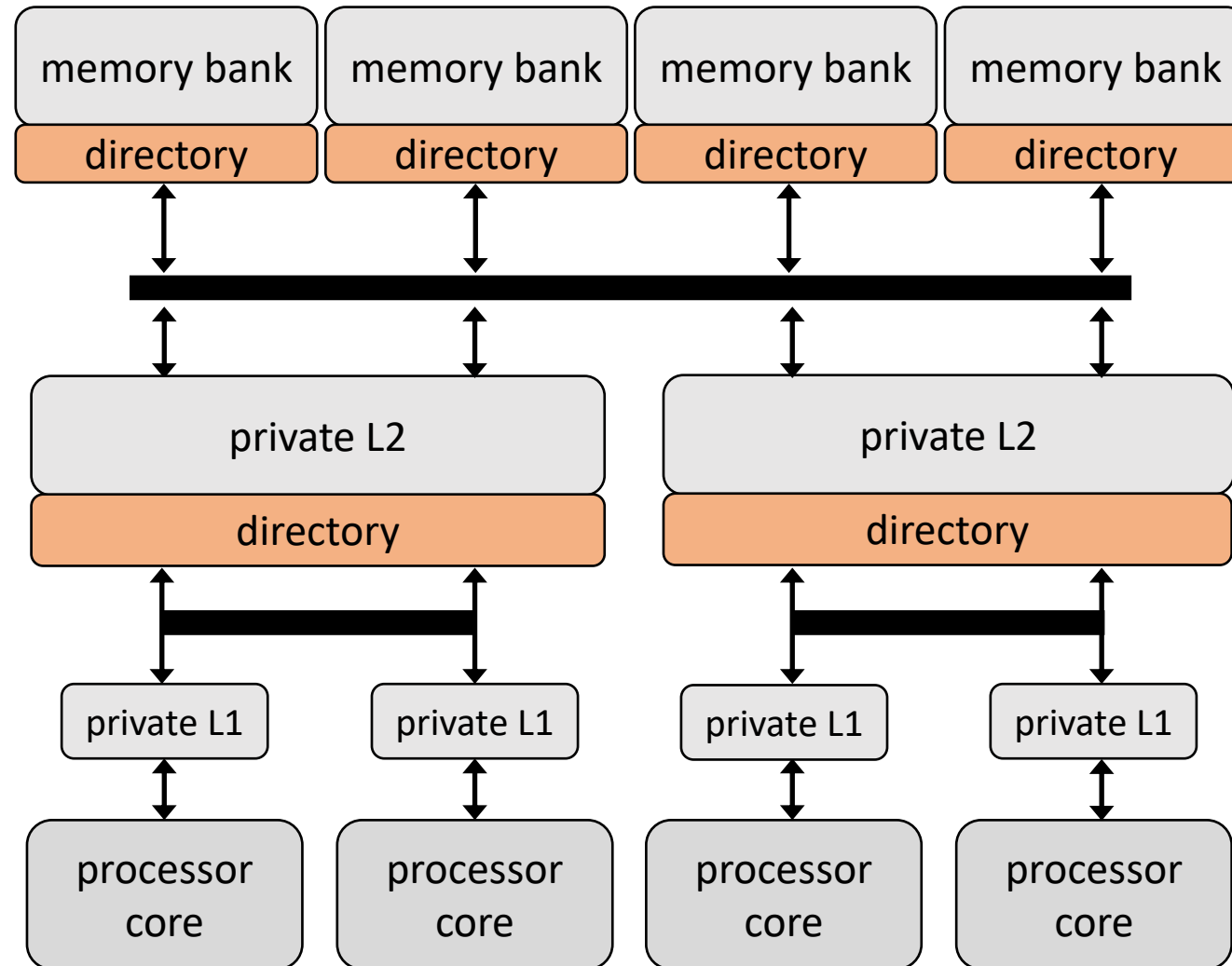
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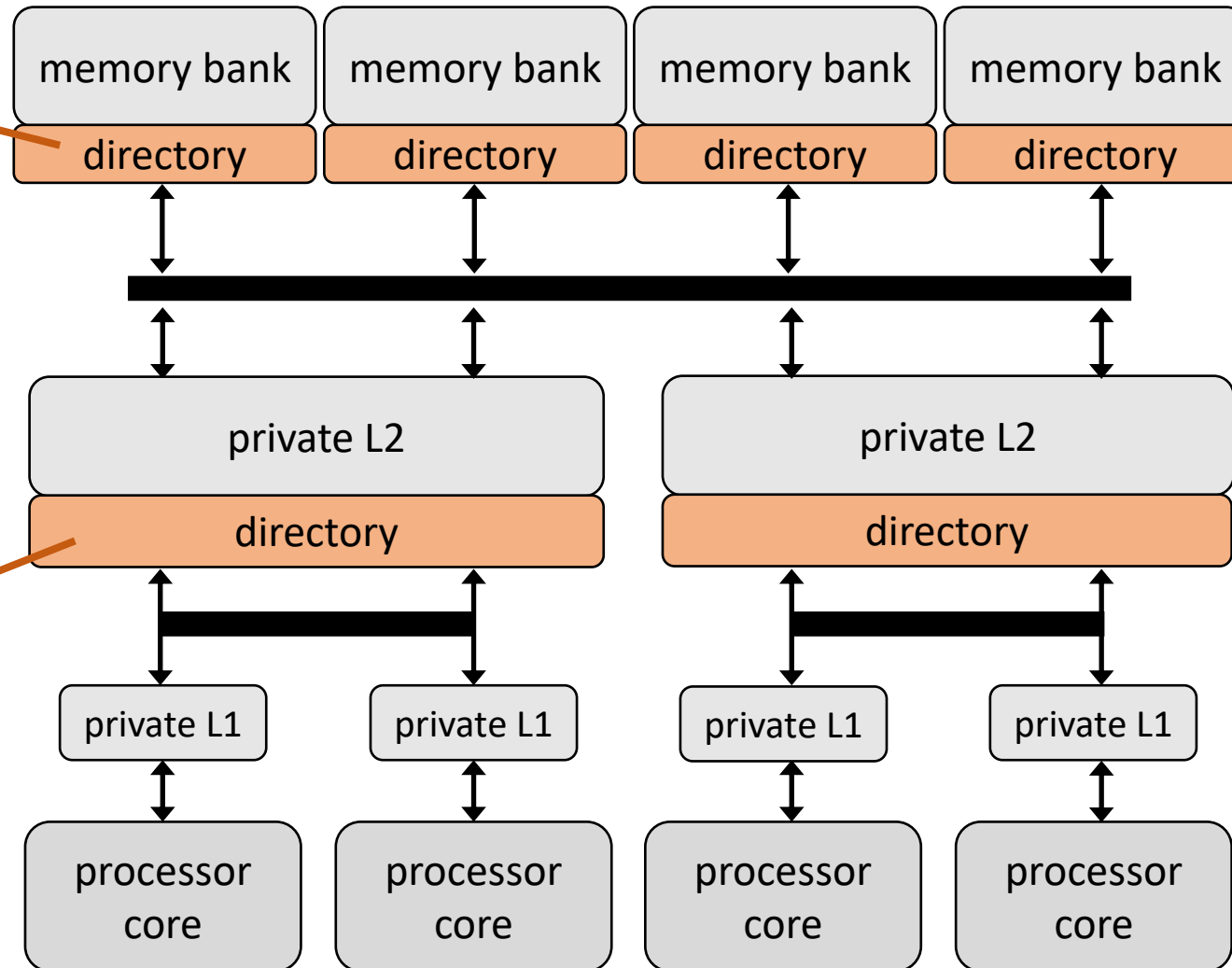
Directories



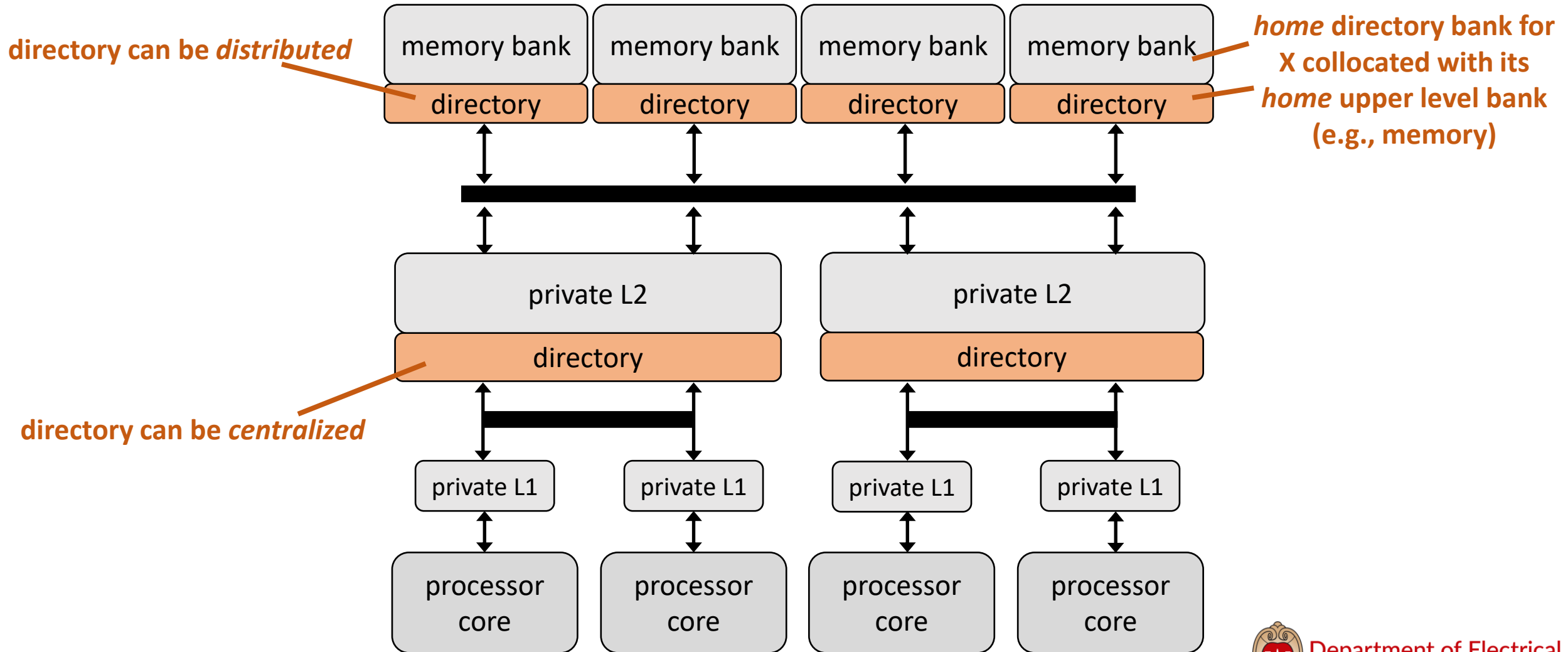
Directories

directory can be *distributed*

directory can be *centralized*



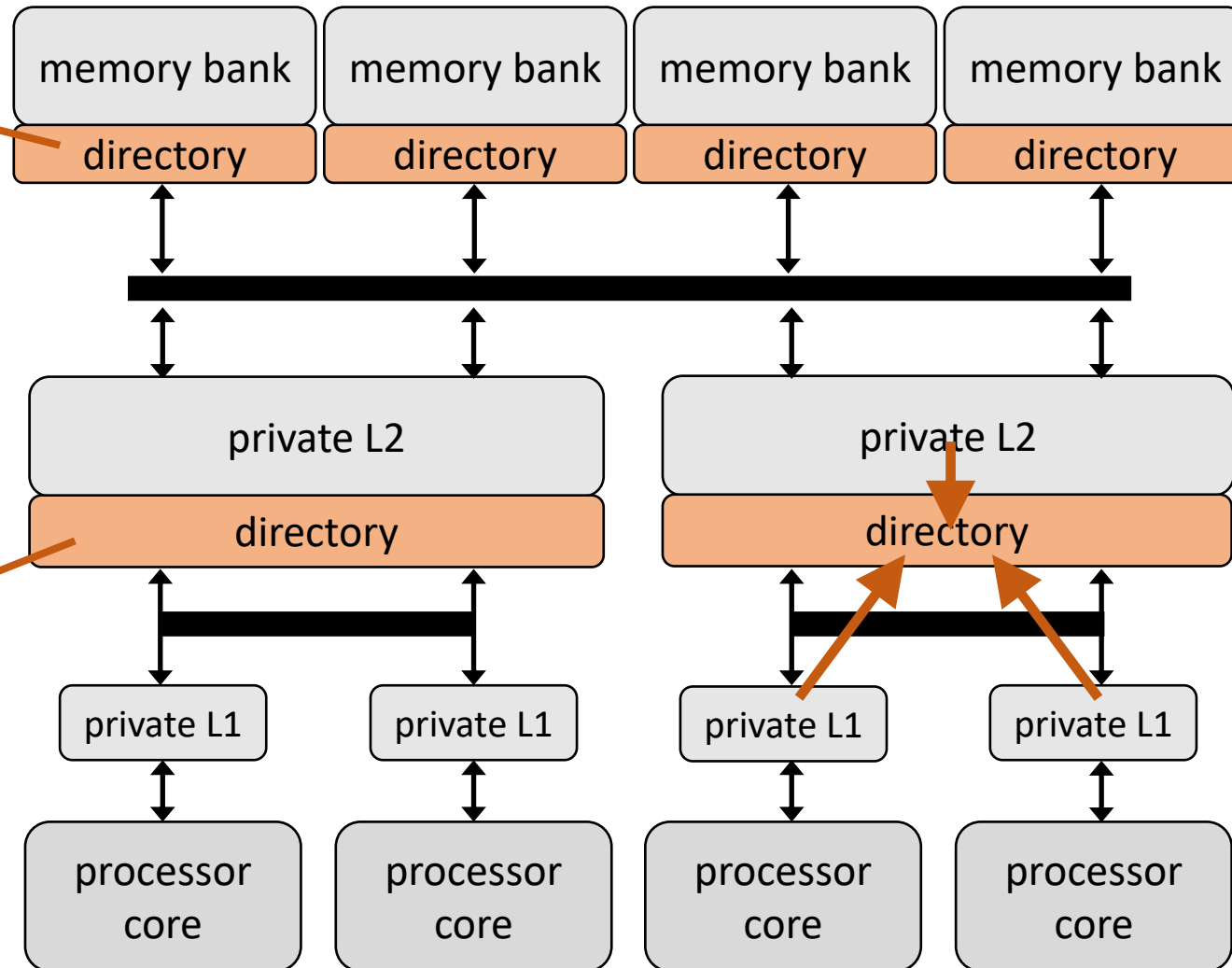
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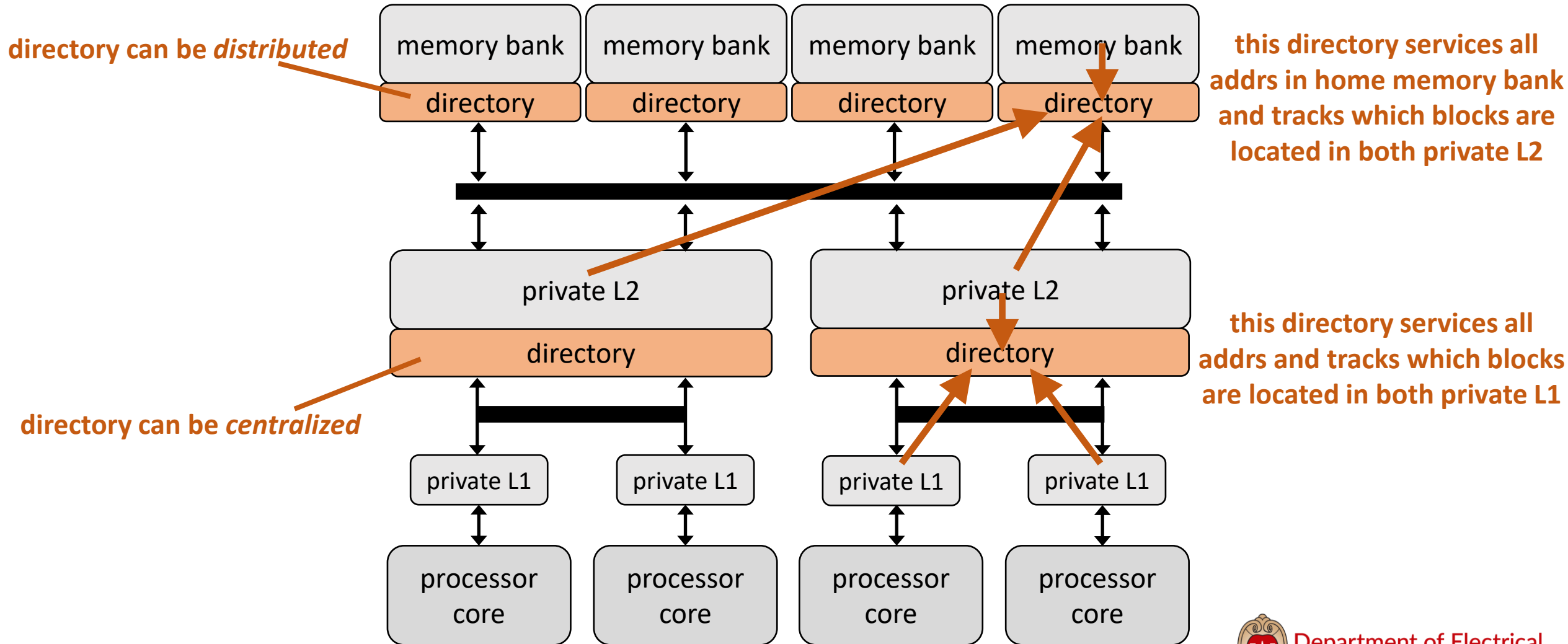
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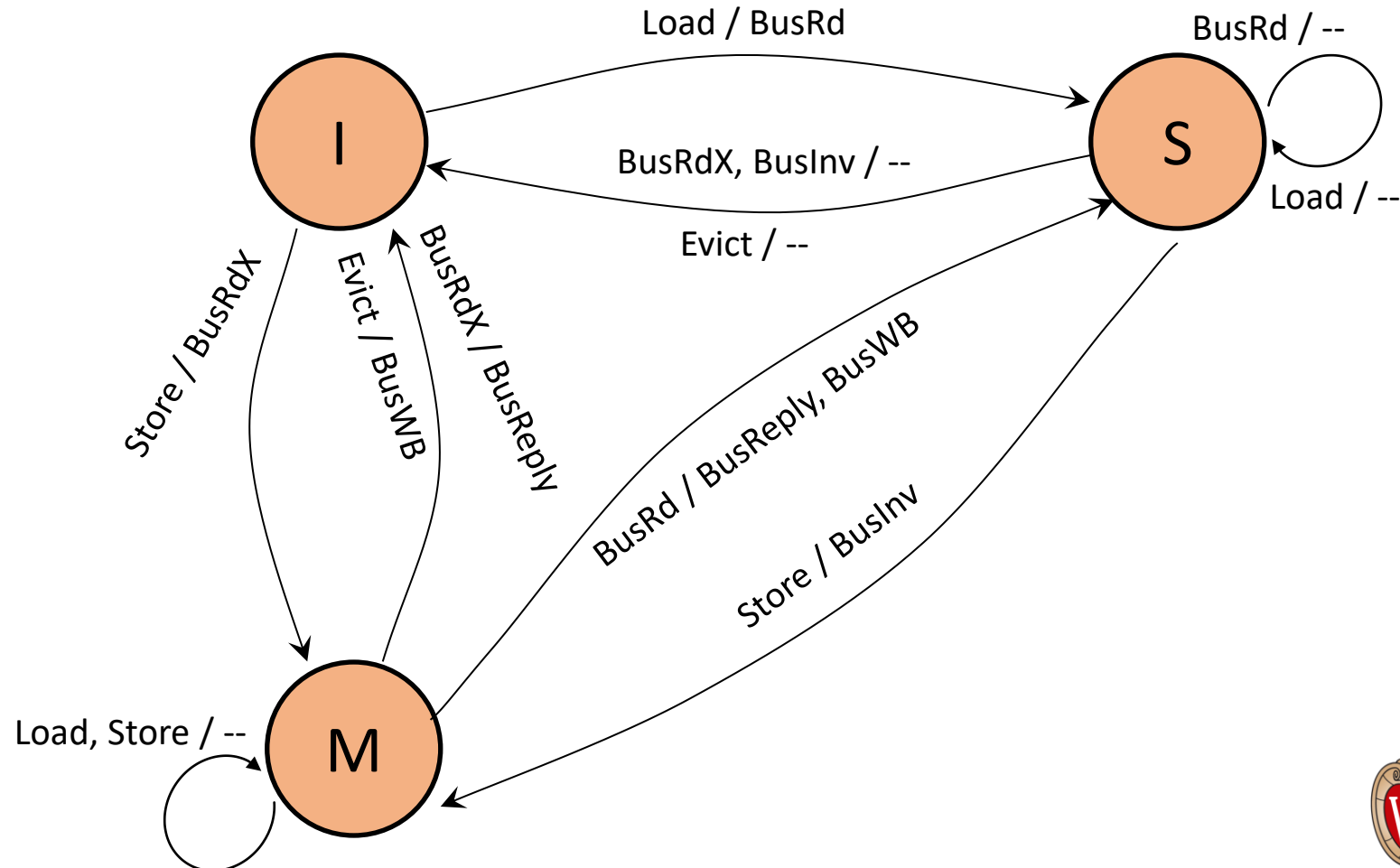
this directory services all
addrs and tracks which blocks
are located in both private L1

Directories



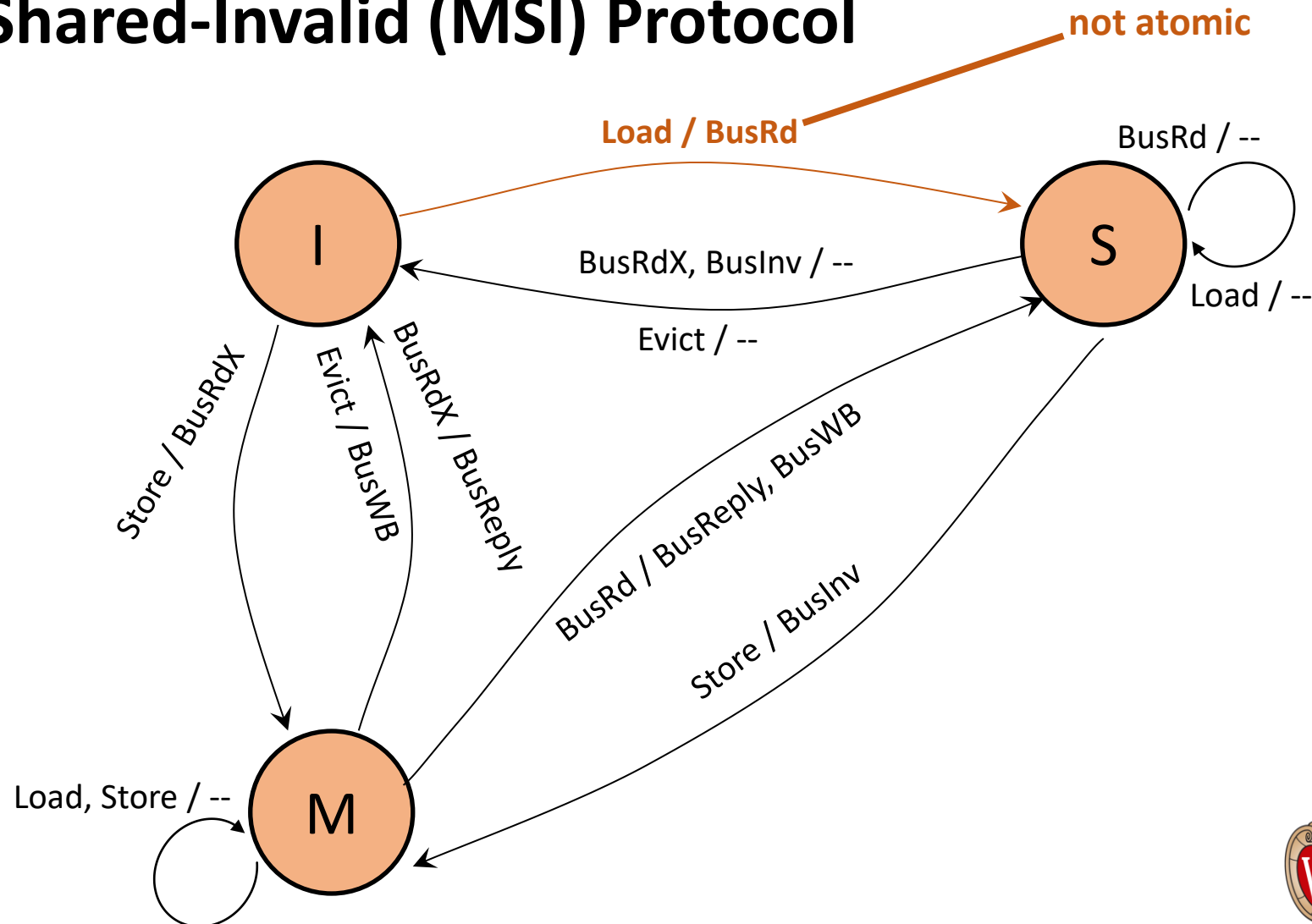
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Modified-Shared-Invalid (MSI) Protocol



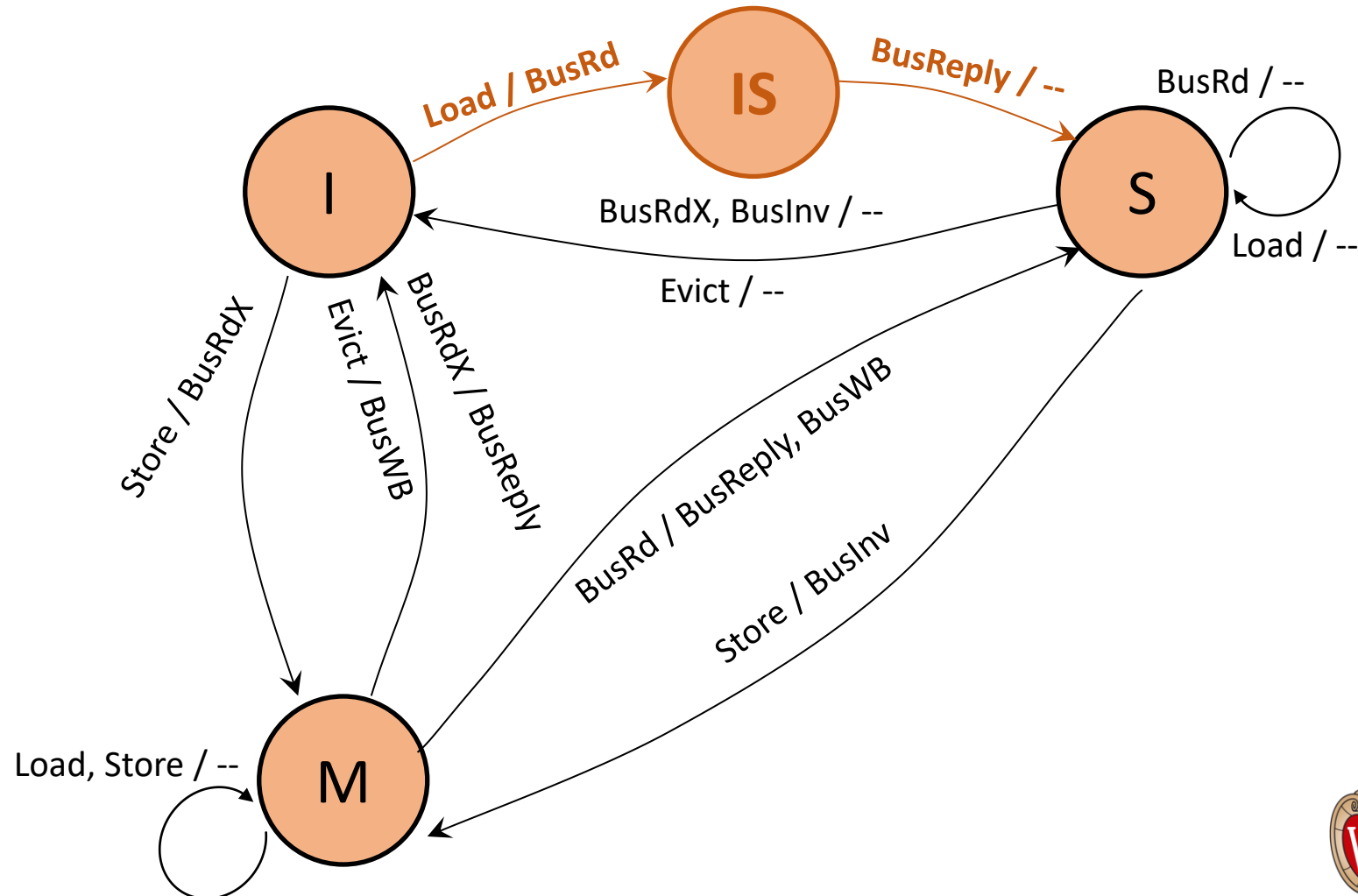
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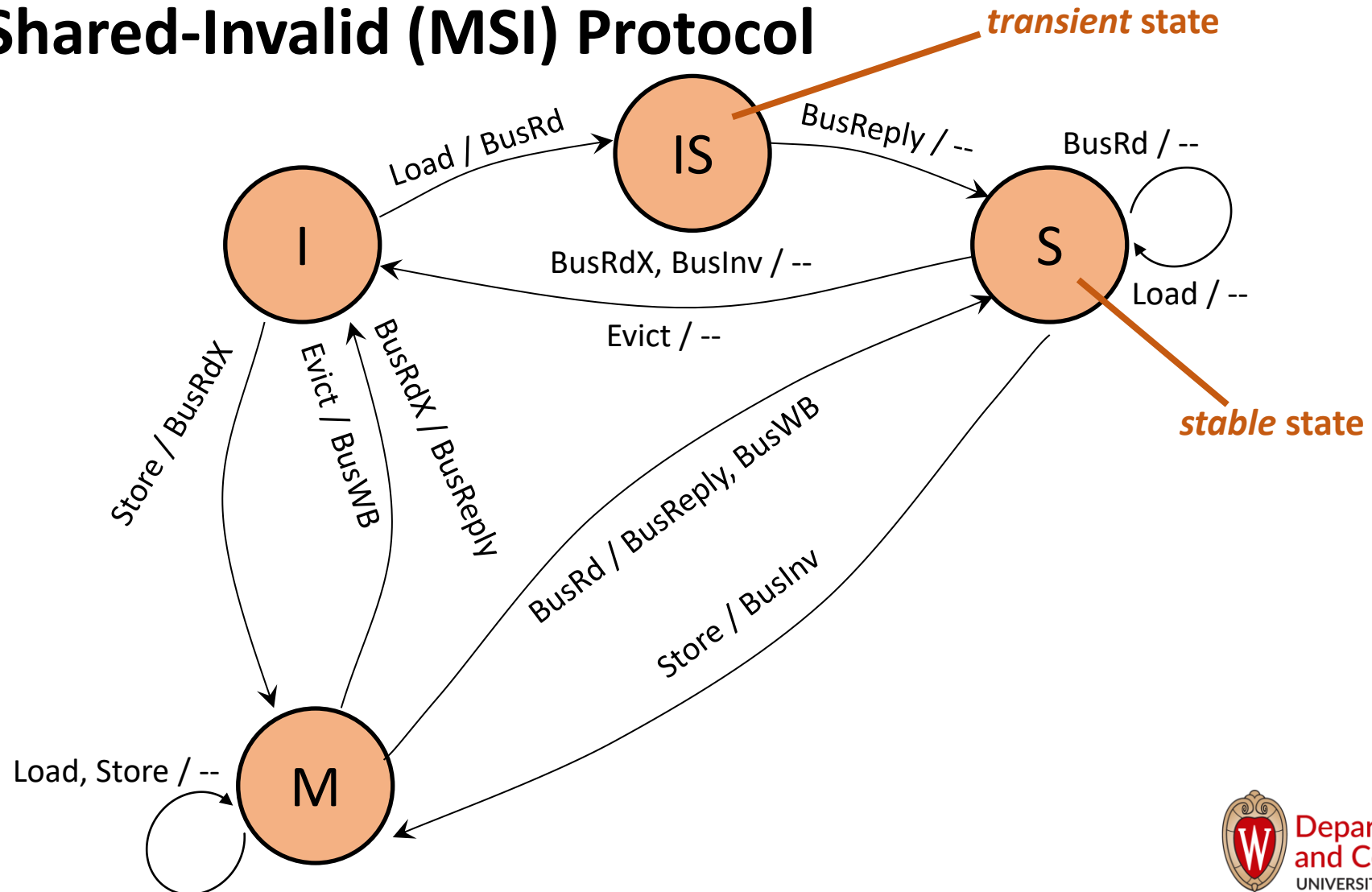
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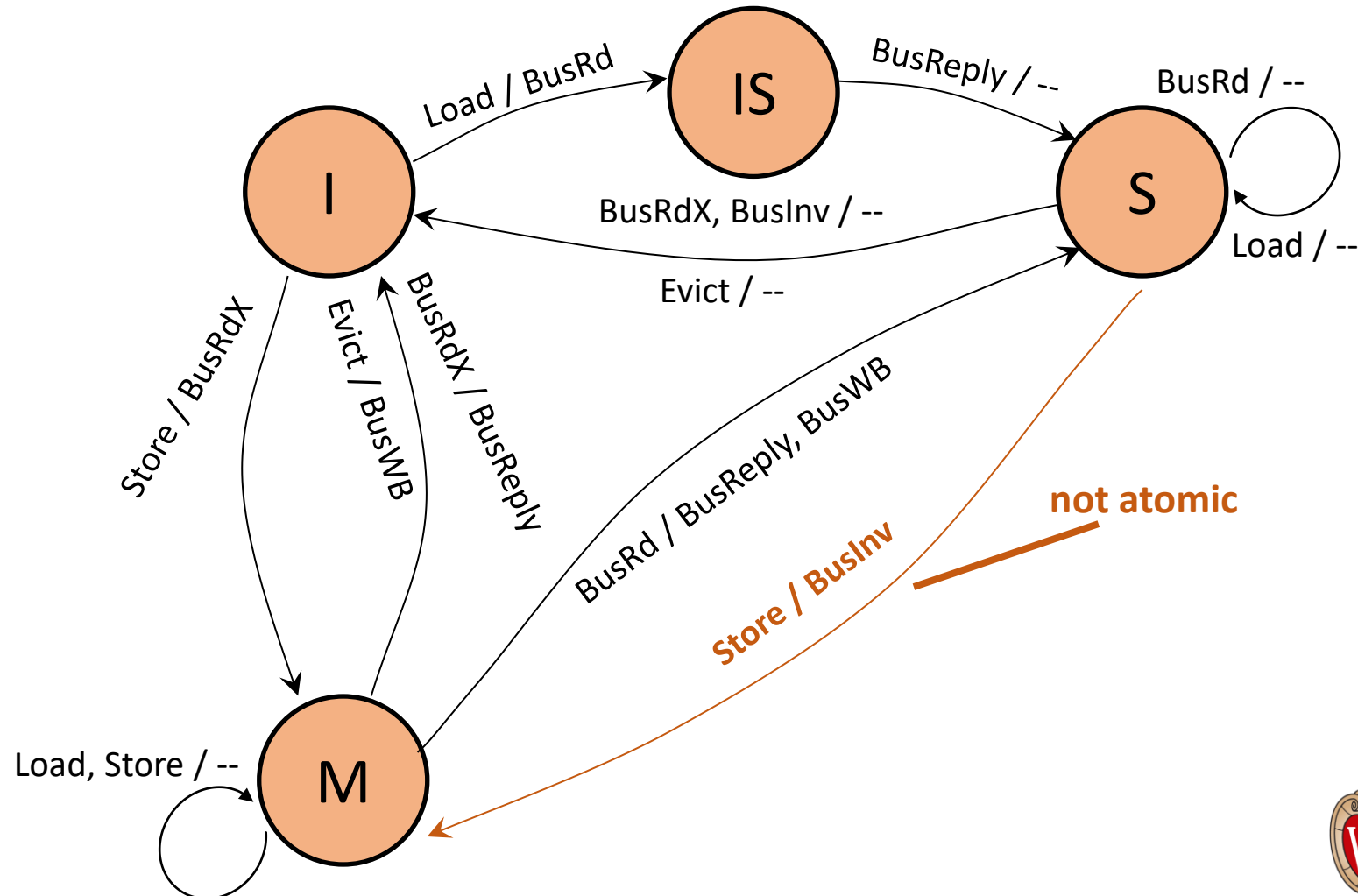
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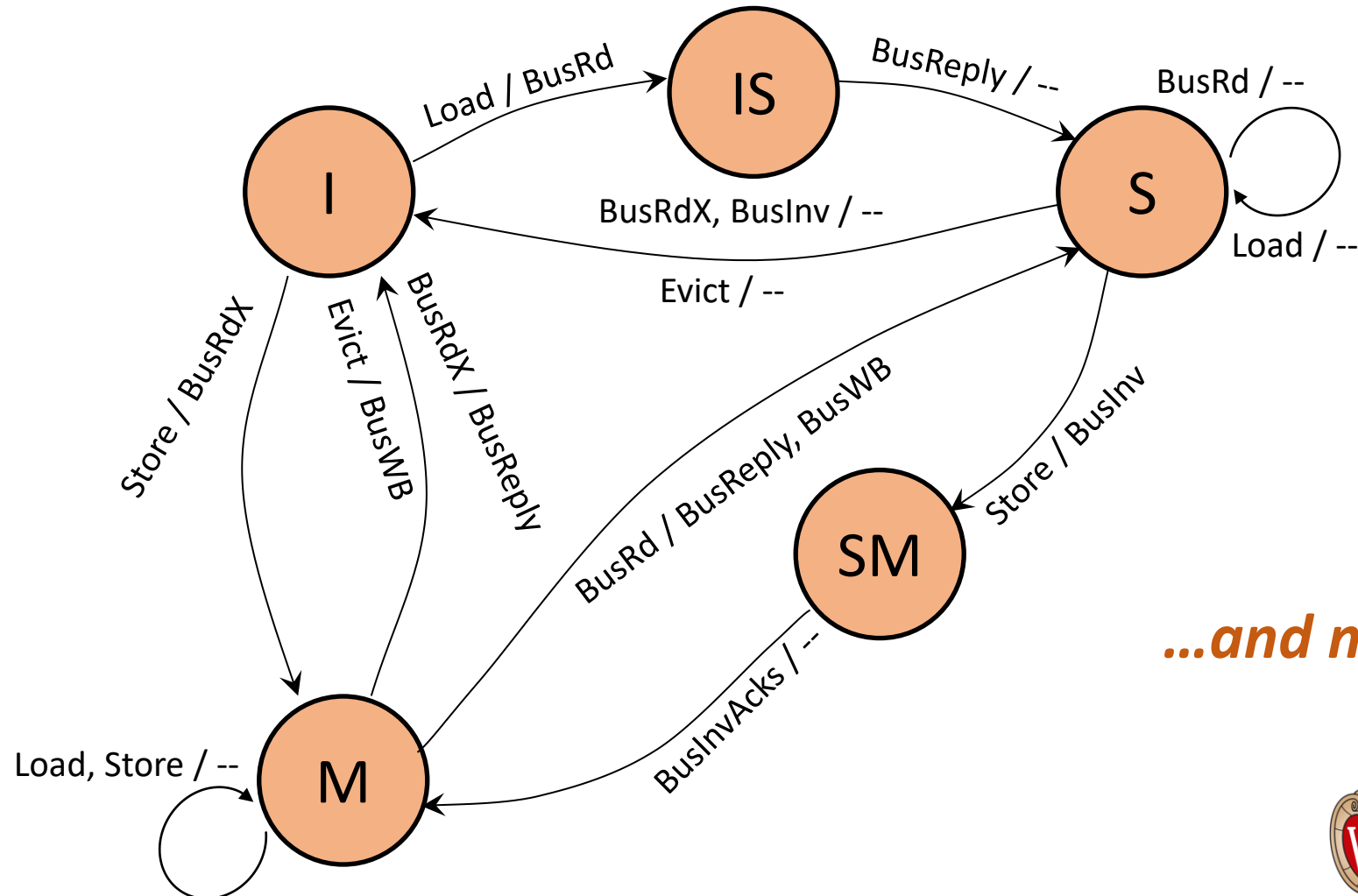
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Directory-Based Coherence

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...and many more!

Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

- Write-back, write-allocate caches

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Modified-Shared-Invalid (MSI) Protocol

- Write-back, write-allocate caches

Processor to cache (i.e., *mandatory* requests):

- *Load*: Read block in cache
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- *GetS*: Request read access to block
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Directory to cache:

- *Fwd-GetS*: Tell owner to send block to sharer
- *Fwd-GetM*: Tell owner to send block to writer
- *Put-Ack*: Notify cache of eviction receipt
- *Inv*: Invalidate sharer cache
- *Data*: Send block to requester

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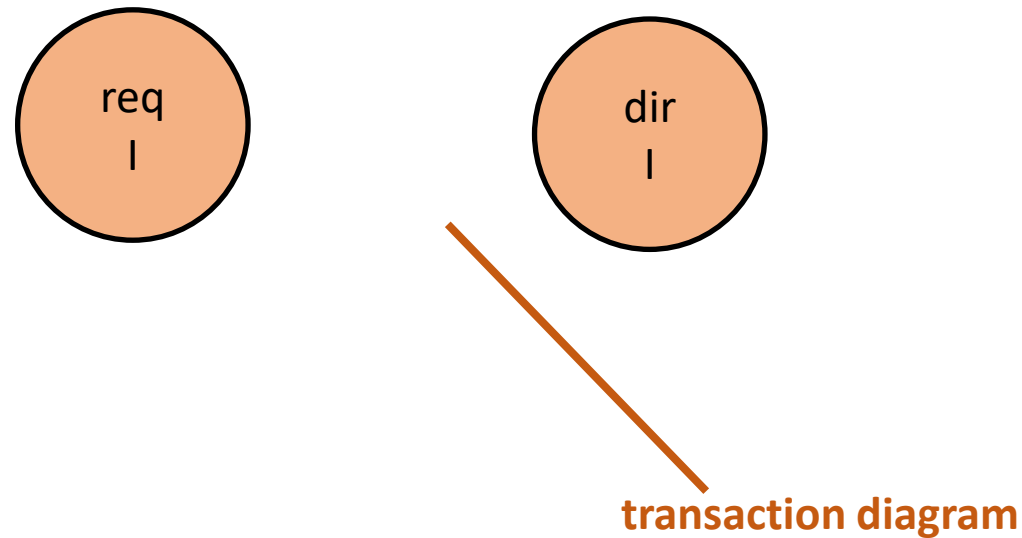
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Cache to cache:

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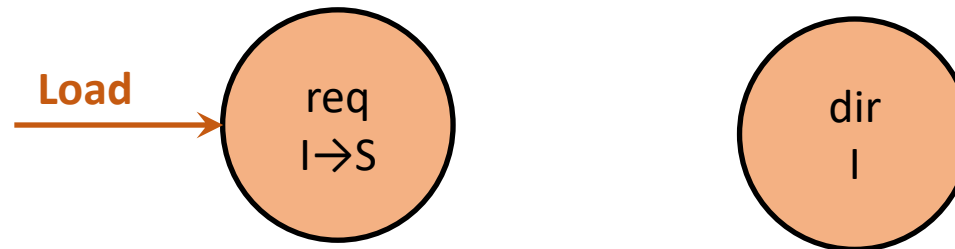
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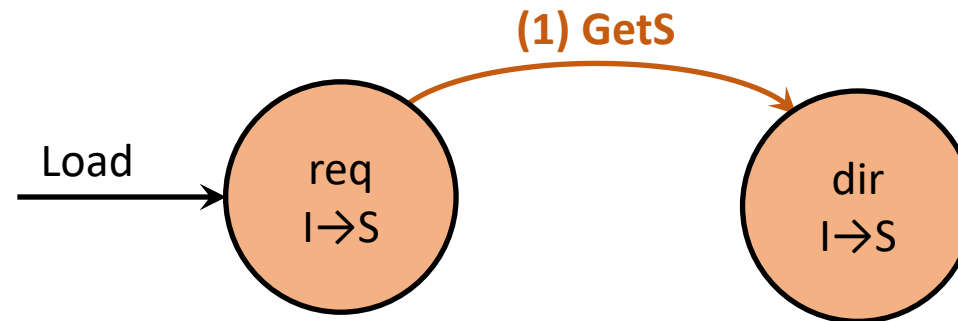
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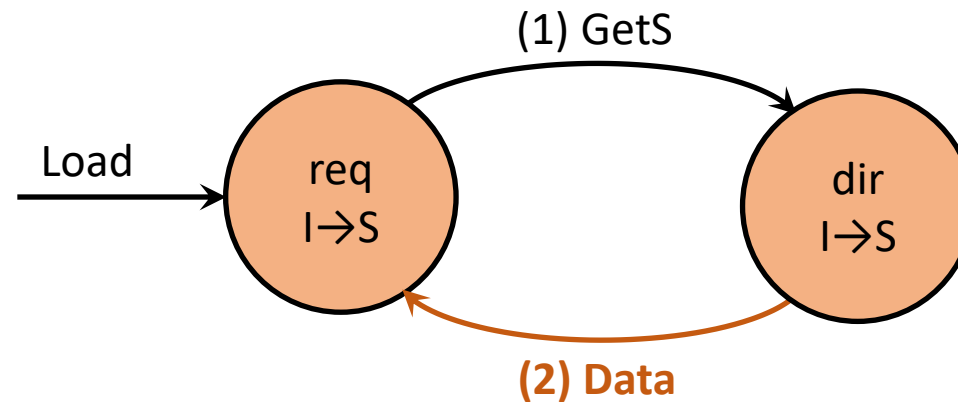
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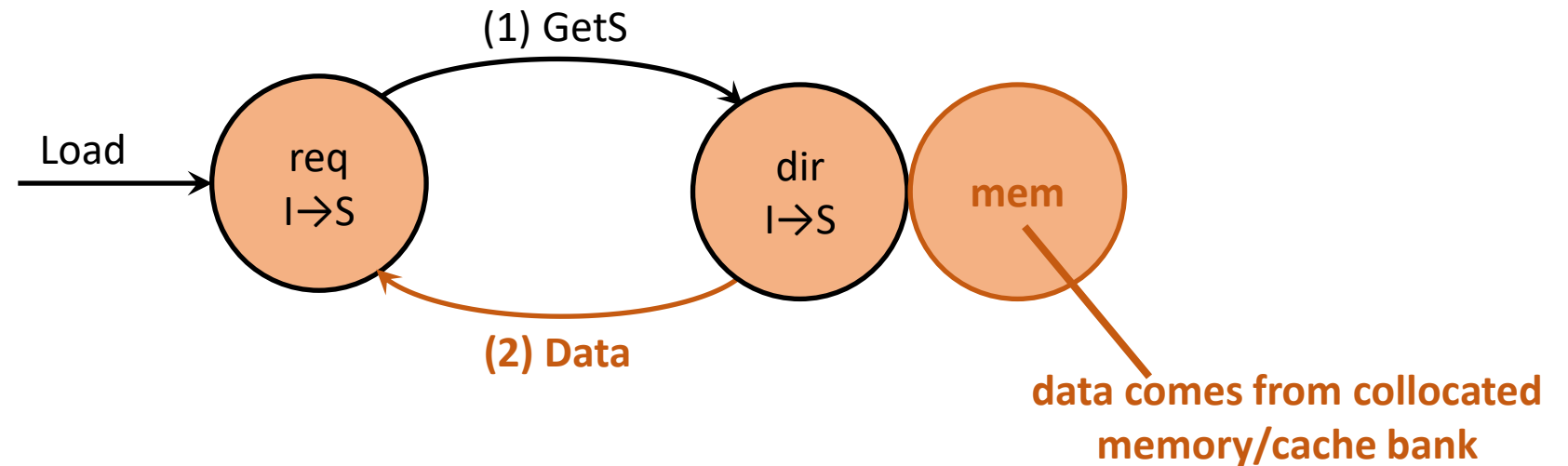
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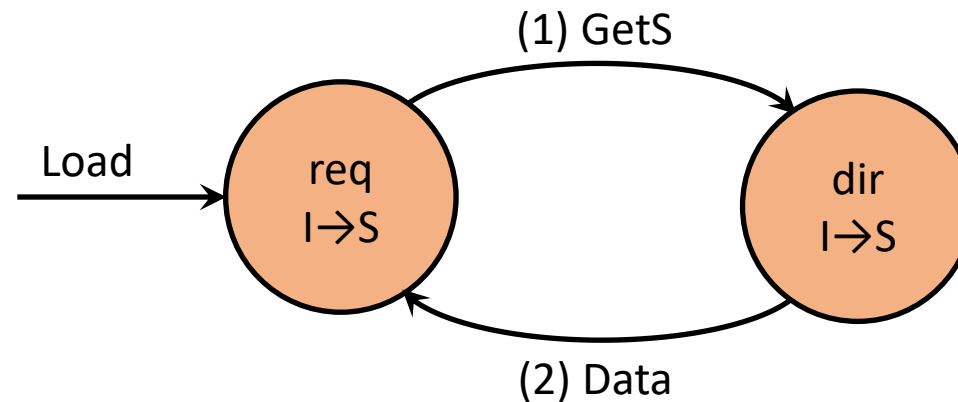
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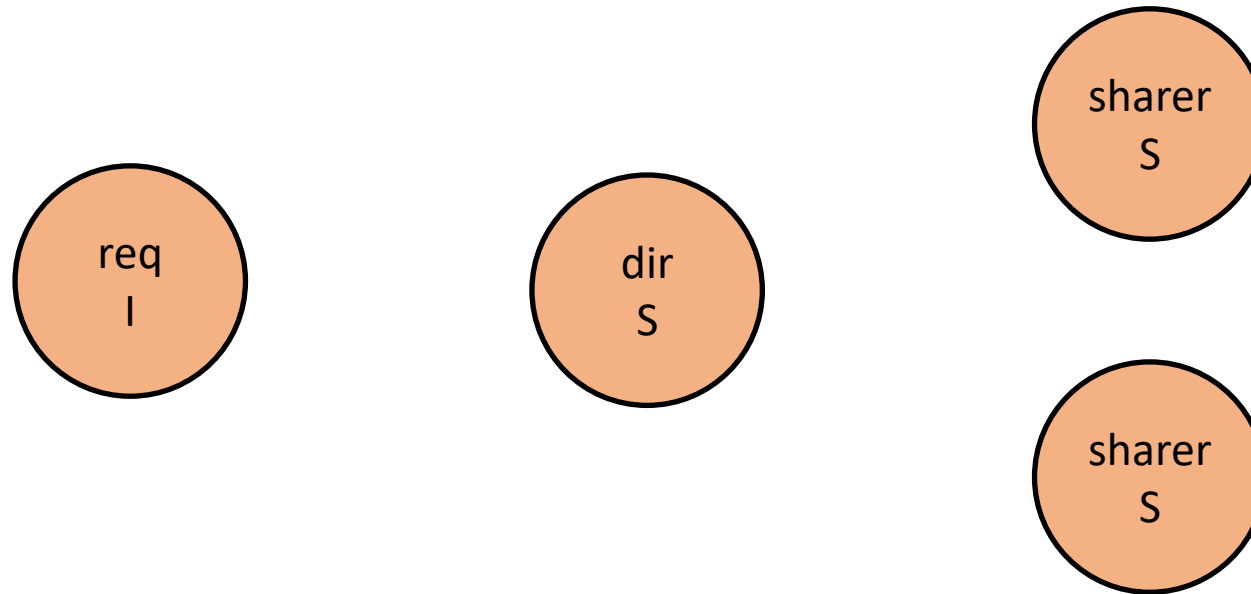
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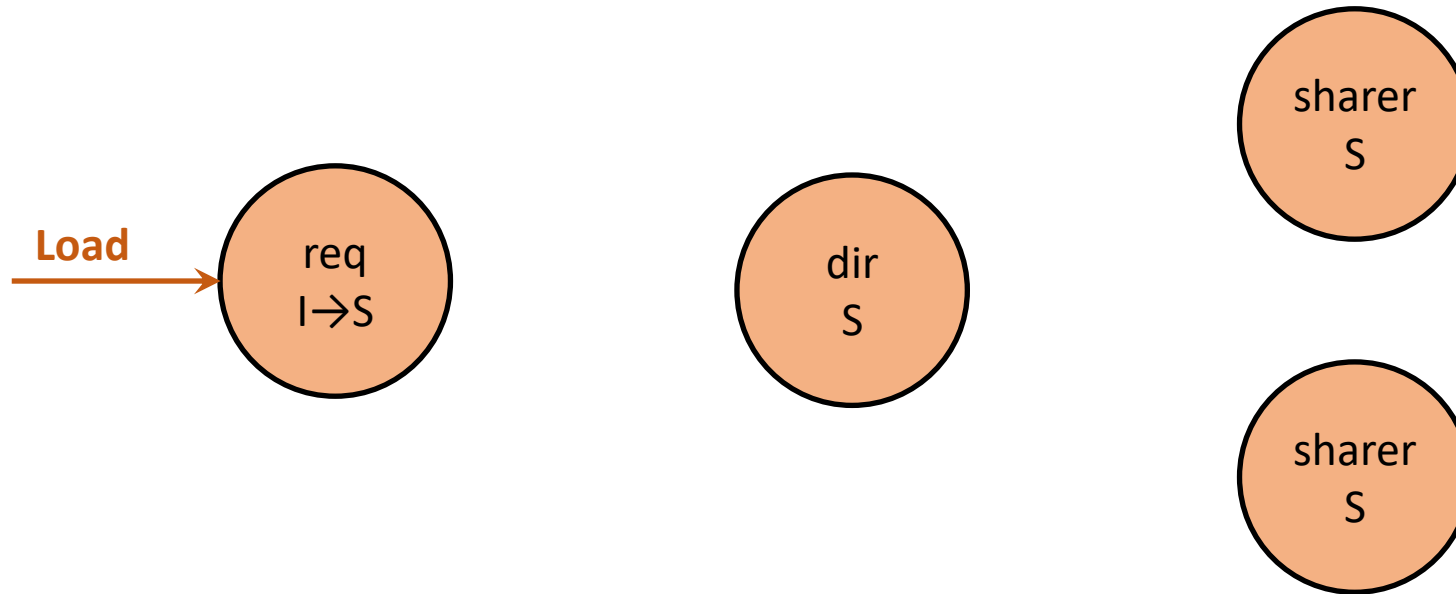
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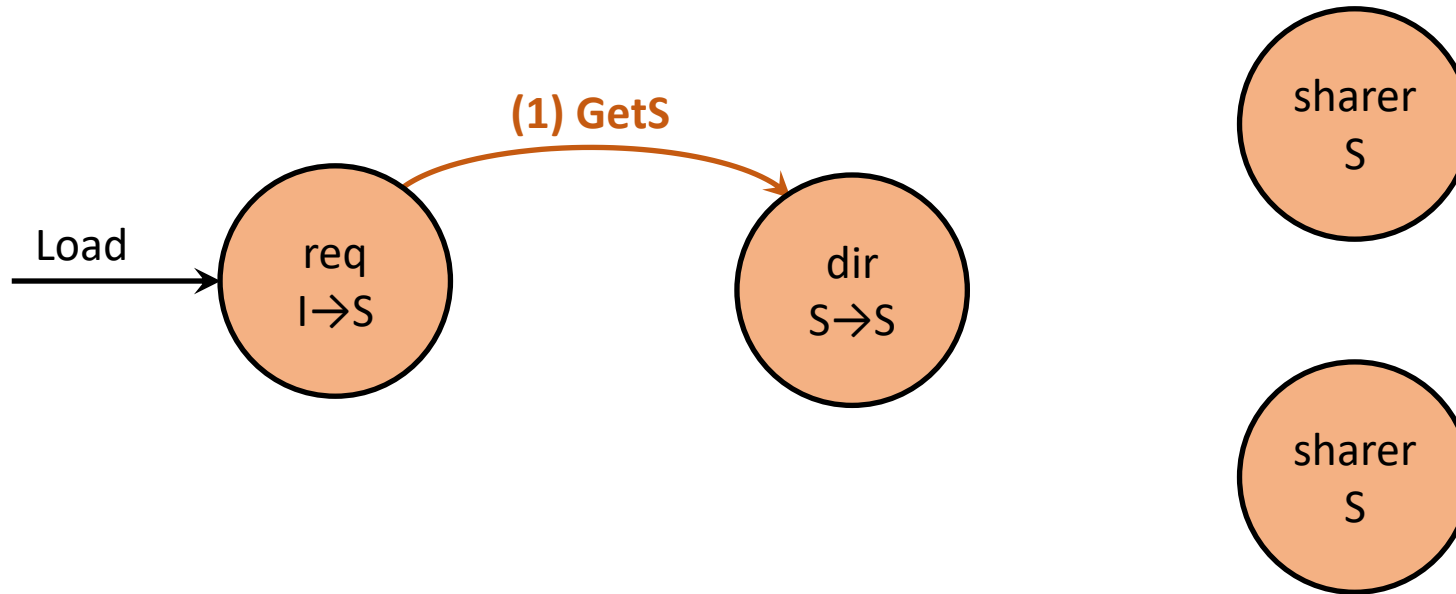
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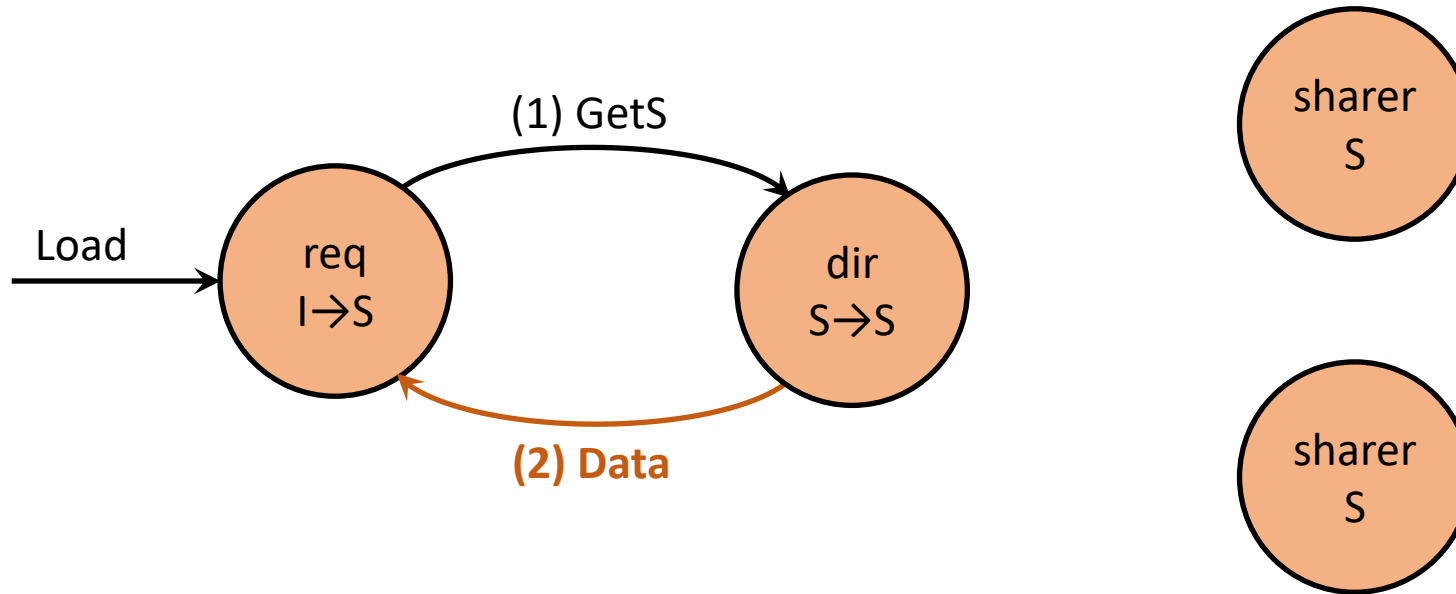
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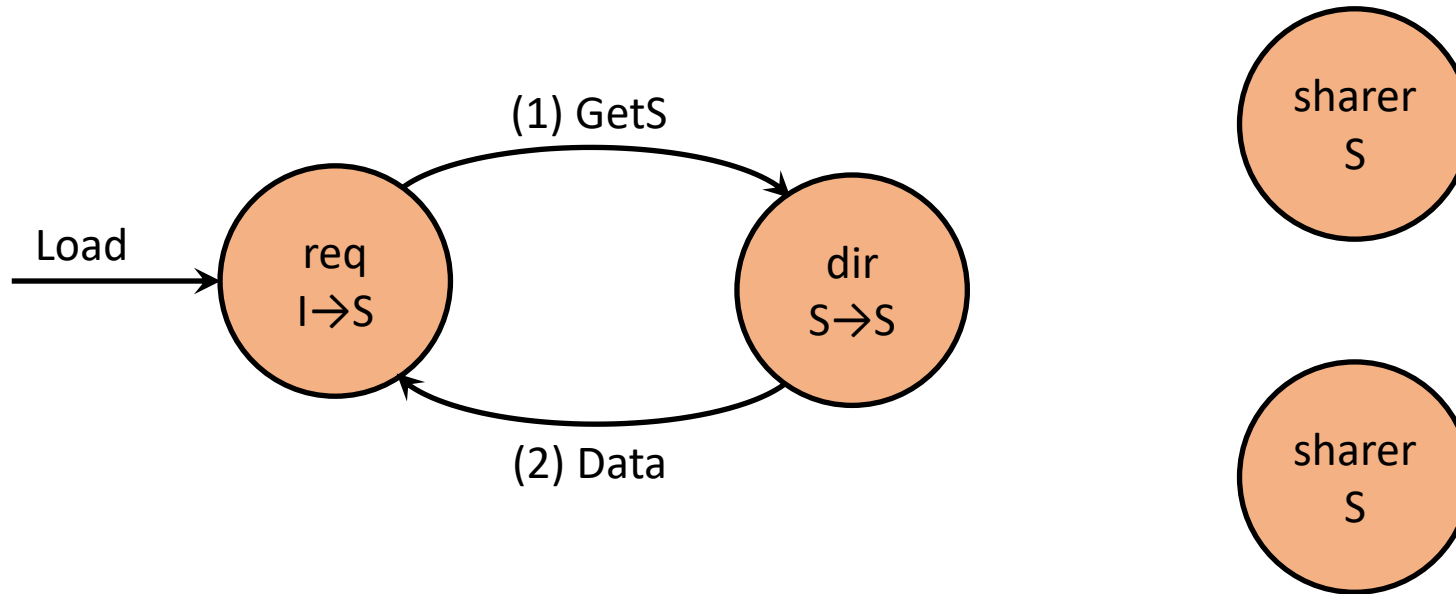
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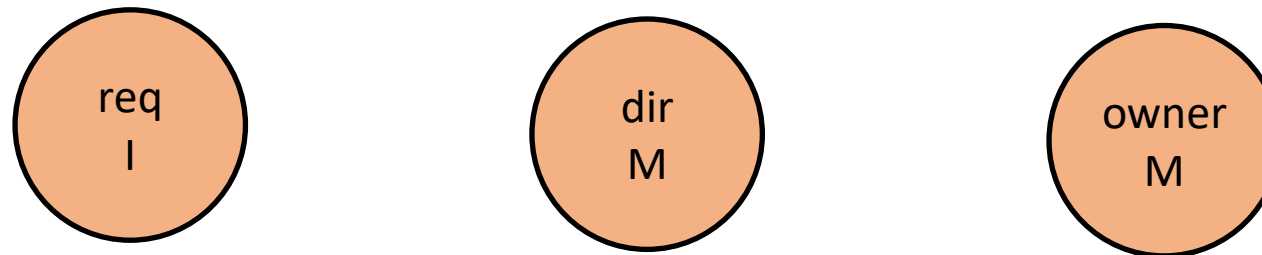
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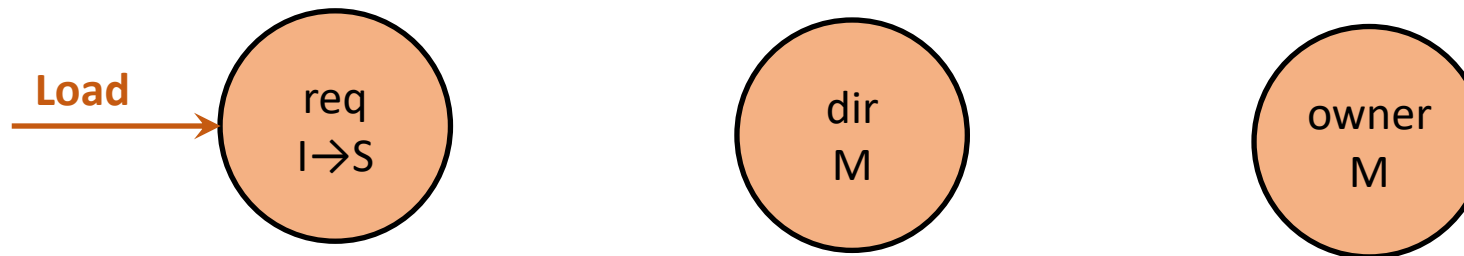
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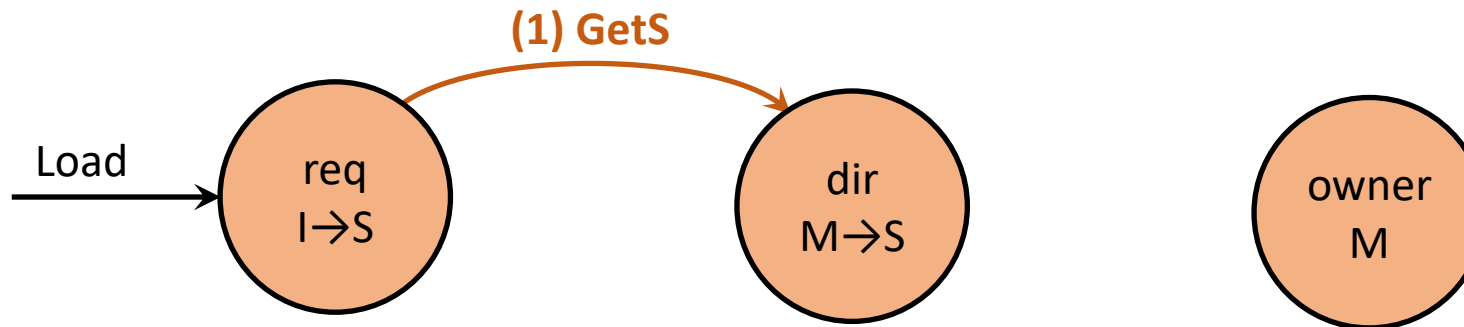
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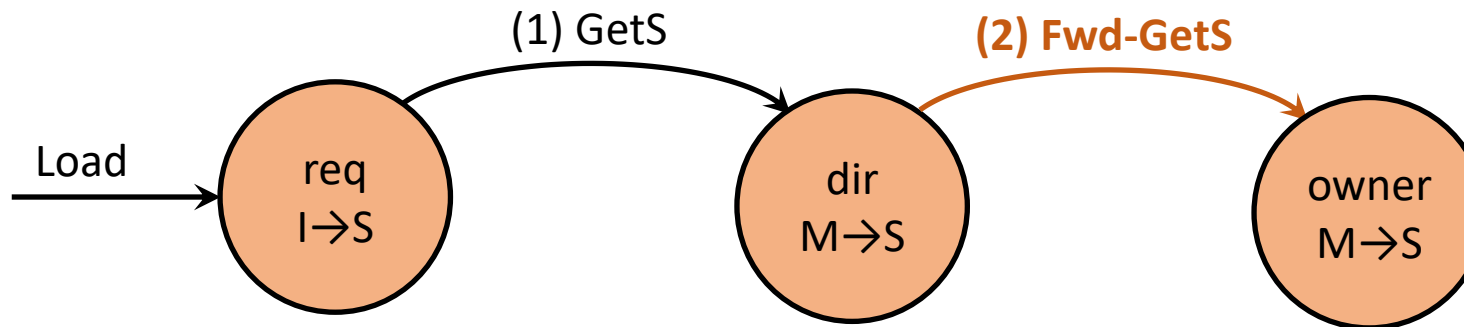
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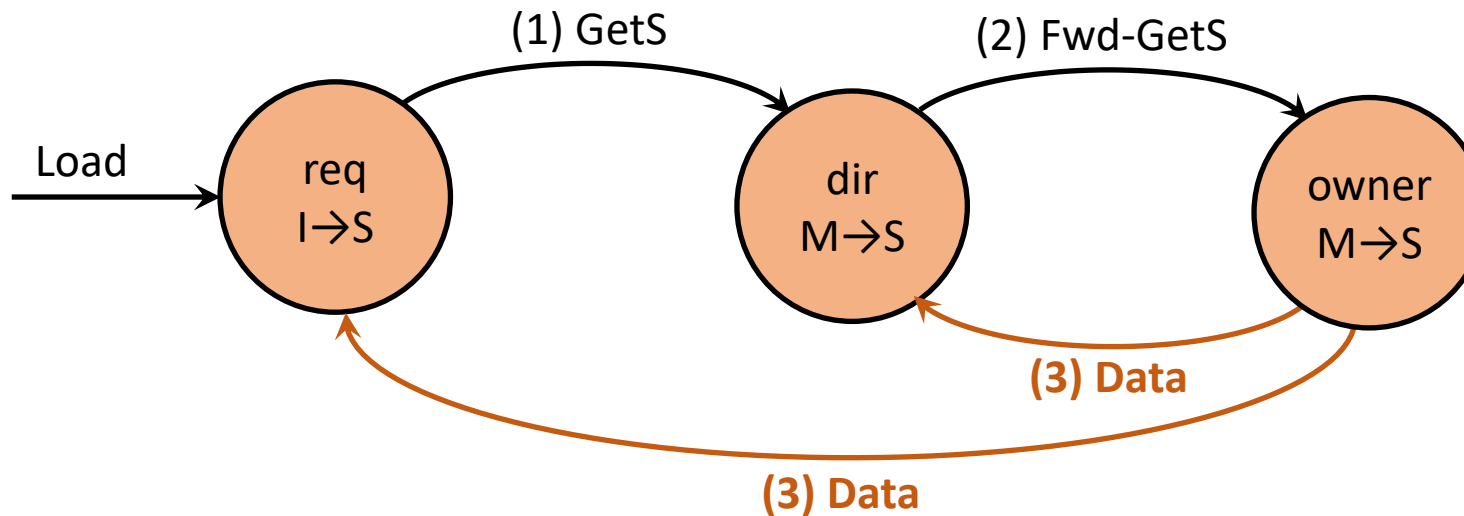
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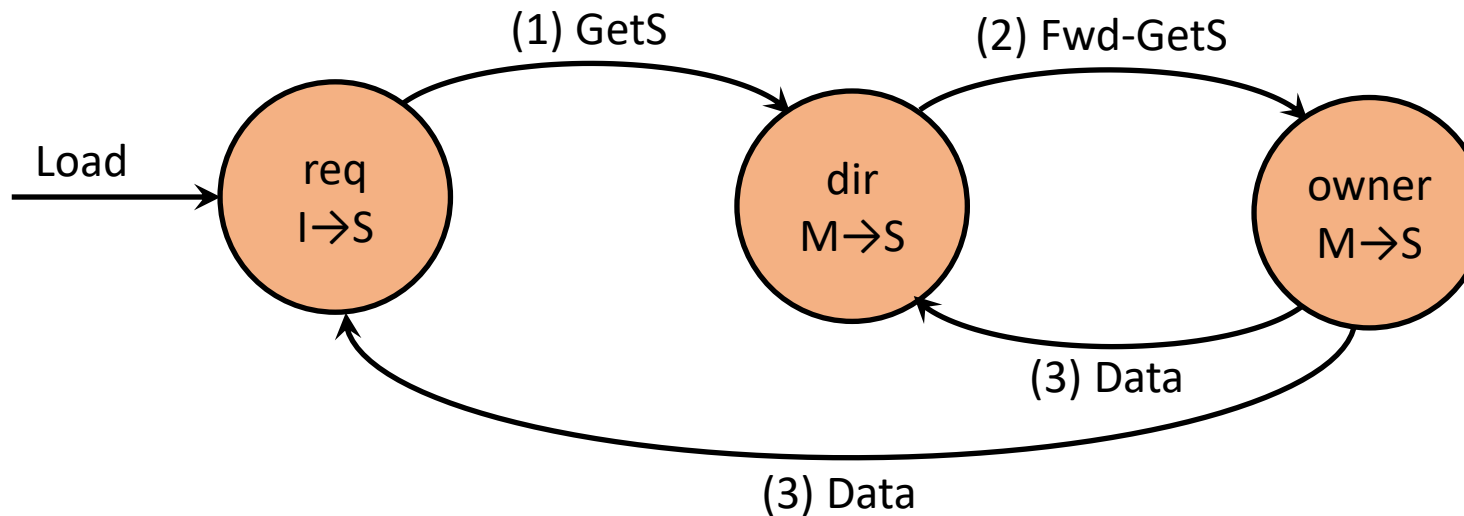
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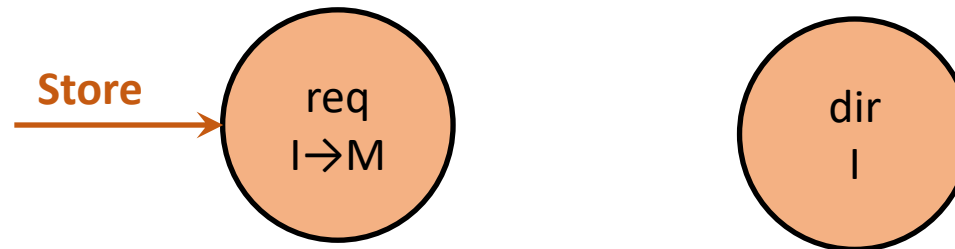
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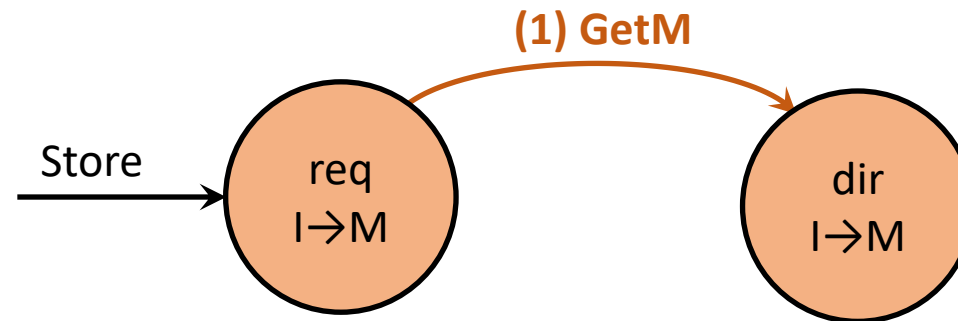
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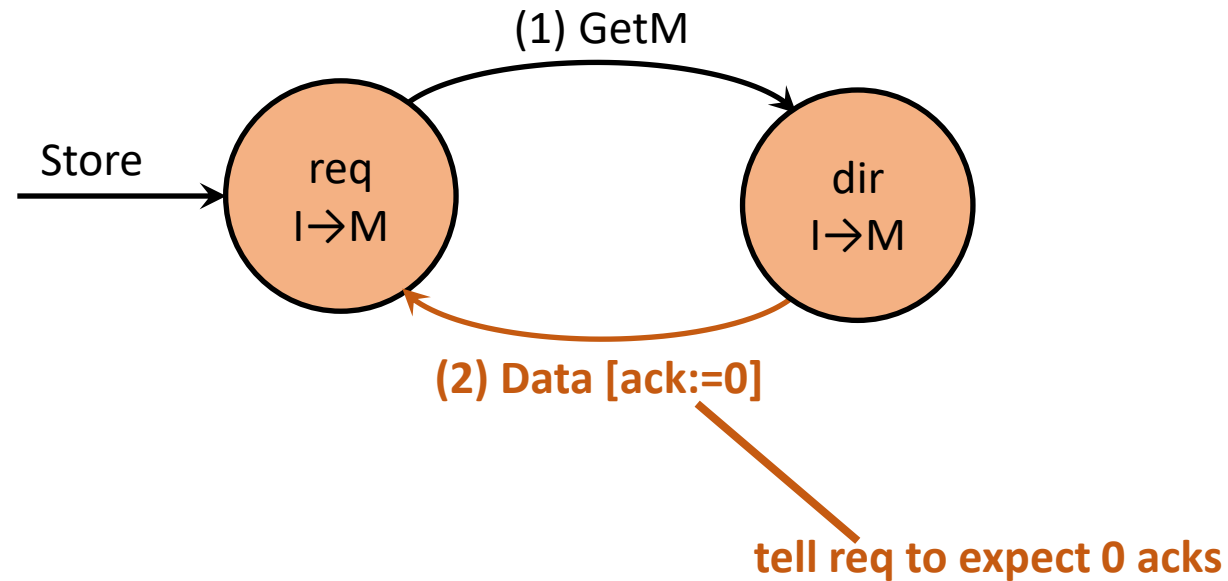
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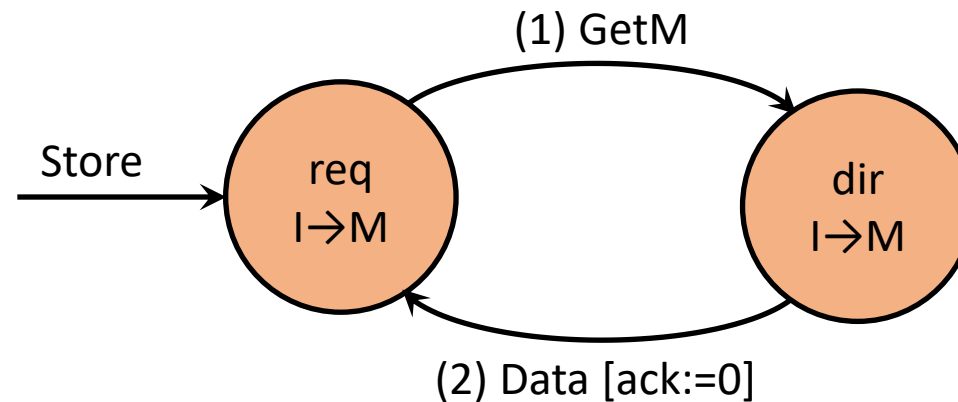
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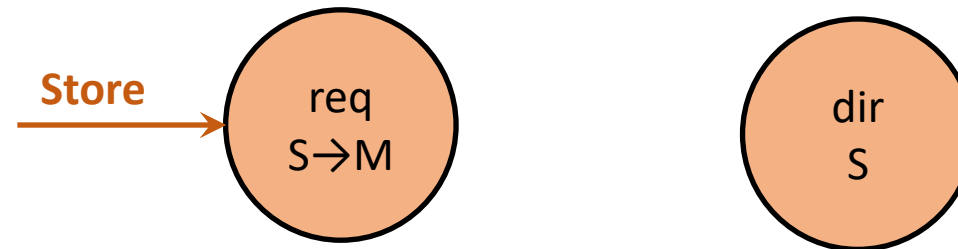
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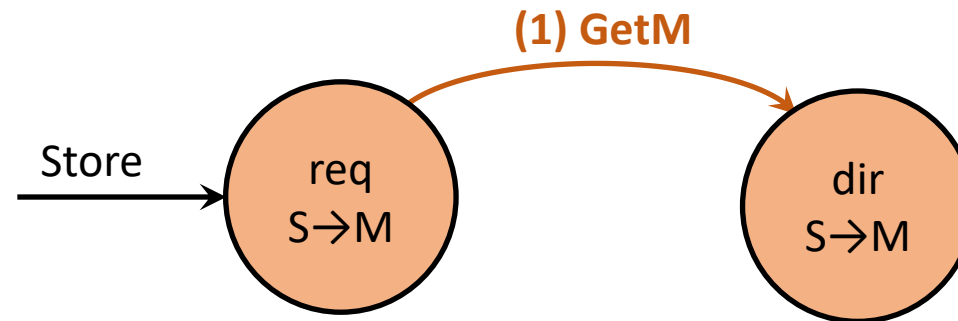
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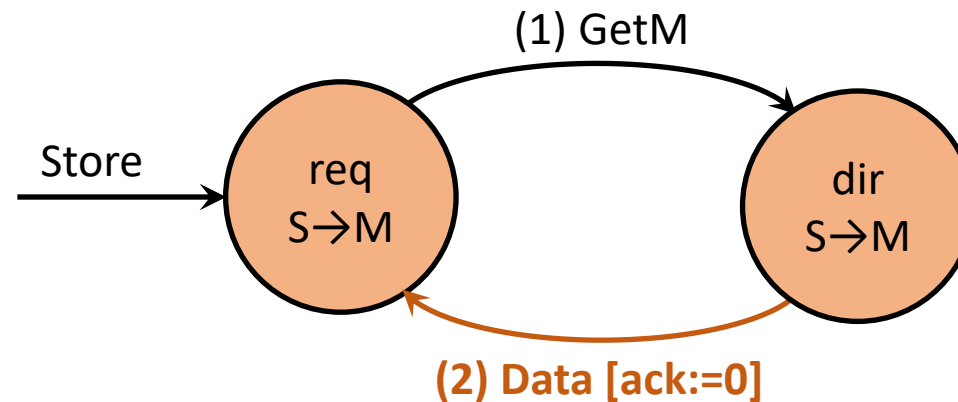
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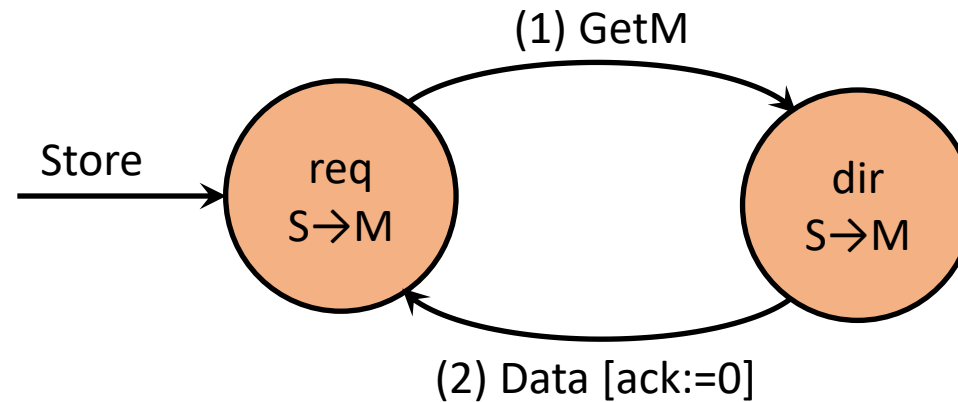
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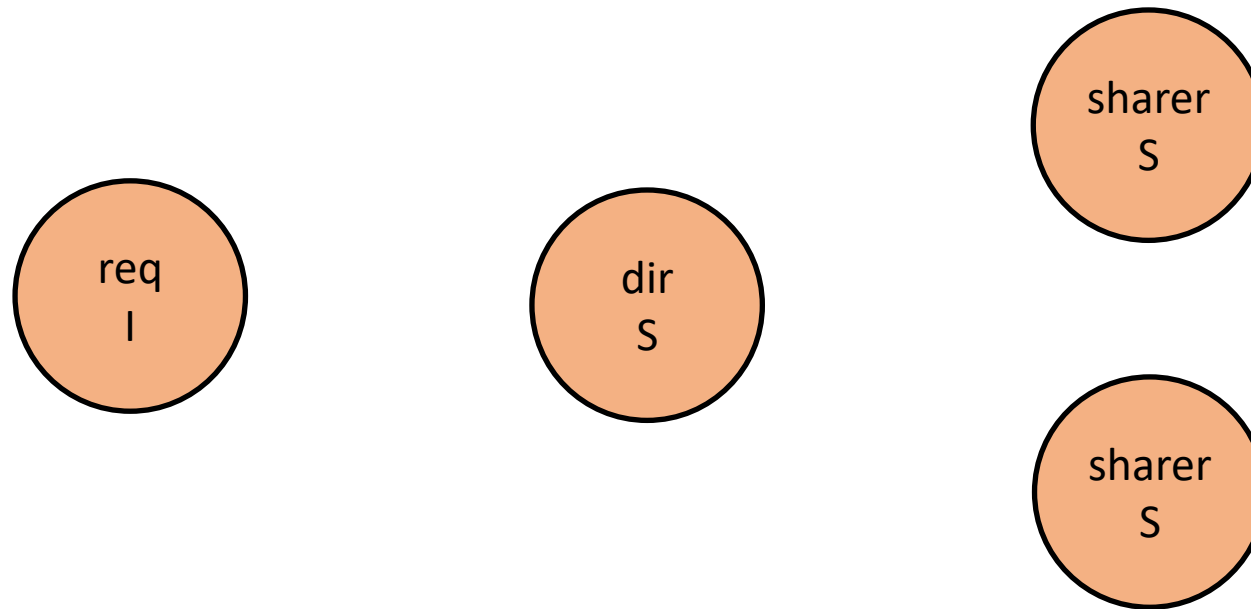
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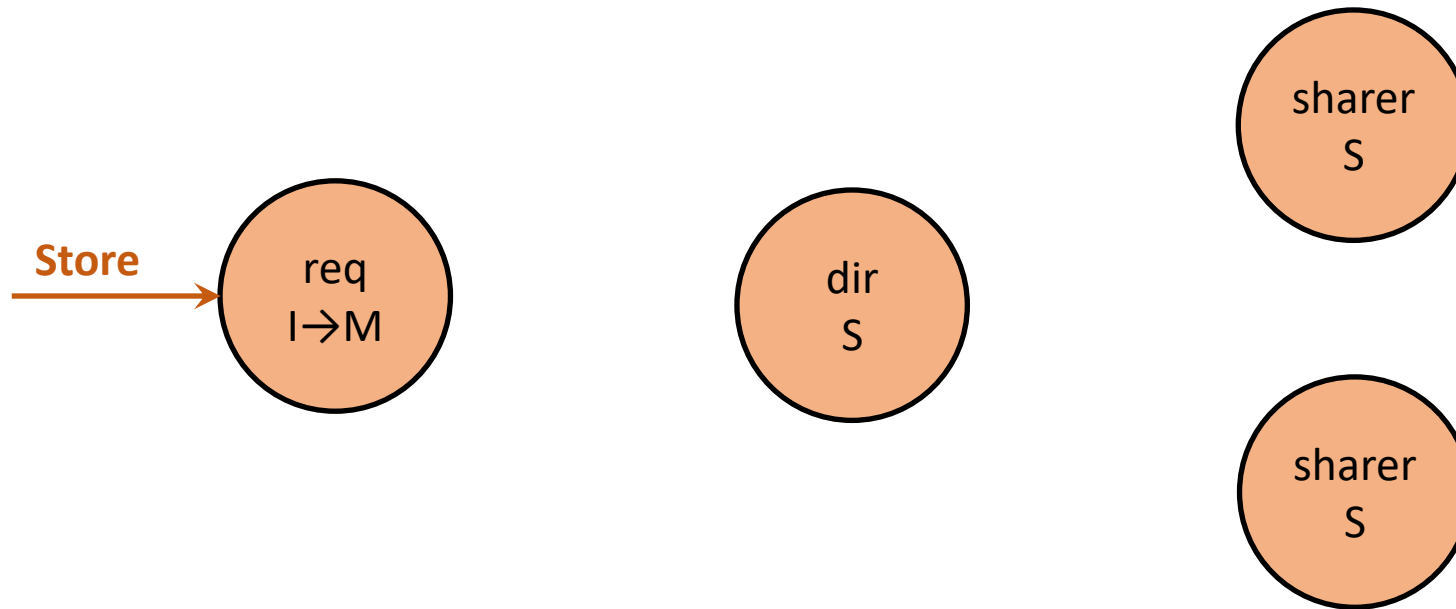
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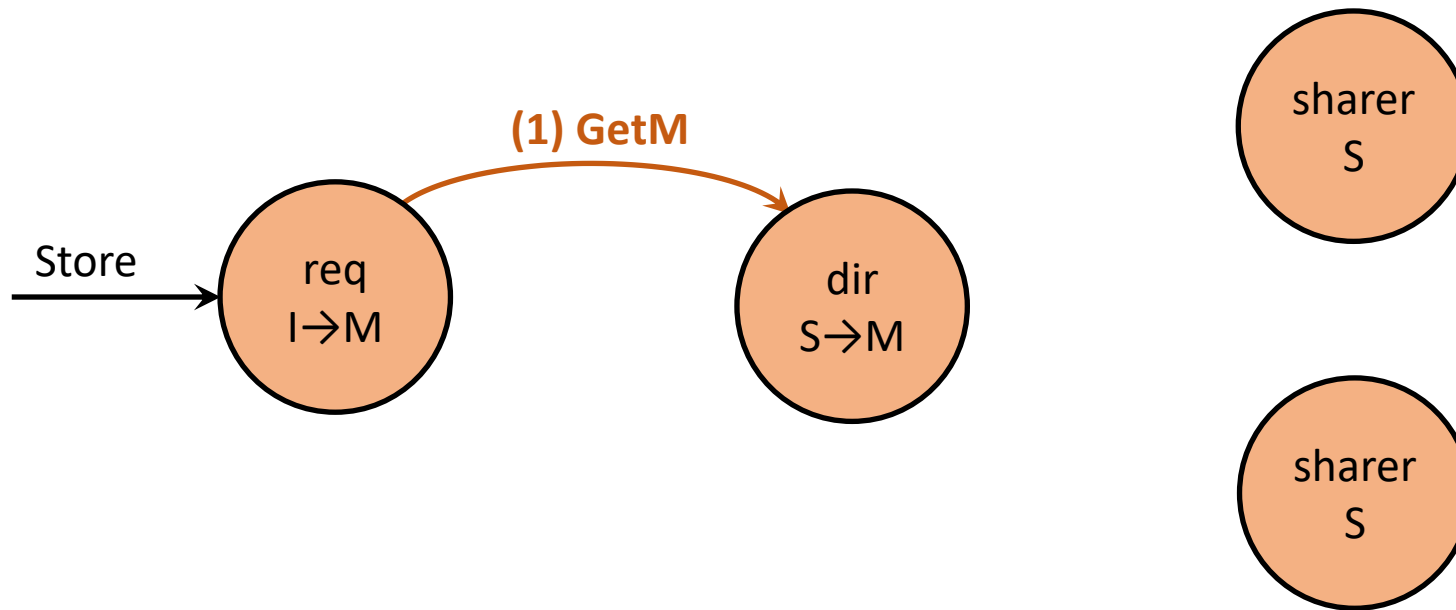
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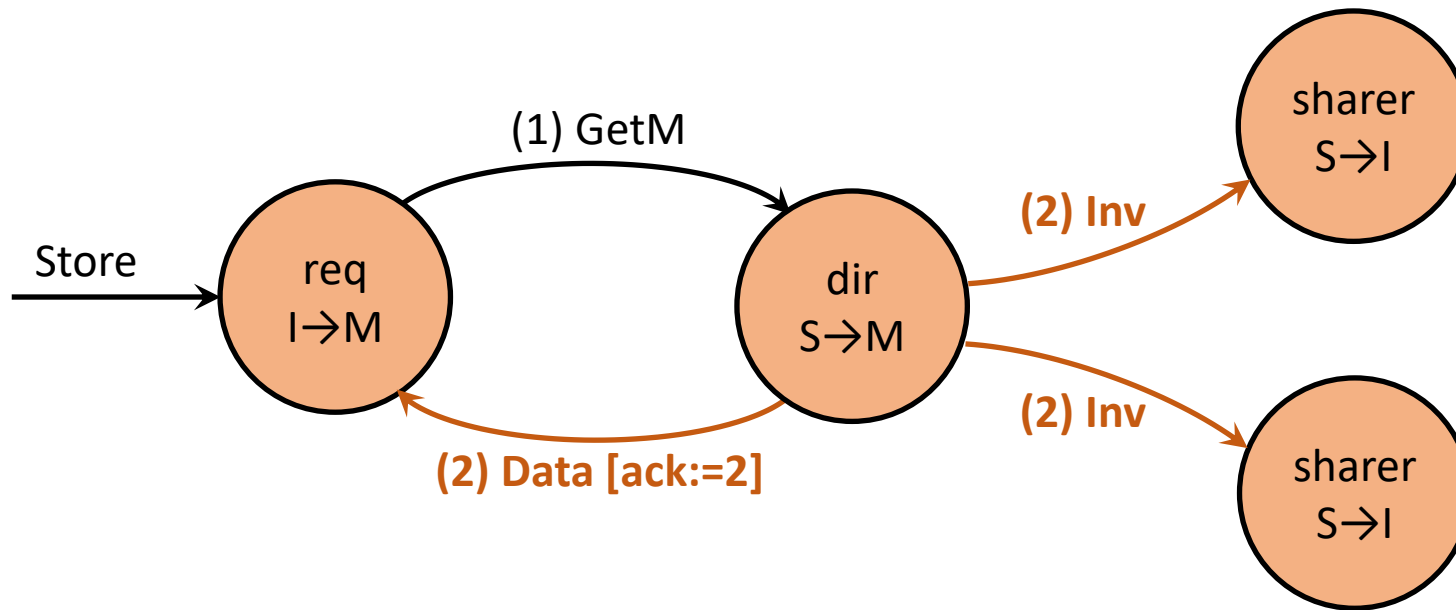
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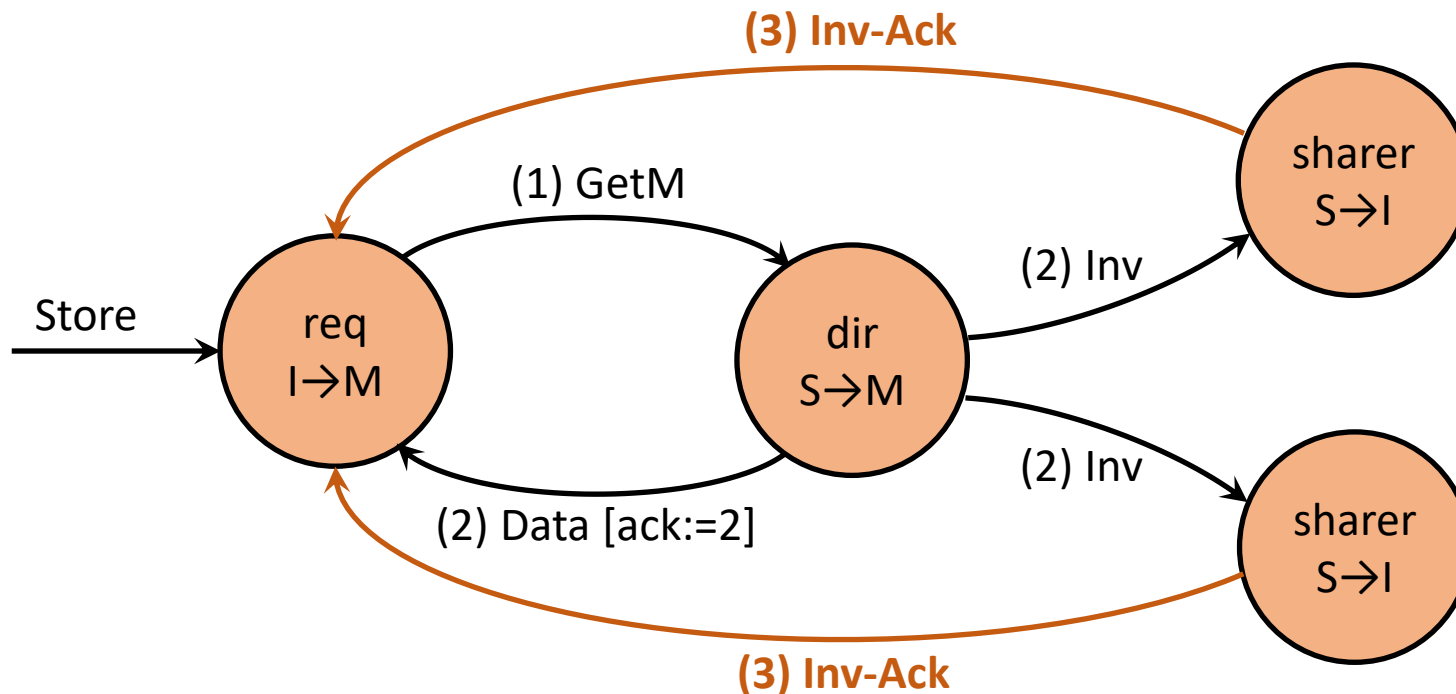
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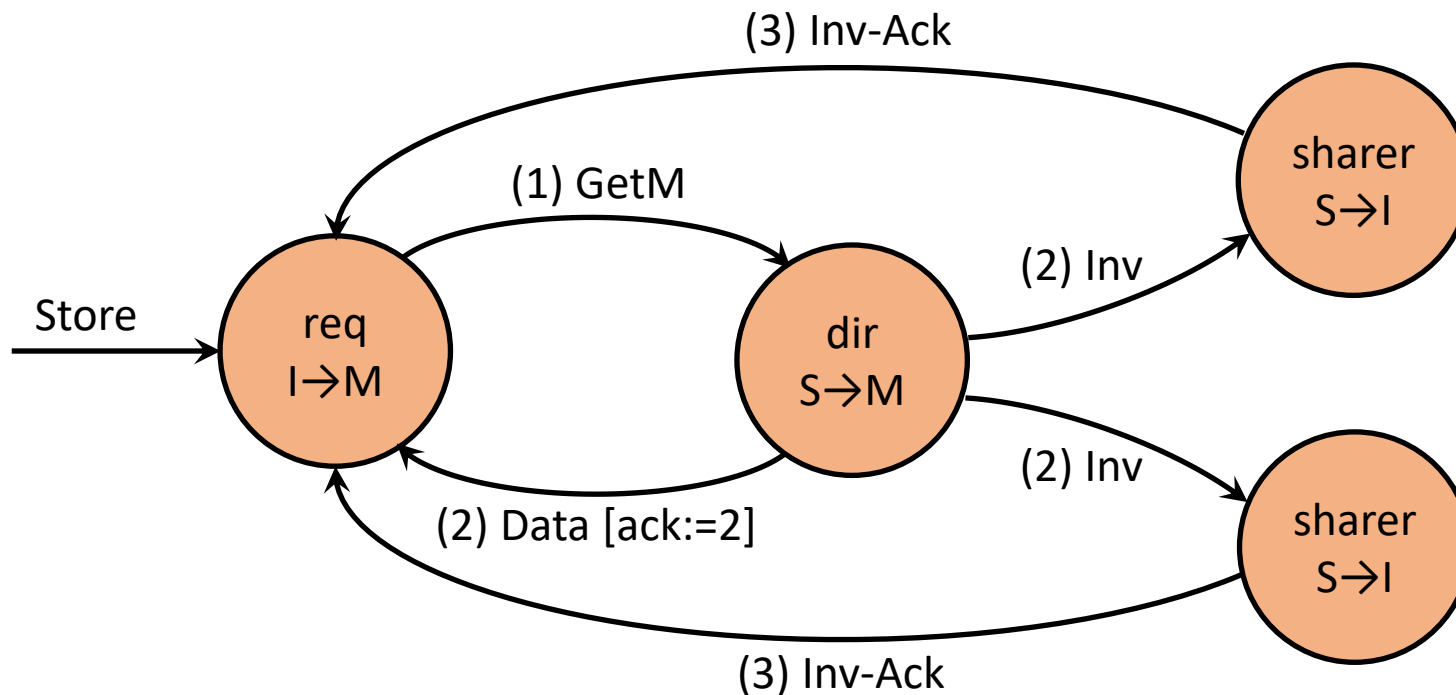
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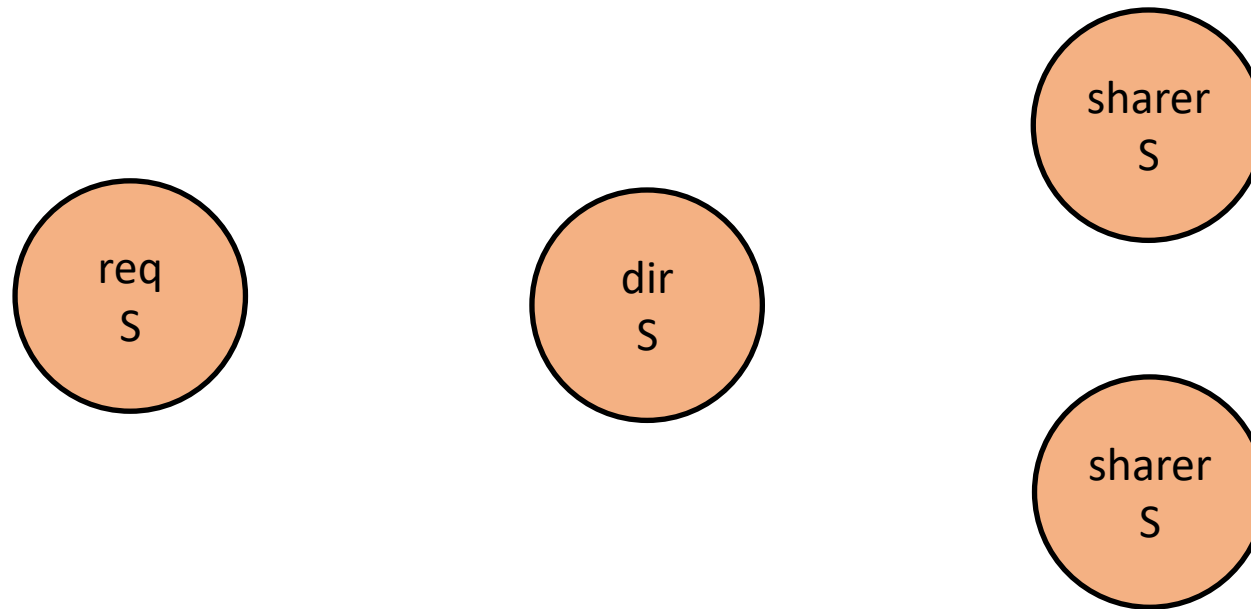
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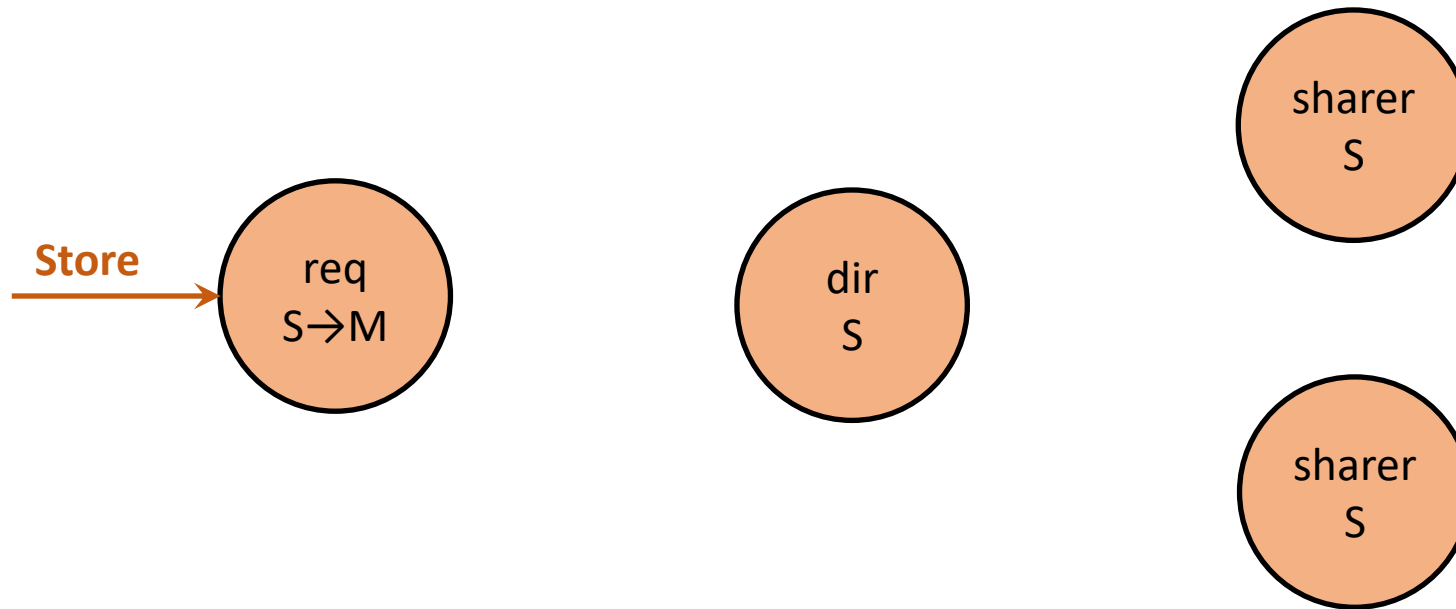
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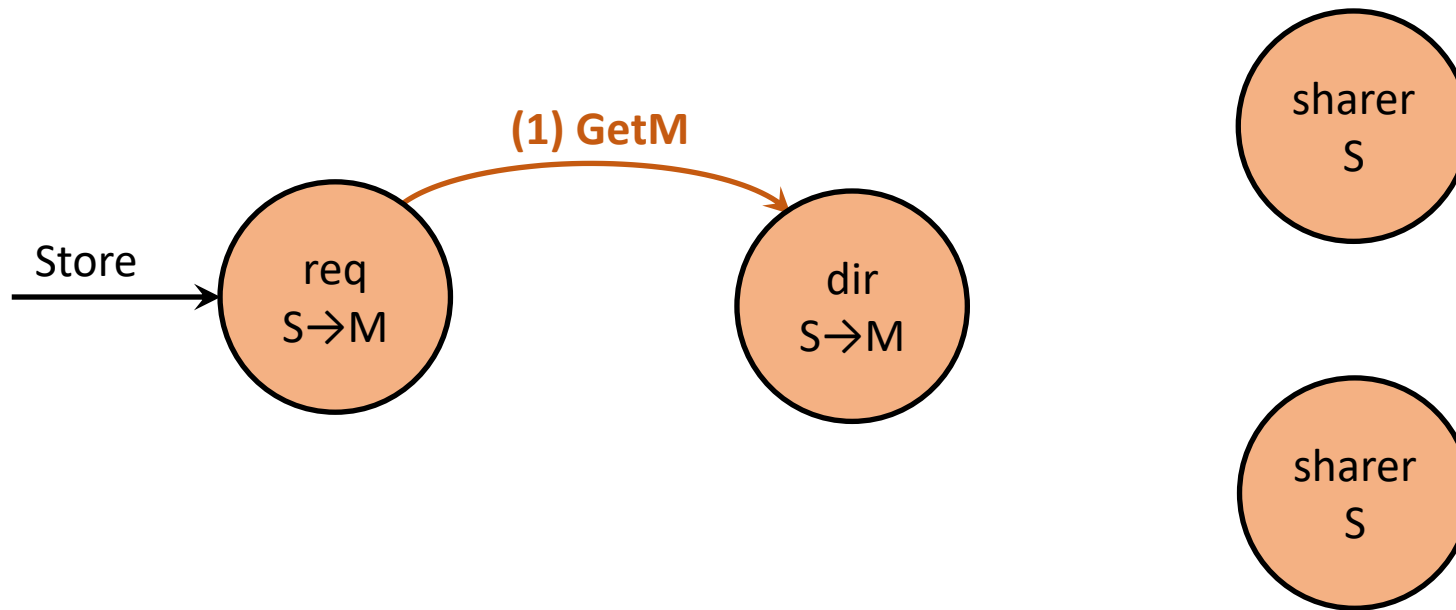
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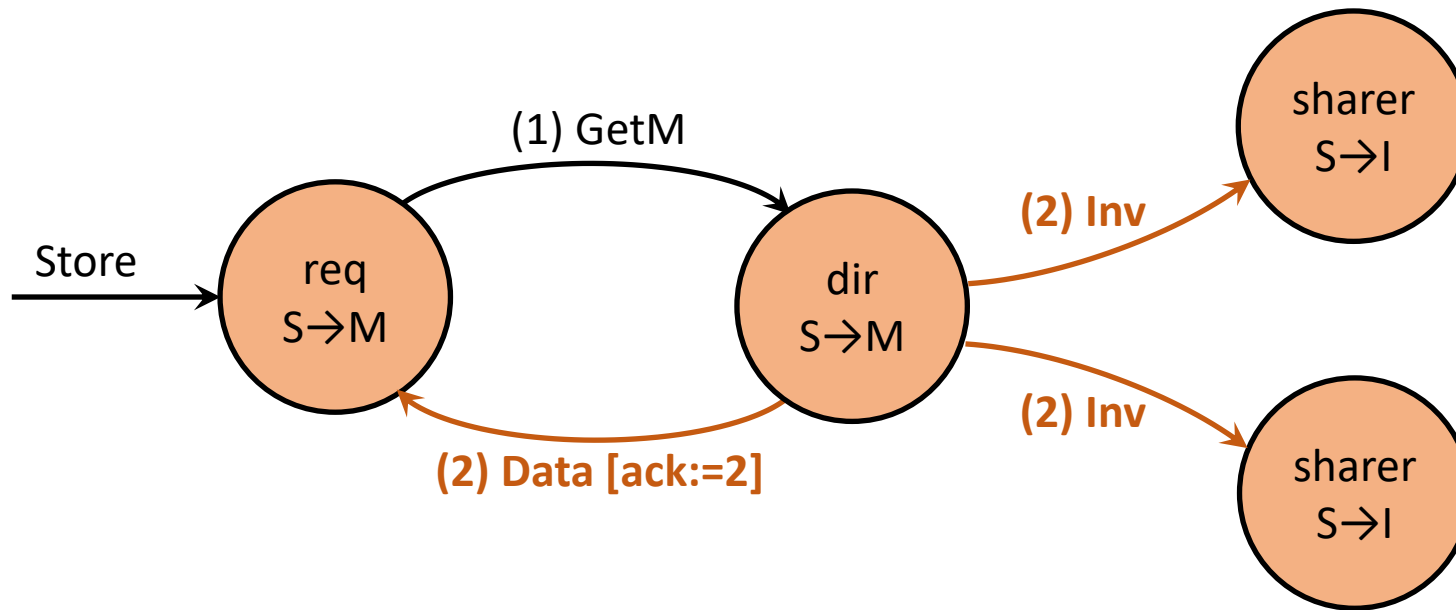
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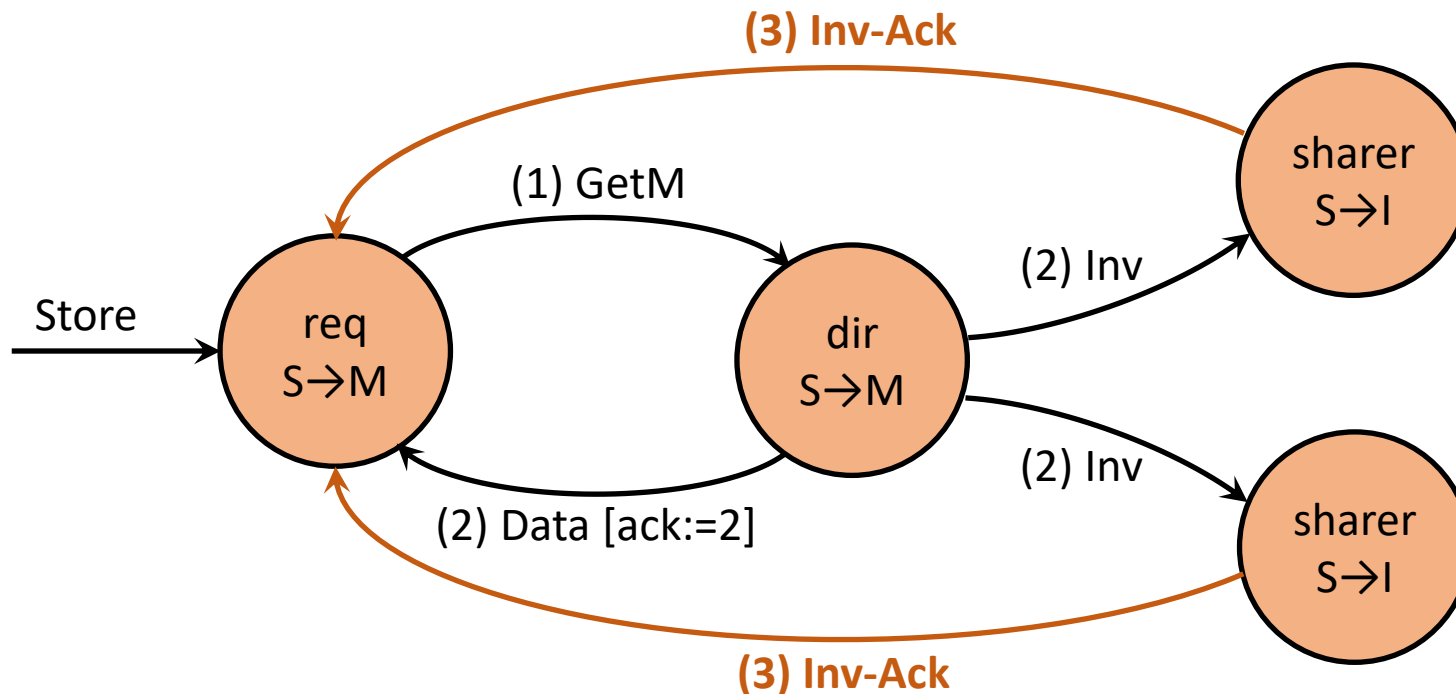
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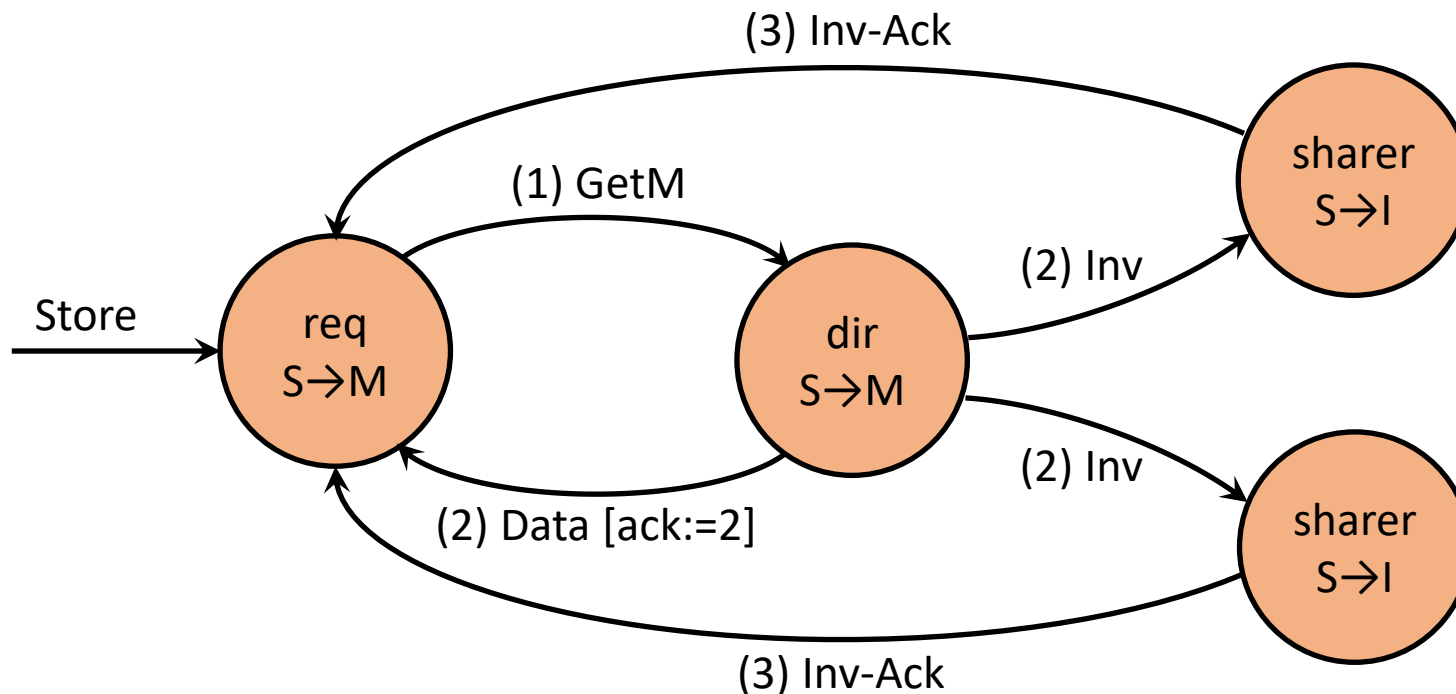
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Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

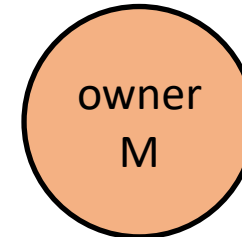
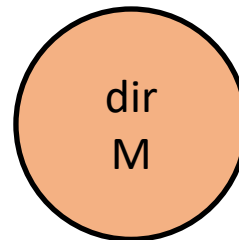
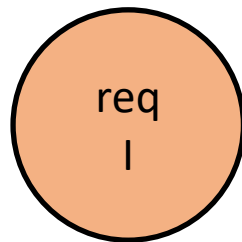
Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

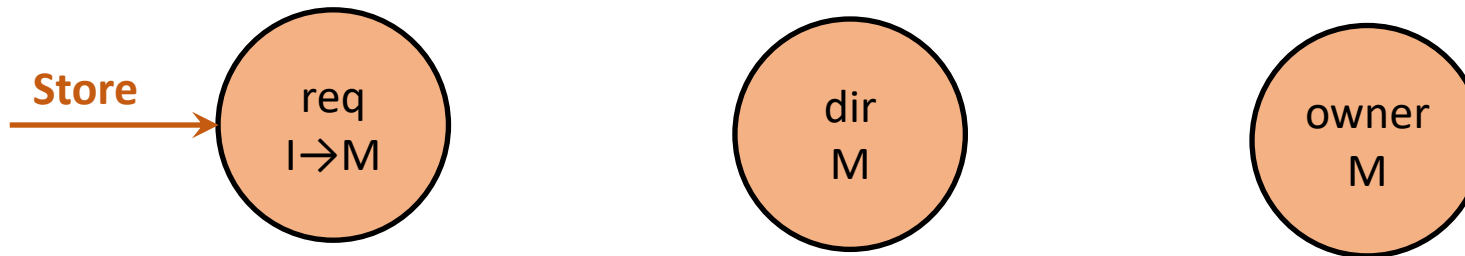
Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

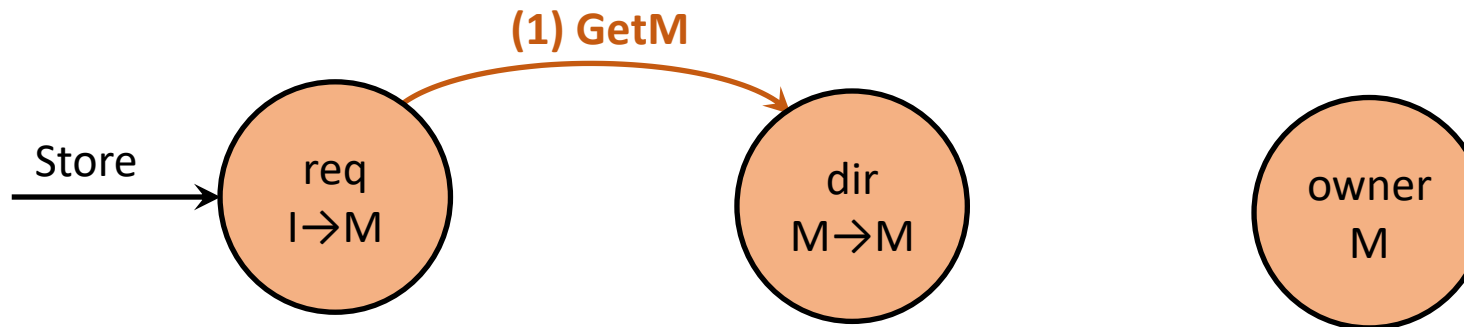
Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

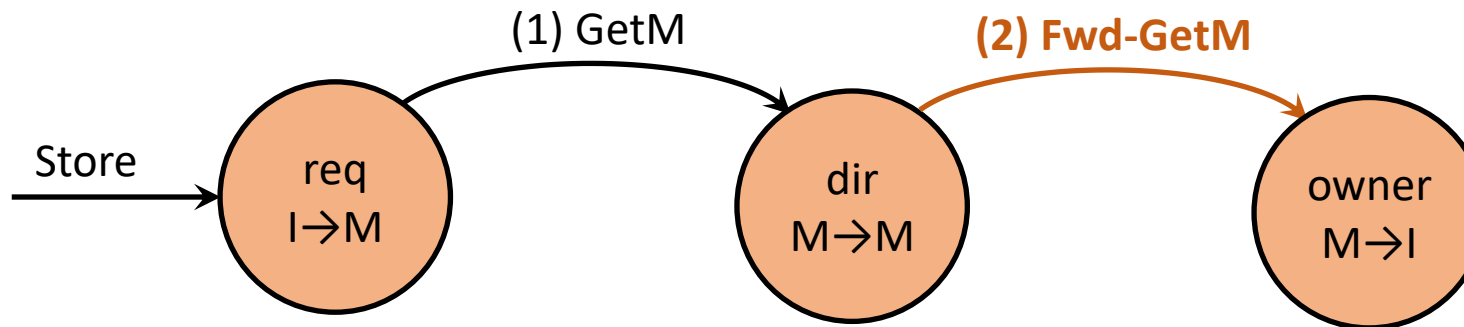
Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

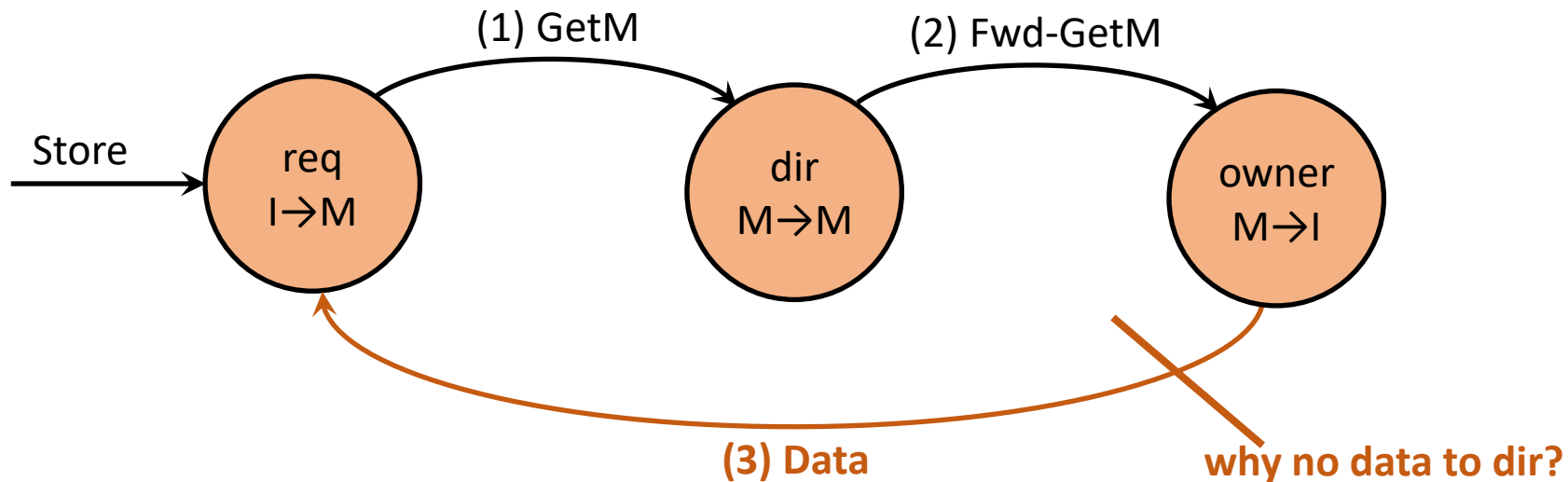
Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

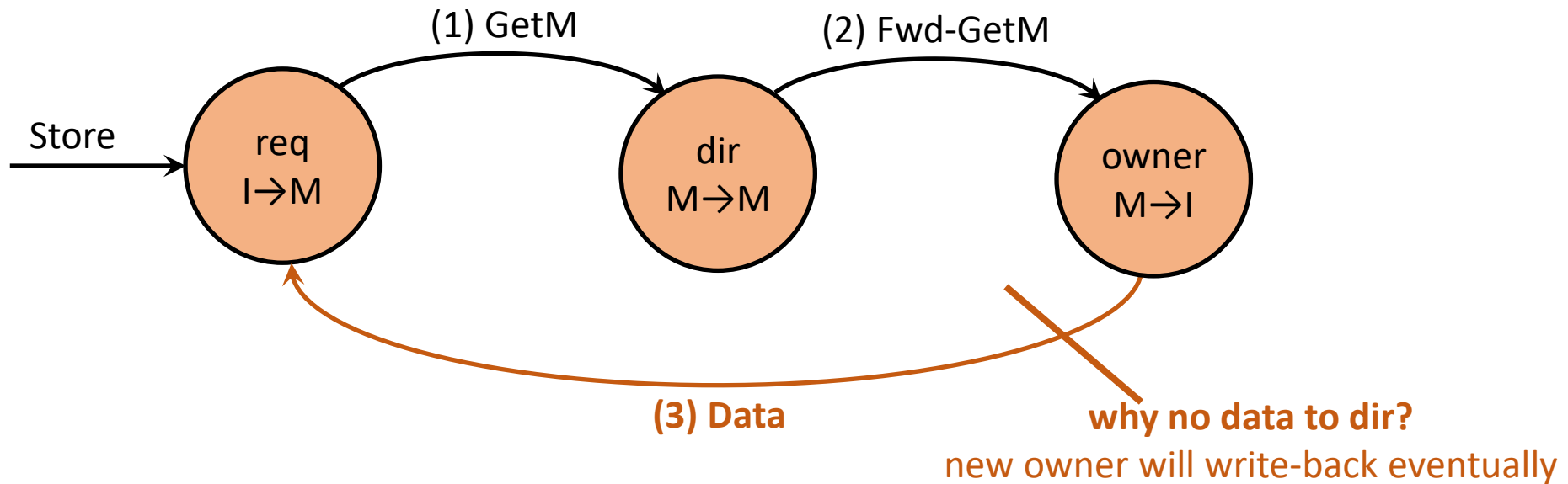
Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

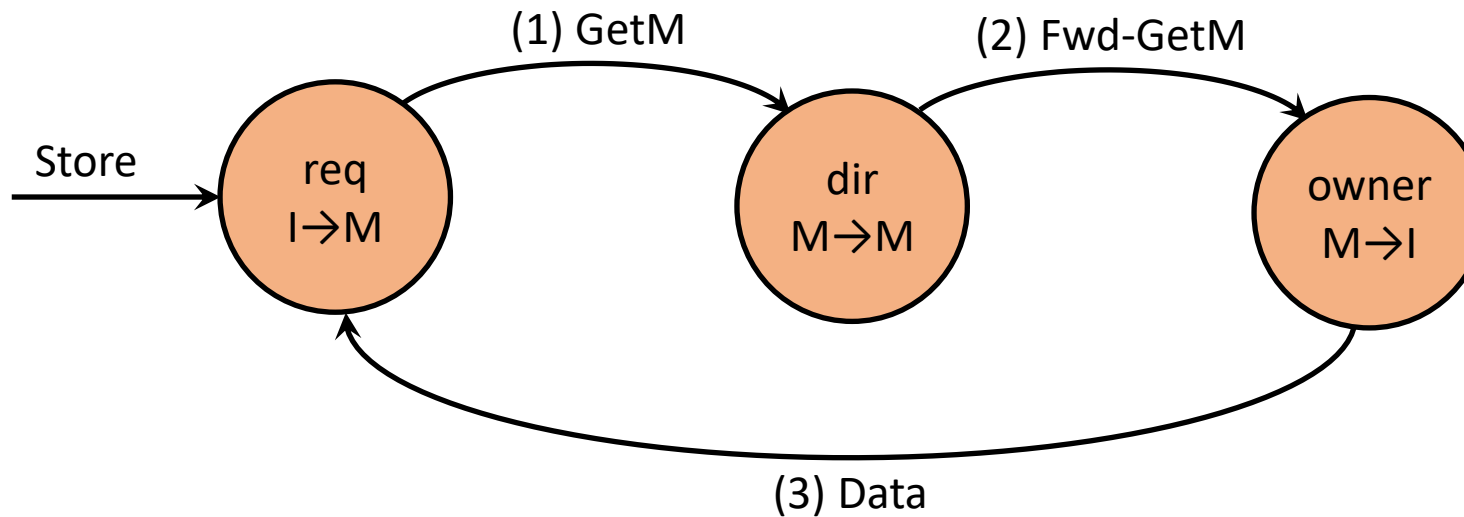
Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

Store?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

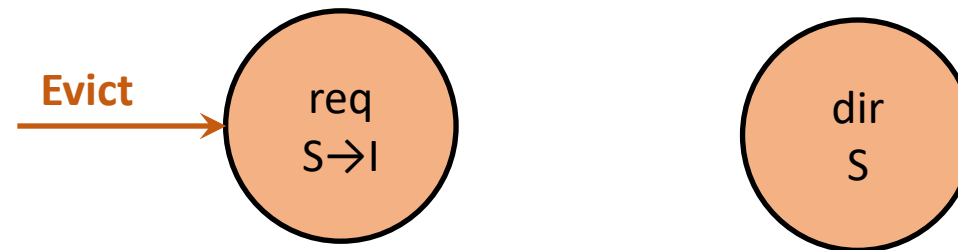
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

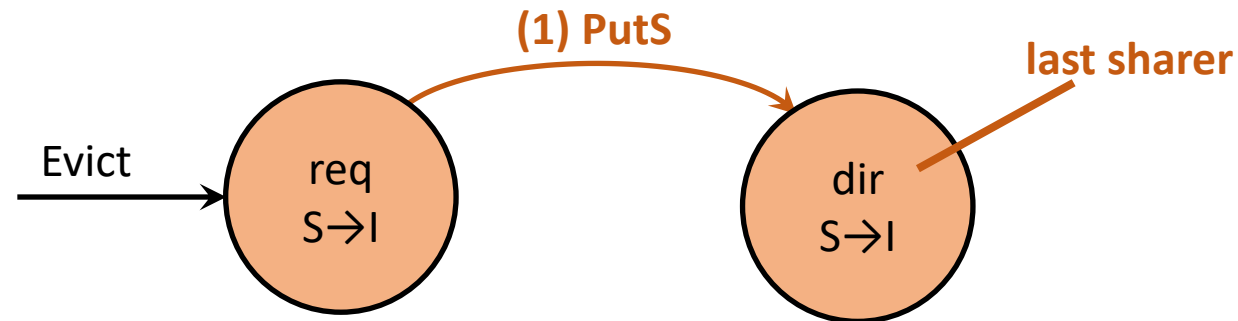
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

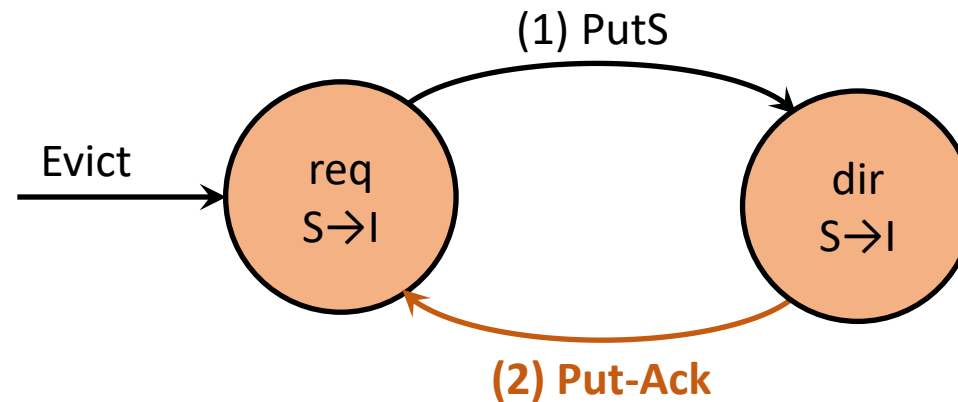
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

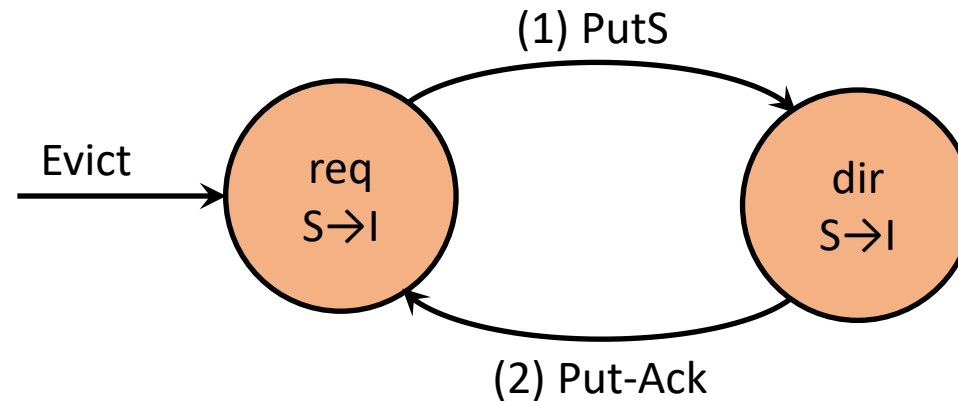
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

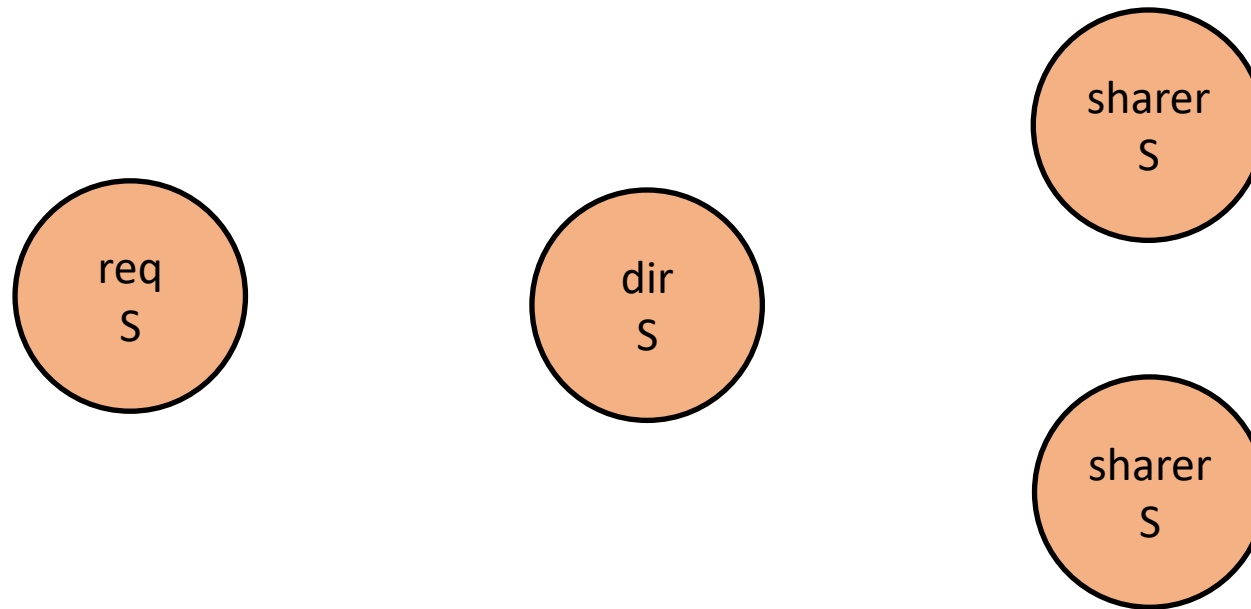
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

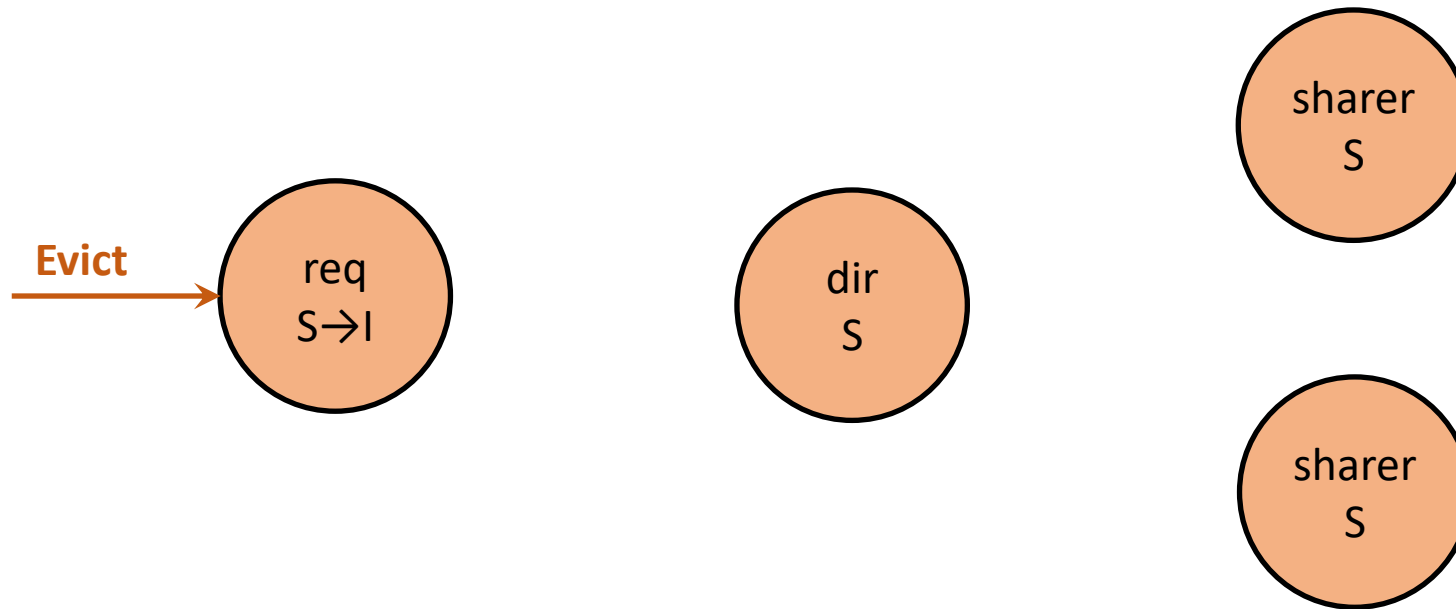
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

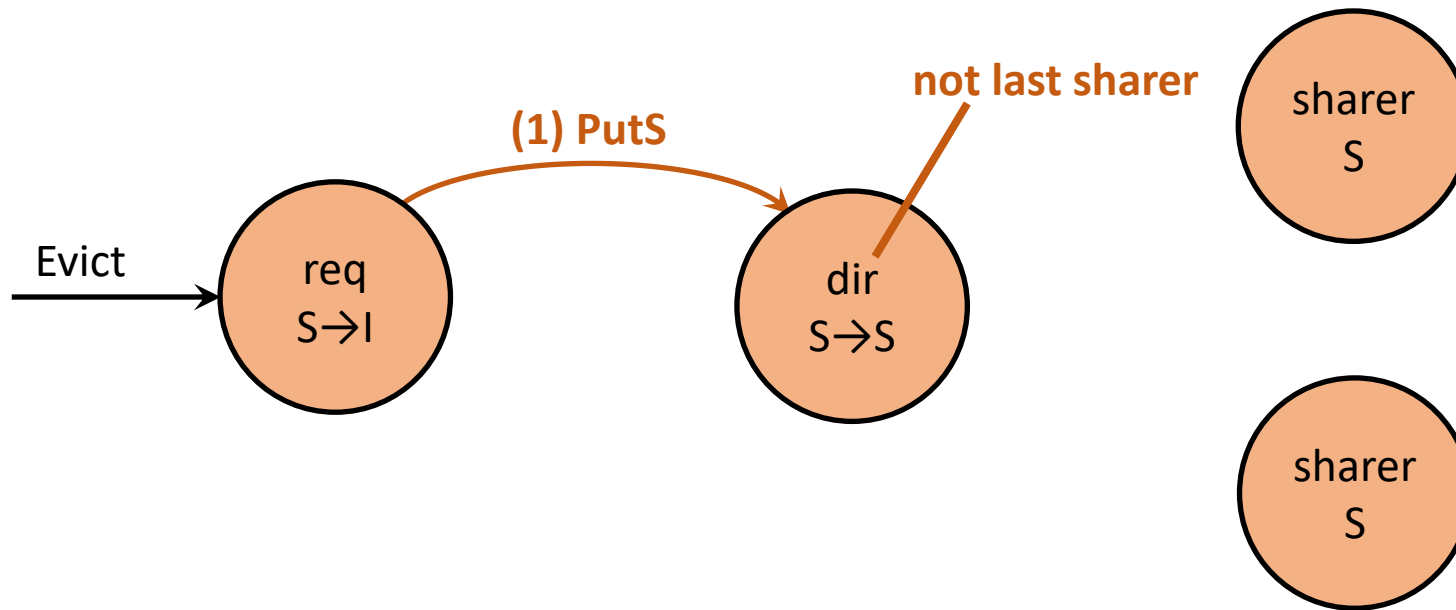
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

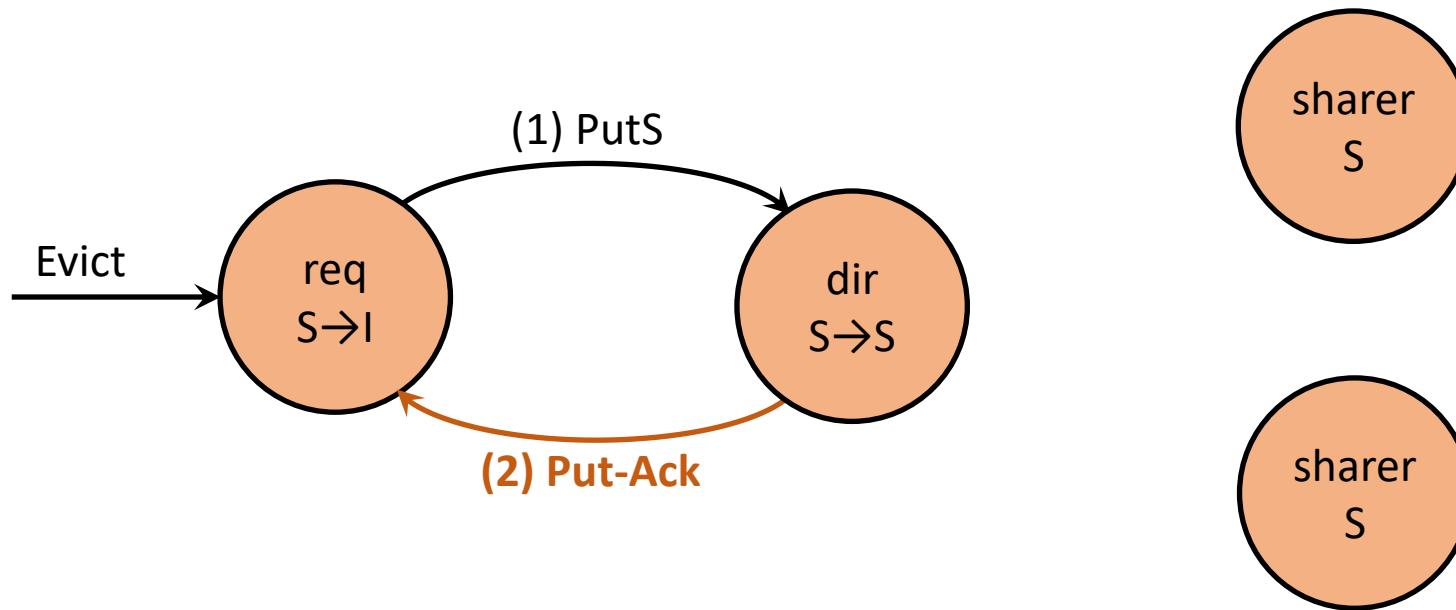
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

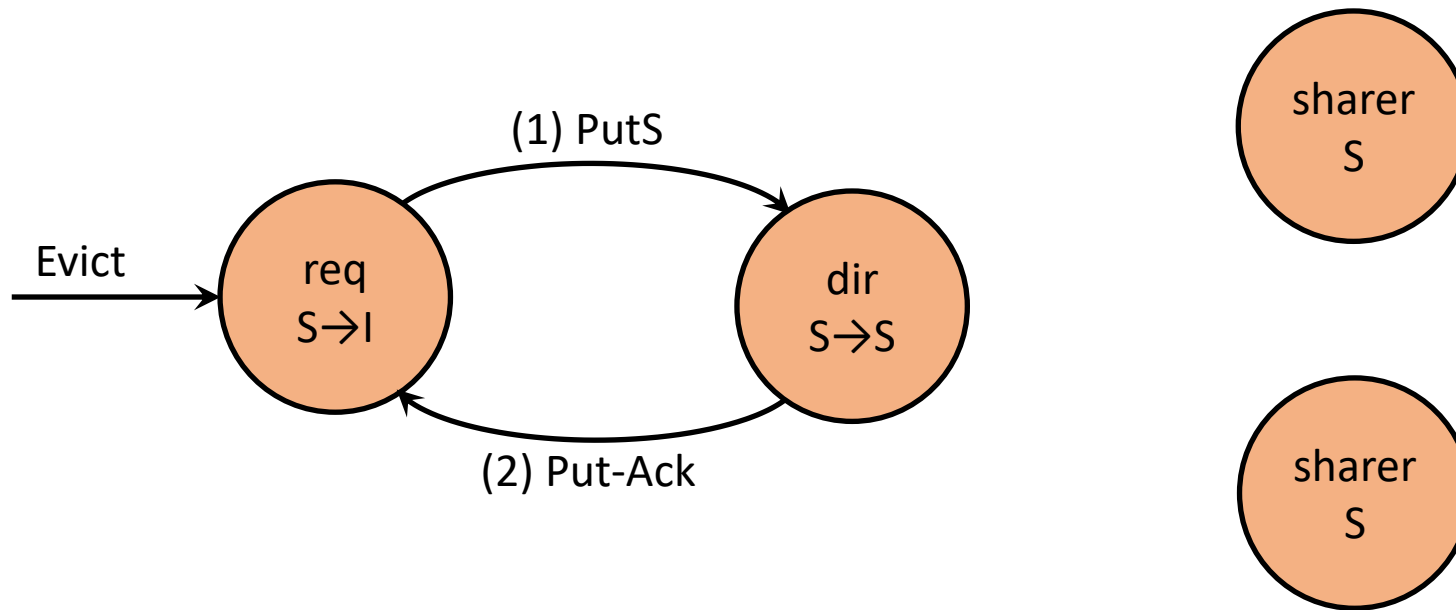
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

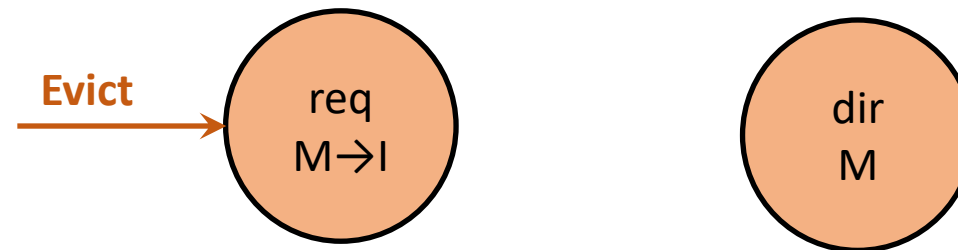
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

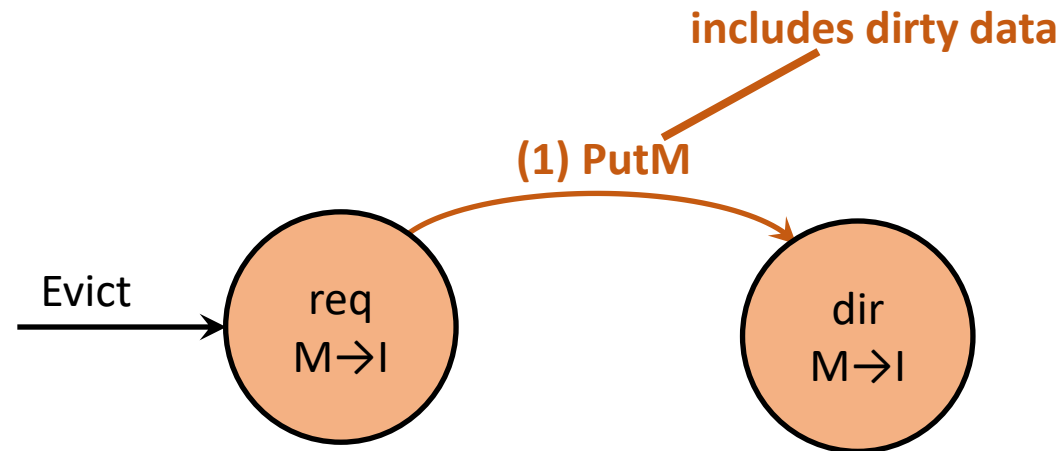
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

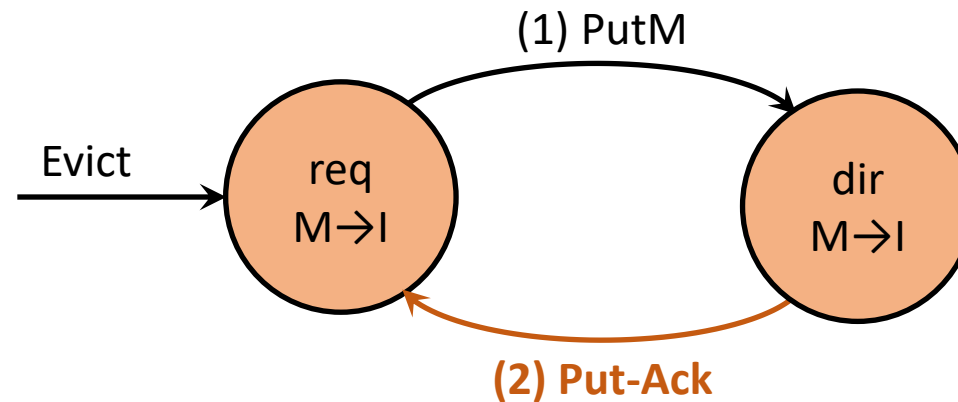
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

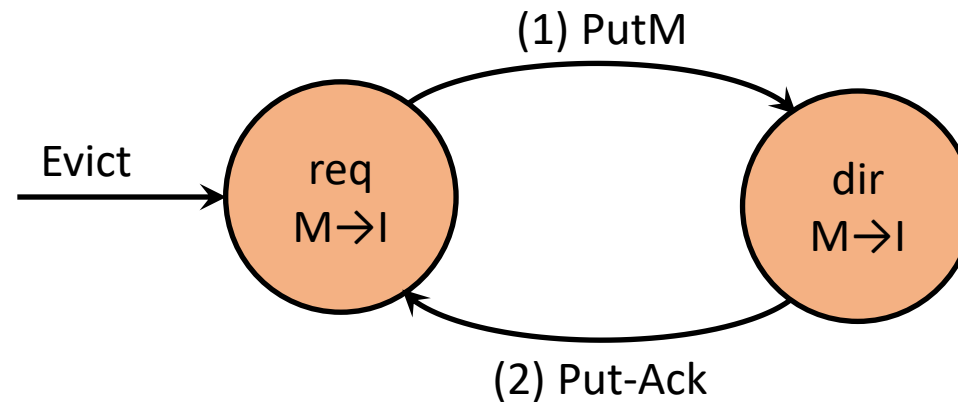
Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

Evict?



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

Cache Controller

Current State	Event					
	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv
I	GetS to dir → S	GetM to dir → M				
S	hit → S	GetM to dir → M	PutS to dir → I			Inv-Ack to req → I
M	hit → M	hit → M	PutM to dir → I	Data to req Data to dir → S	Data to req → I	

Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

Directory Controller

Current State	Event					
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	Data
I	Data to req → S	Data to req → M				
S	Data to req → S	Data to req Inv to sharers → M	Put-Ack to req → S	Put-Ack to req → I		Data to mem → S
M	Fwd-GetS to owner → S	Fwd-GetM to owner → M			Data to mem Put-Ack to req → I	

Directory-Based Coherence

Cache Controller

Current State	Event					
	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv
I	GetS to dir → S	GetM to dir → M				
S	hit → S	GetM to dir → M	PutS to dir → I			Inv-Ack to req → I
M	hit → M	hit → M	PutM to dir → I	Data to req Data to dir → S	Data to req → I	

Directory Controller

Current State	Event					
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	Data
I	Data to req → S	Data to req → M				
S	Data to req → S	Data to req Inv to sharers → M	Put-Ack to req → S	Put-Ack to req → I		Data to mem → S
M	Fwd-GetS to owner → S	Fwd-GetM to owner → M			Data to mem Put-Ack to req → I	



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

...but transactions are not atomic!

Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

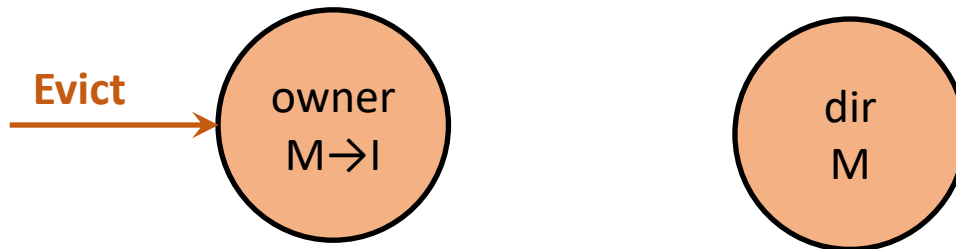
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

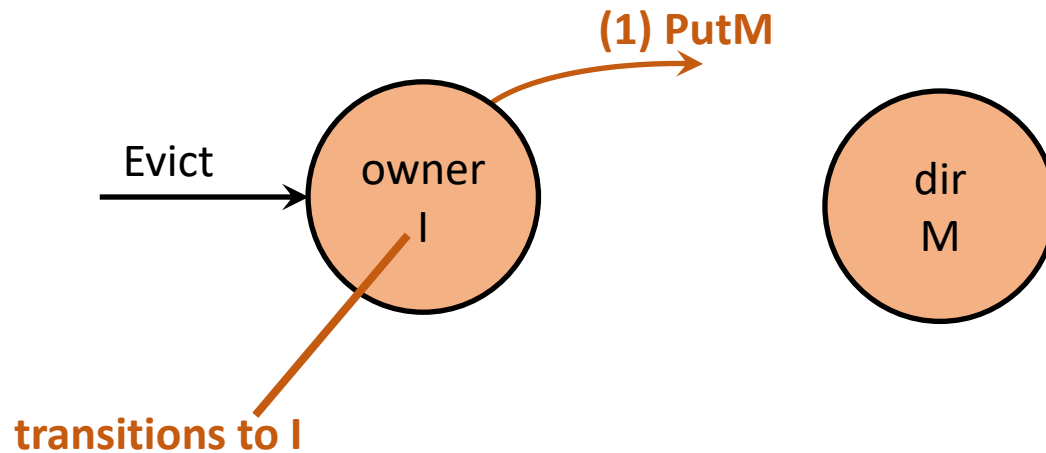
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

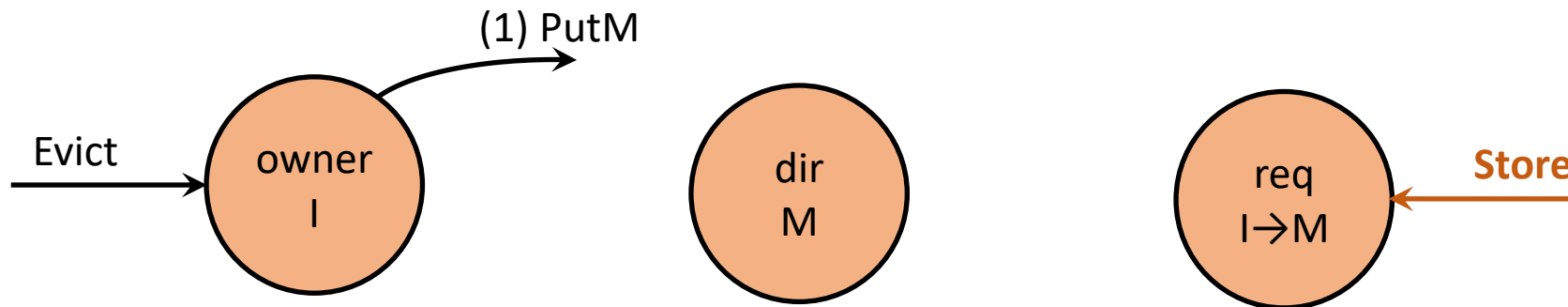
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

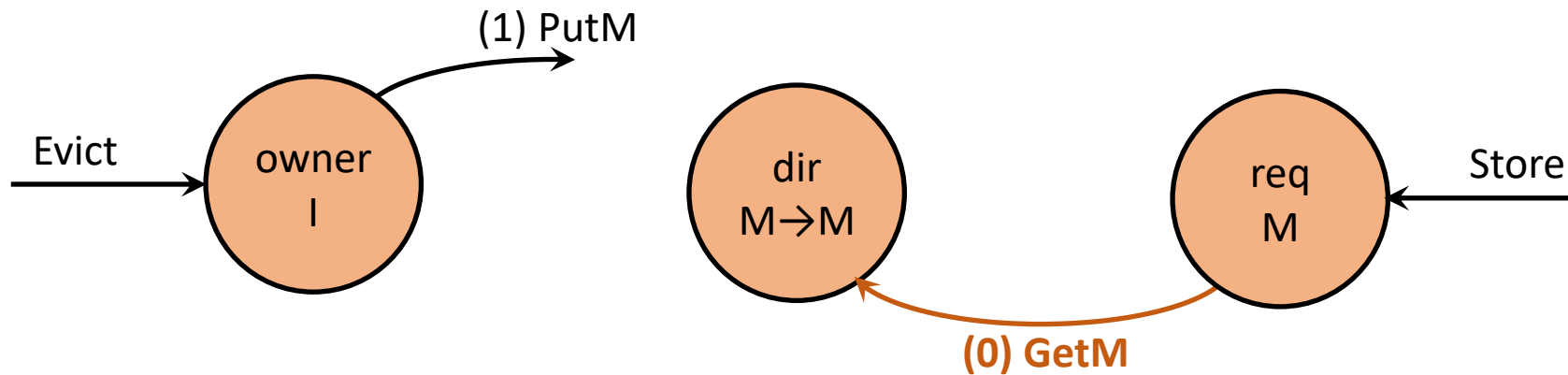
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

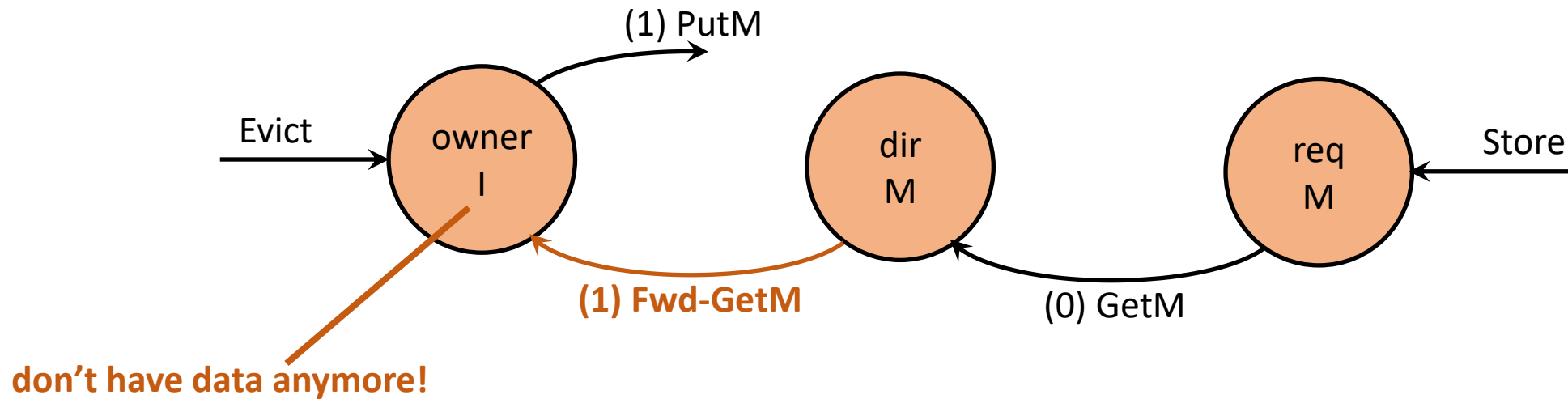
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

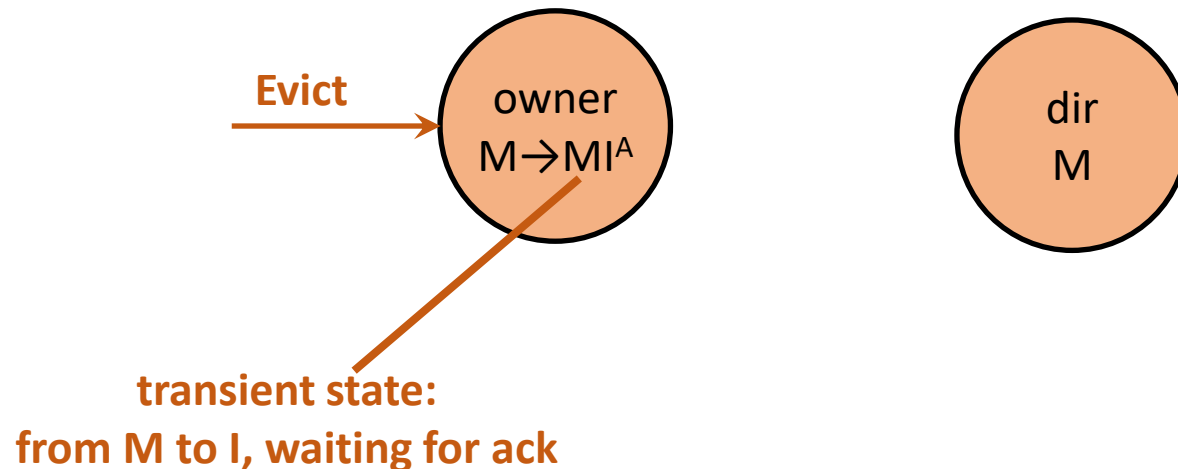
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

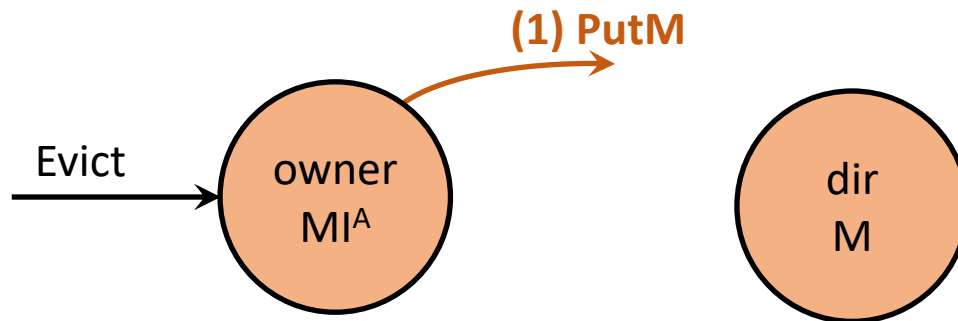
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

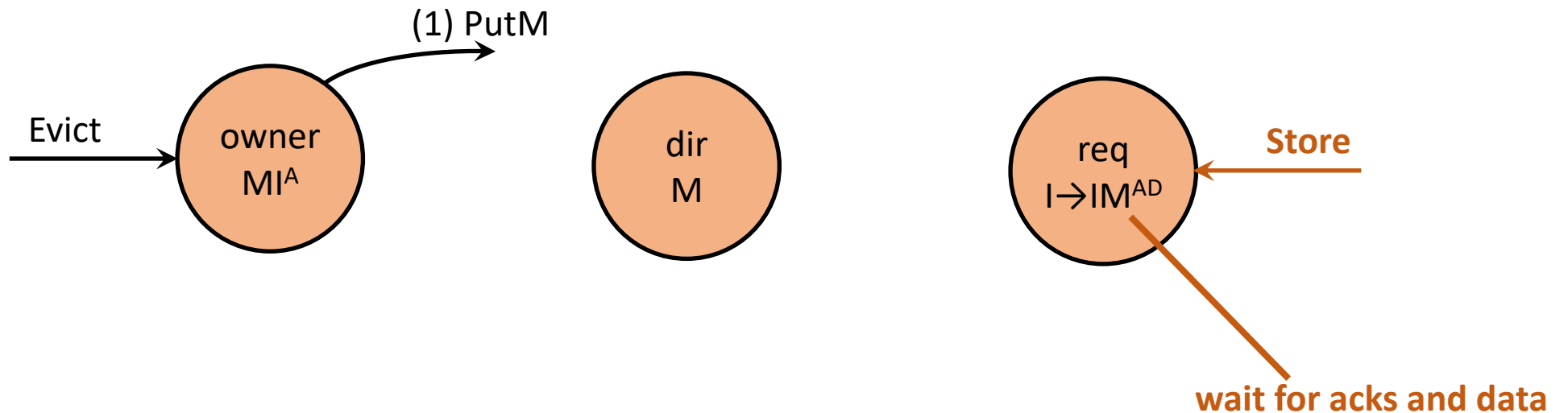
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

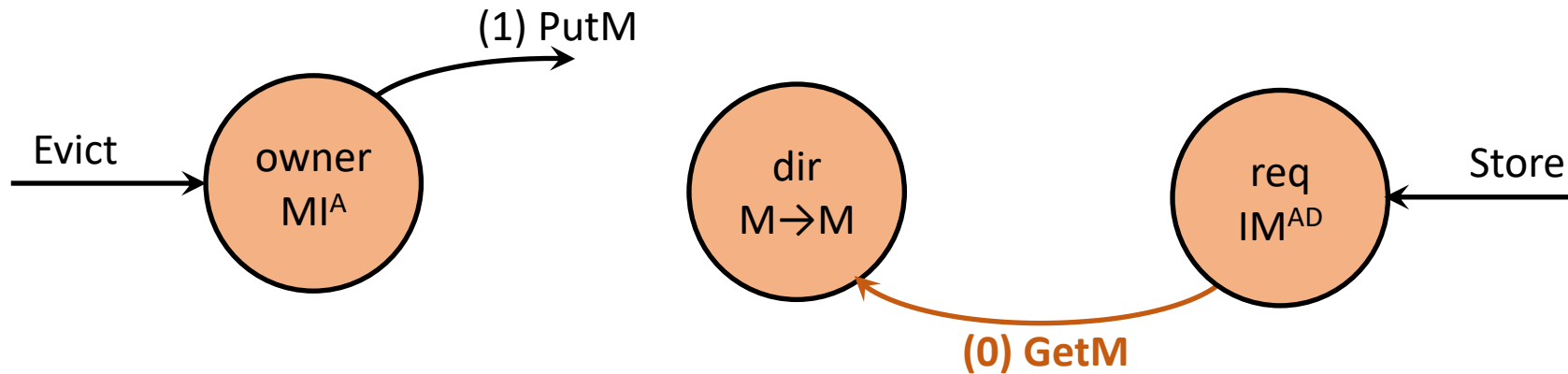
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

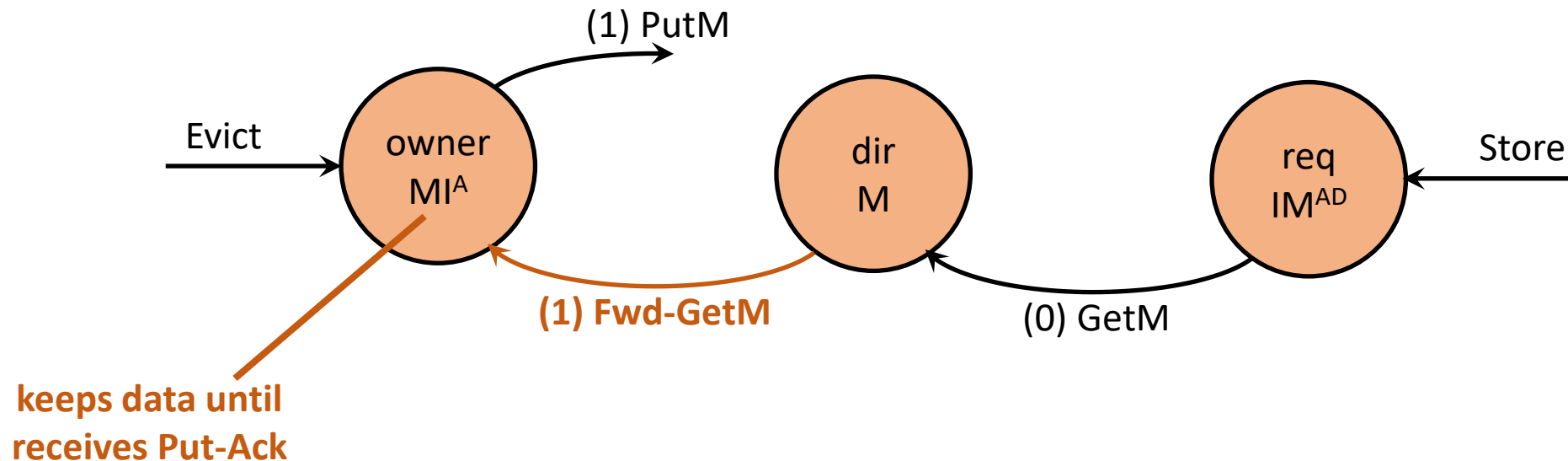
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

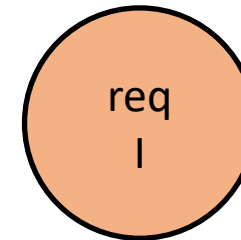
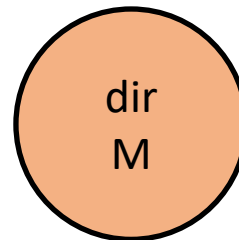
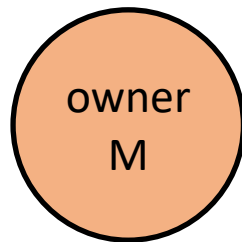
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

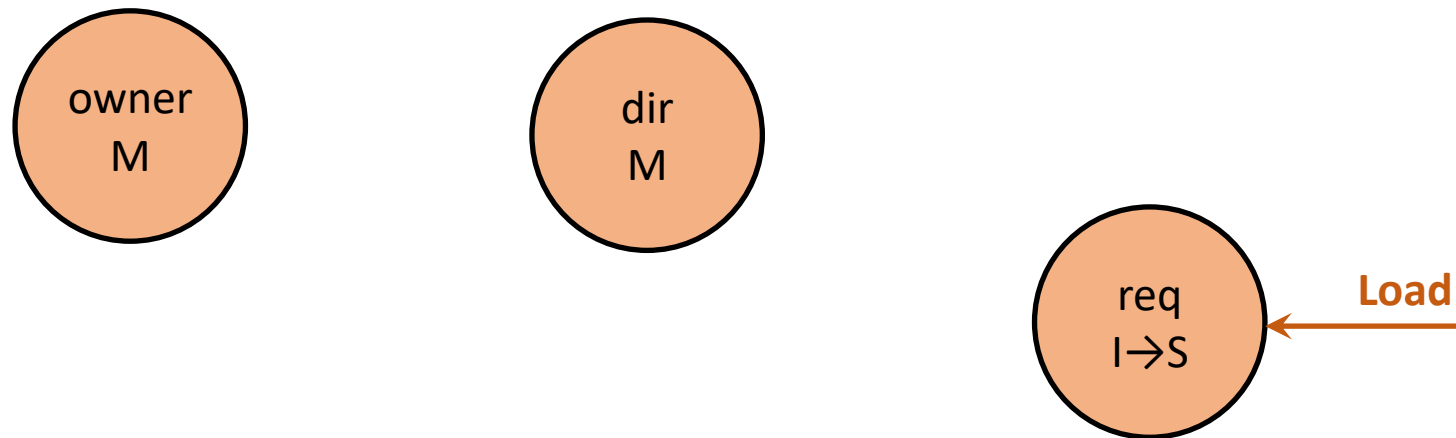
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

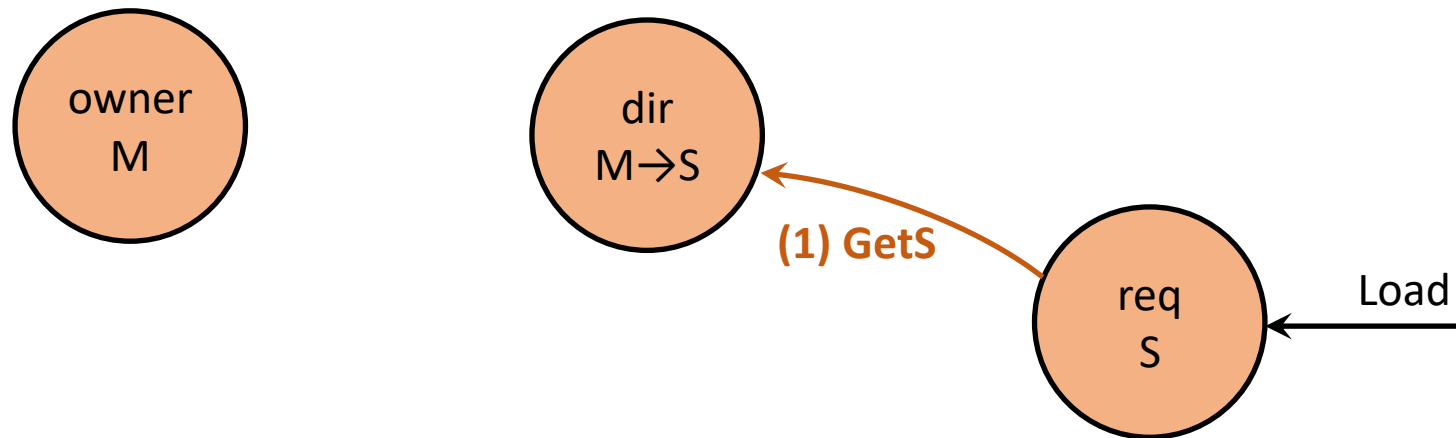
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

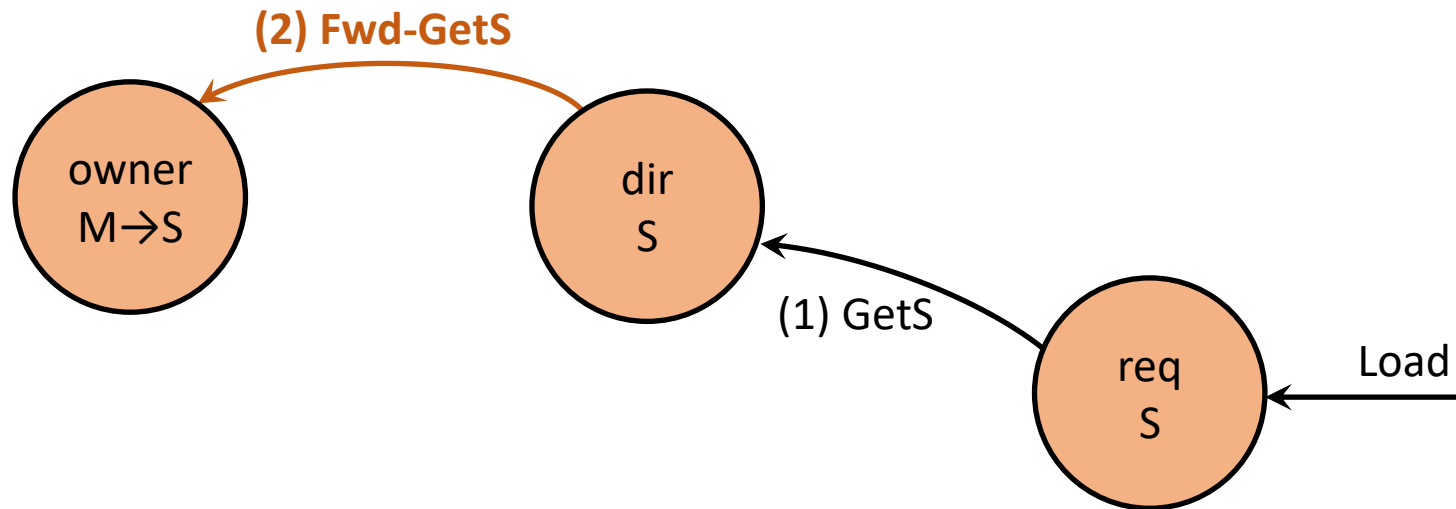
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

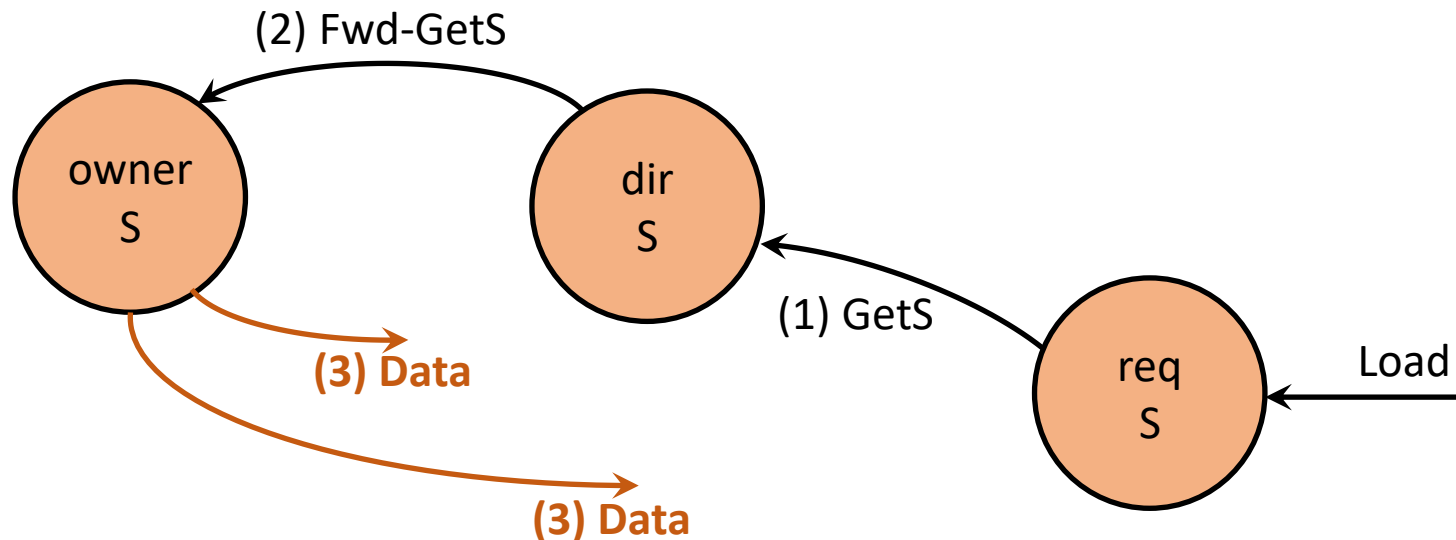
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

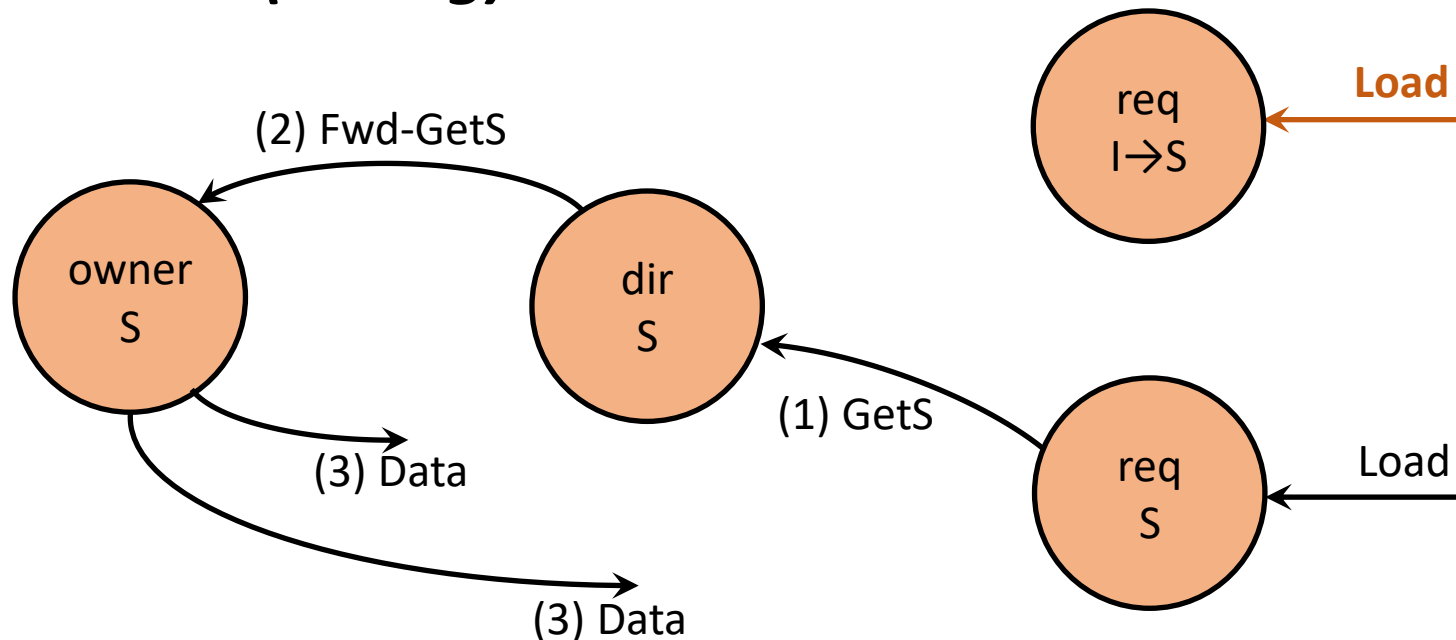
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

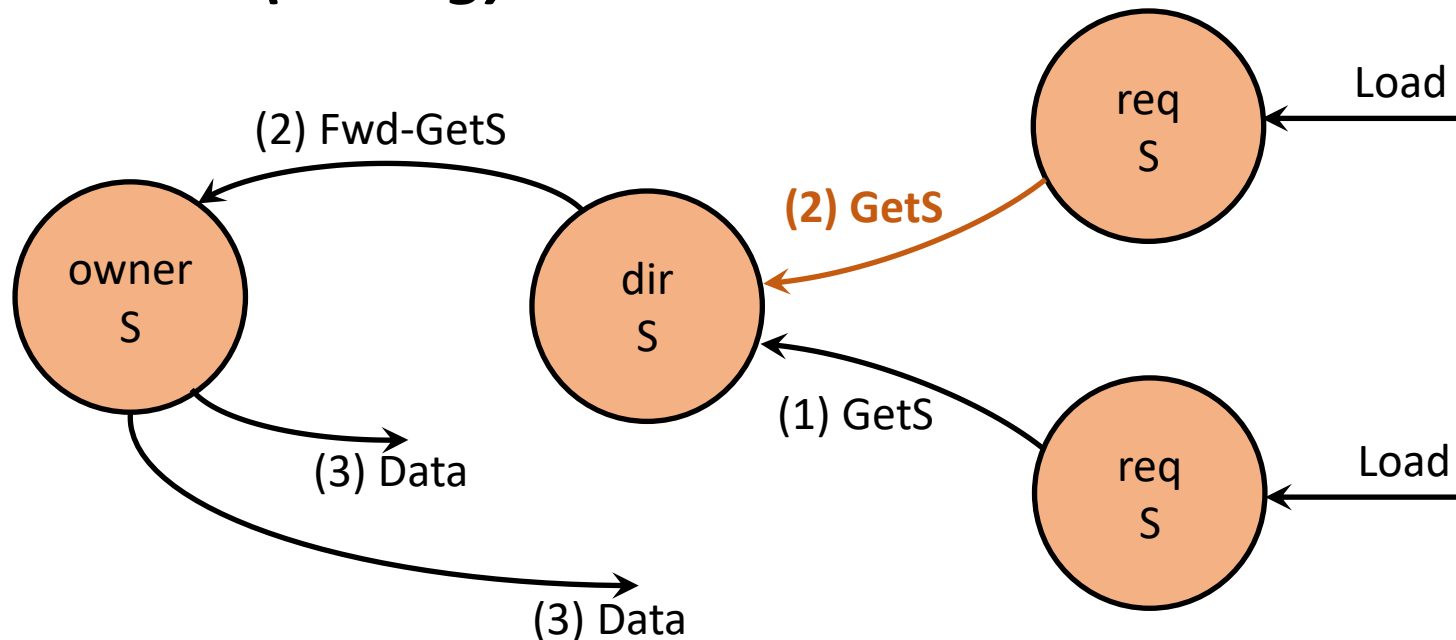
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

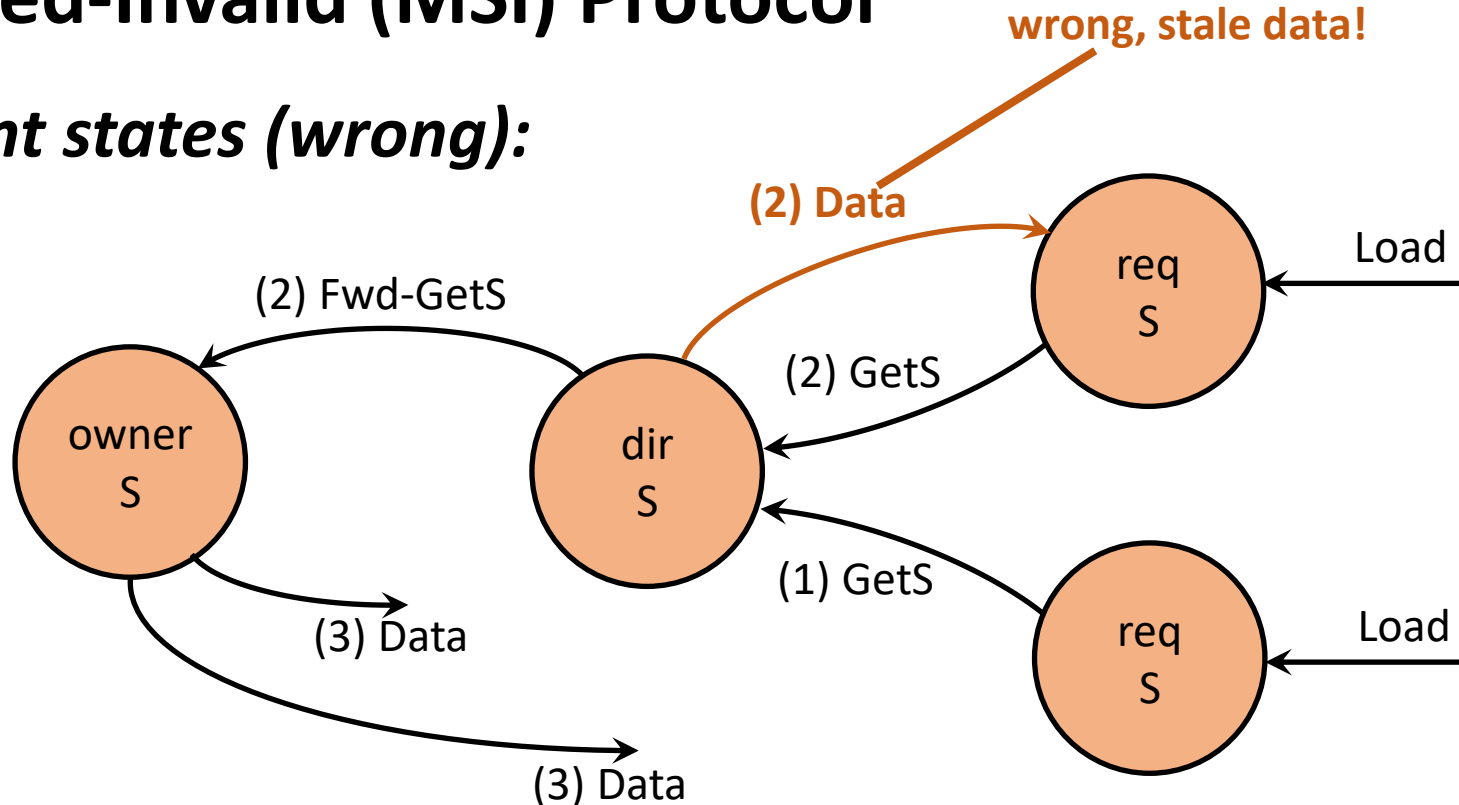
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

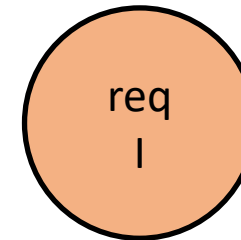
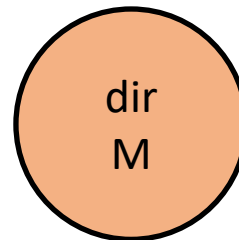
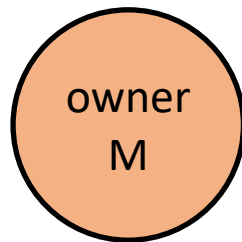
Without transient states (wrong):



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

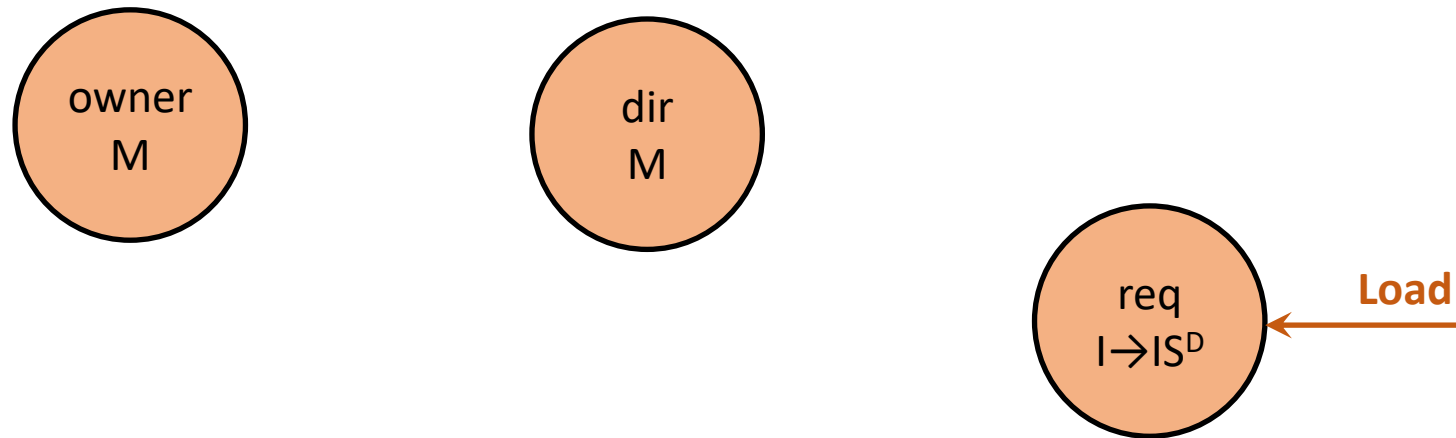
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

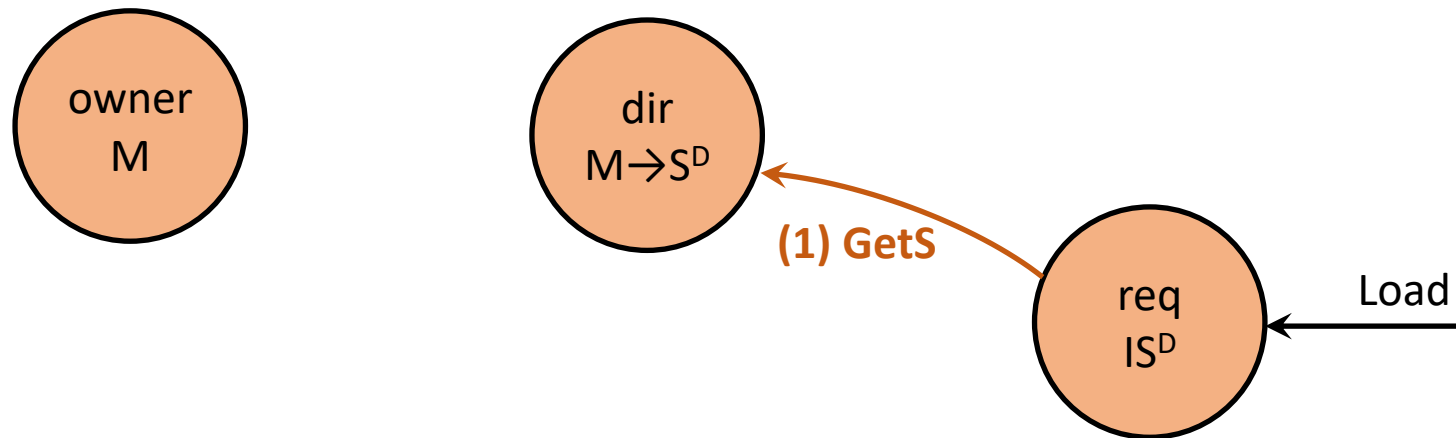
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

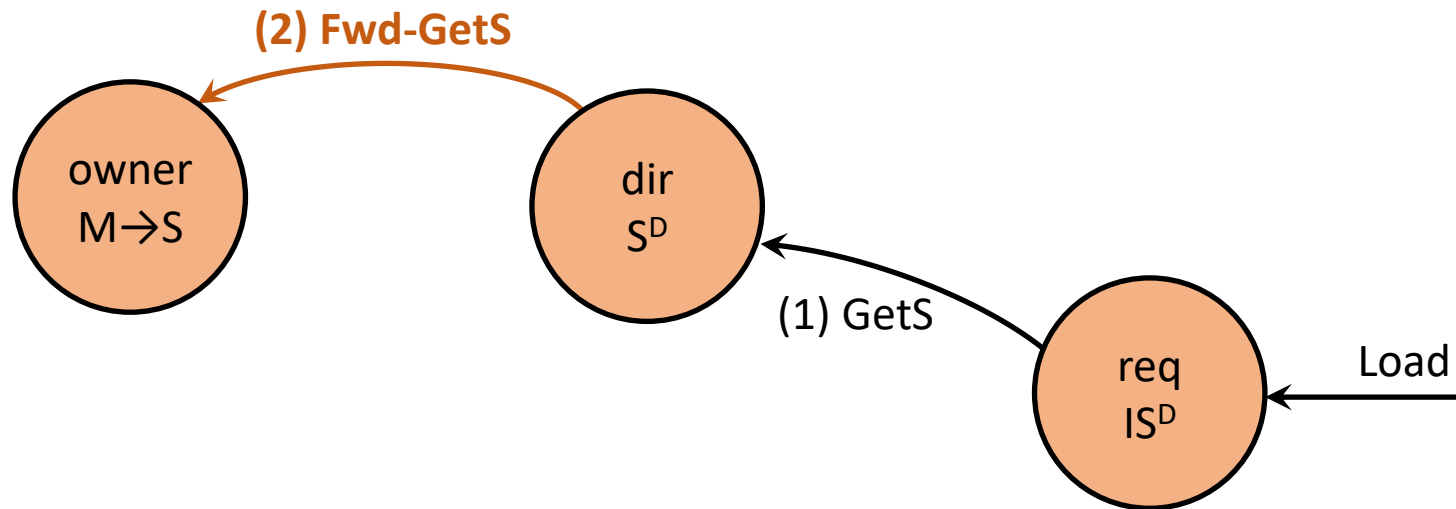
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

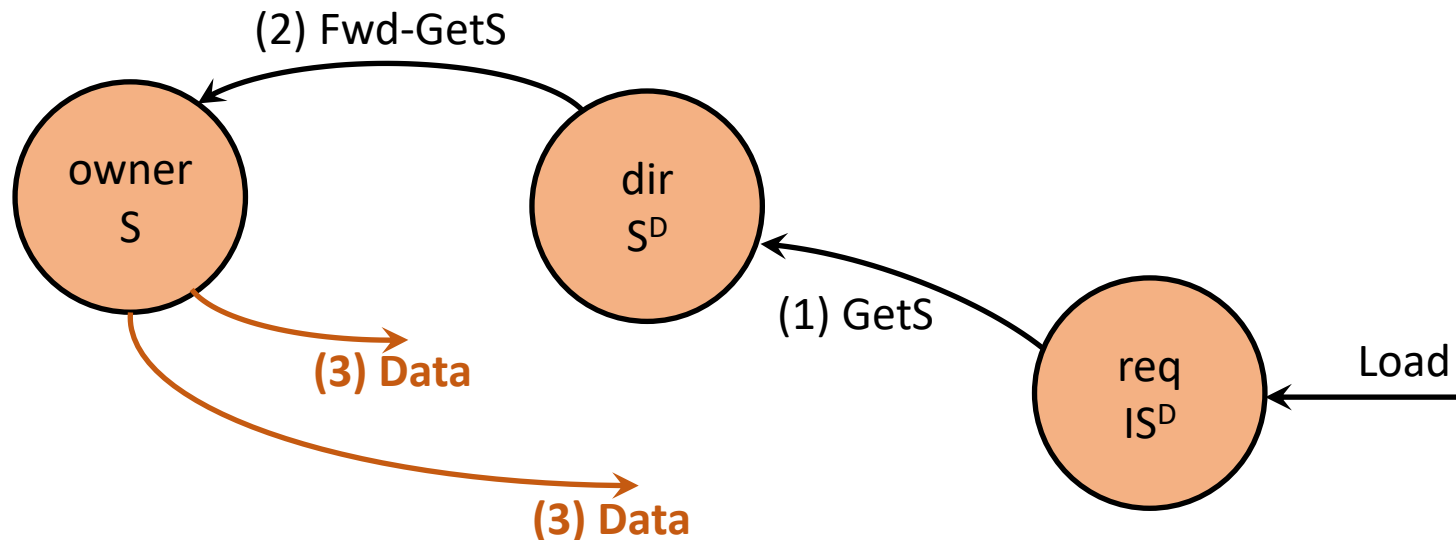
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

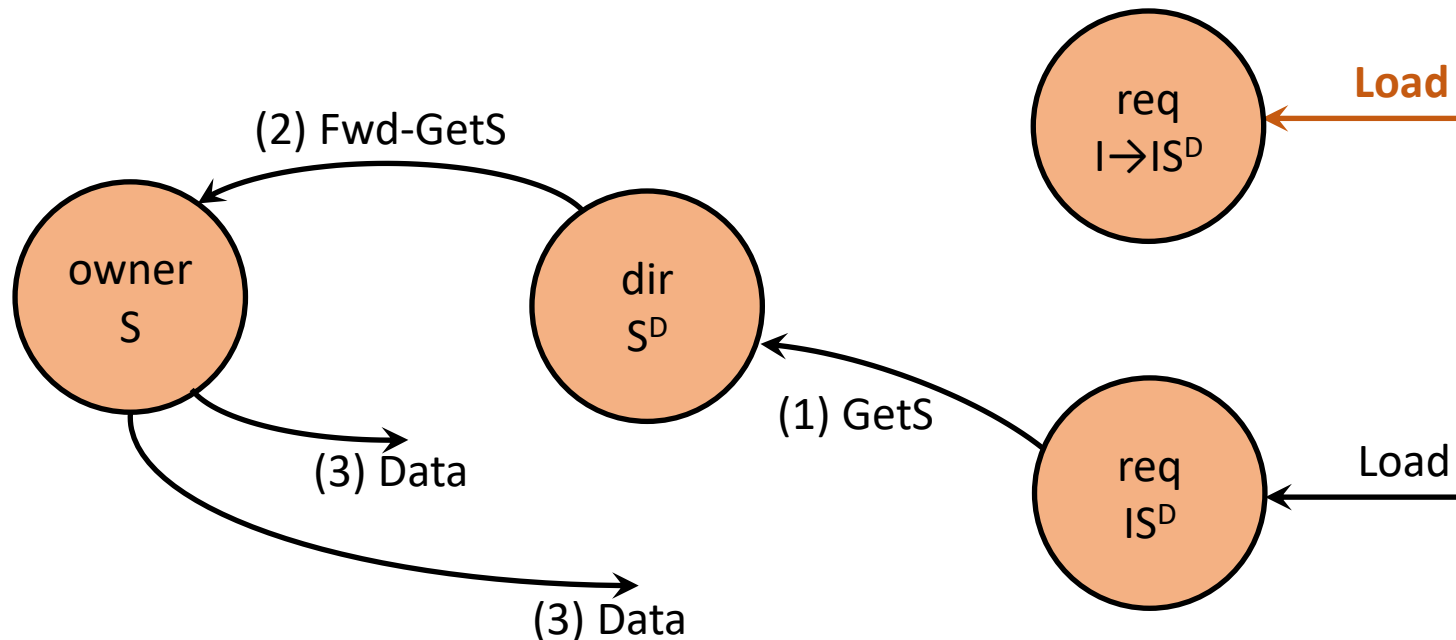
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

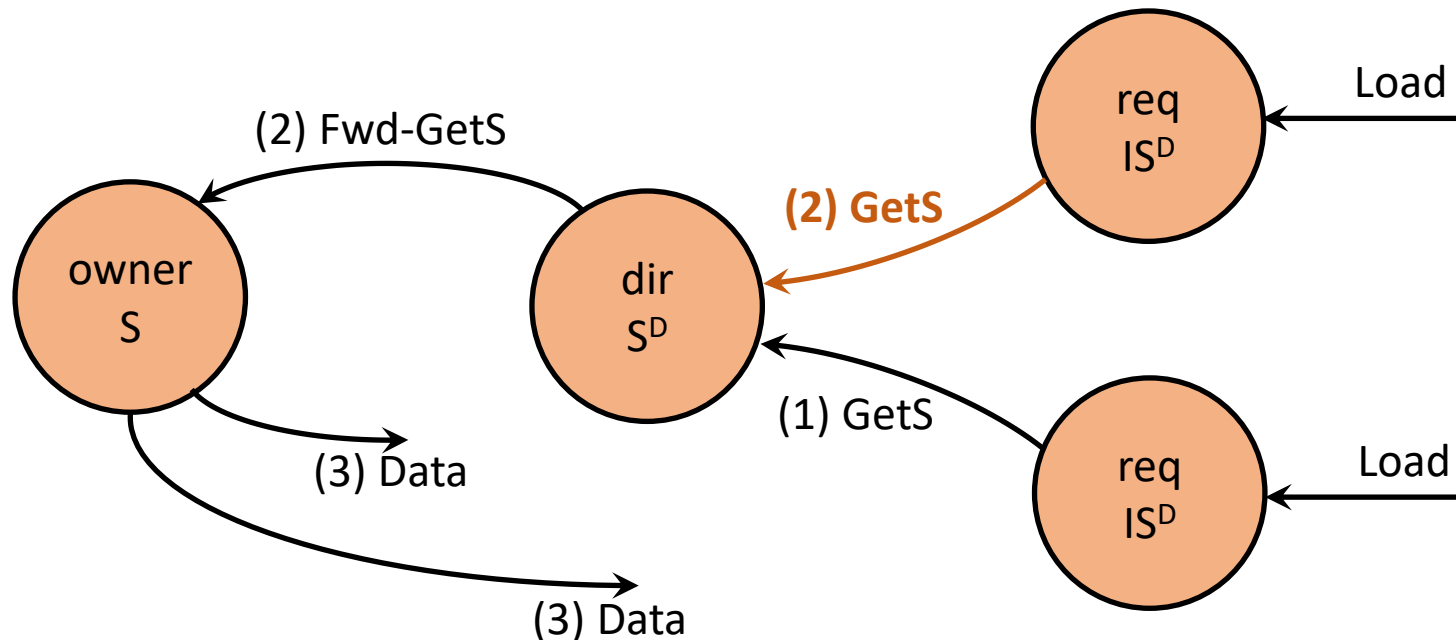
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

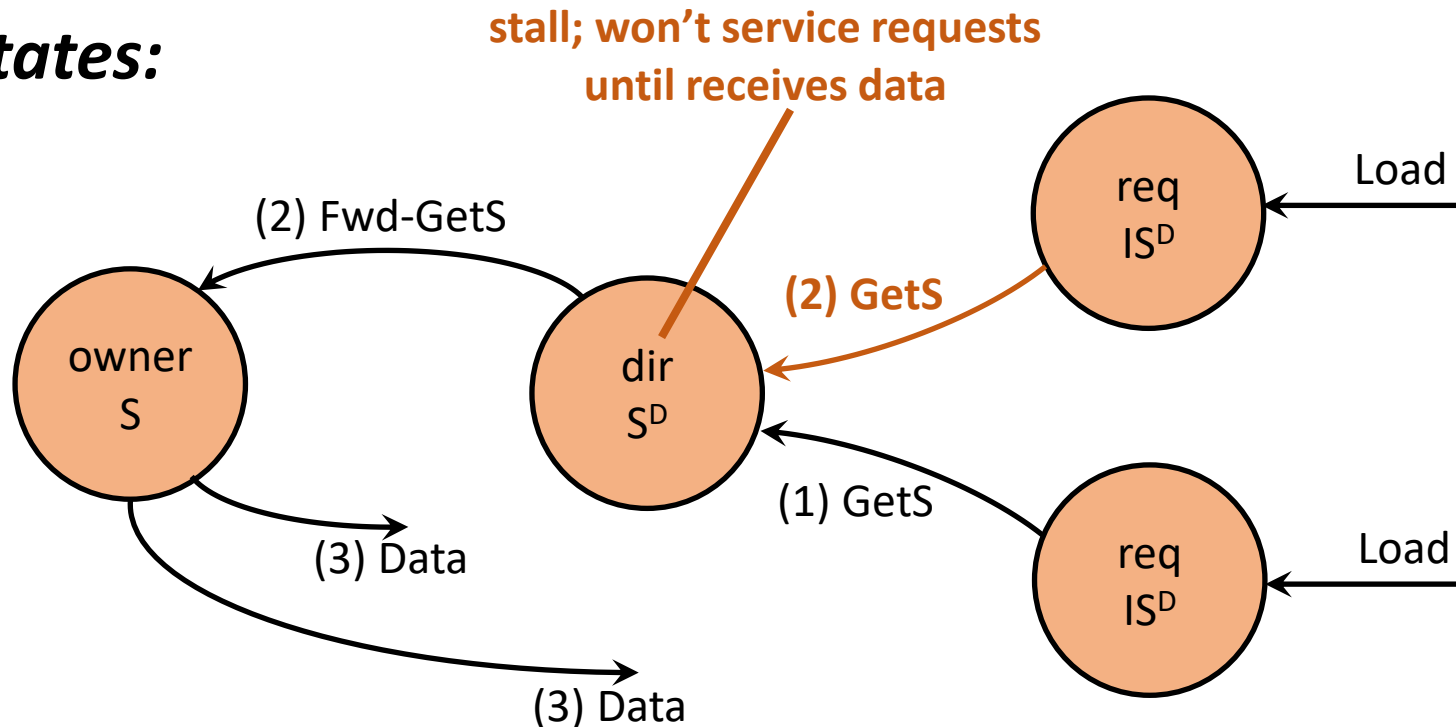
With transient states:



Directory-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

With transient states:



Directory-Based Coherence

Cache Controller

Current State	Event					
	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv
I	GetS to dir → S	GetM to dir → M				
S	hit → S	GetM to dir → M	PutS to dir → I			Inv-Ack to req → I
M	hit → M	hit → M	PutM to dir → I	Data to req Data to dir → S	Data to req → I	

...if atomic transactions (i.e., no transient states)

Directory-Based Coherence

Cache Controller

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M
S	hit → S	GetM to dir → SM ^{AD}	PutS to dir → SI ^A			Inv-Ack to req → I						
SM ^{AD}	hit → SM ^{AD}	stall	stall	stall	stall	Inv-Ack to req → IM ^{AD}		→ M	→ SM ^A		ack-- → SM ^{AD}	
SM ^A	hit → SM ^A	stall	stall	stall	stall						ack-- → SM ^A	→ M
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

Directory-Based Coherence

Directory Controller

Current State	Event					
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	Data
I	Data to req → S	Data to req → M				
S	Data to req → S	Data to req Inv to sharers → M	Put-Ack to req → S	Put-Ack to req → I		Data to mem → S
M	Fwd-GetS to owner → S	Fwd-GetM to owner → M			Data to mem Put-Ack to req → I	

...if atomic transactions (i.e., no transient states)



Directory-Based Coherence

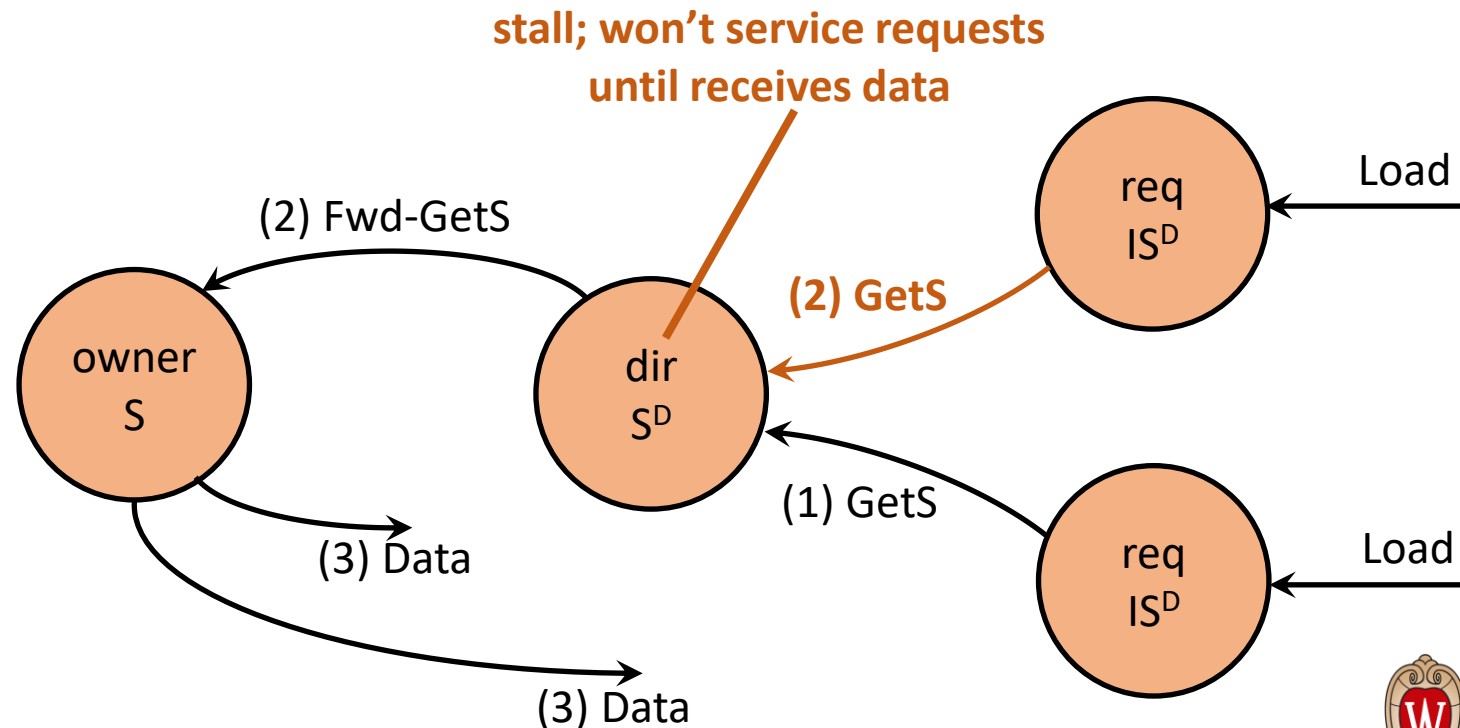
Directory Controller

Current State	Event						
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
I	Data to req → S	Data to req → M	Put-Ack to req → I	Put-Ack to req → I		Put-Ack to req → I	
S	Data to req → S	Data to req Inv to sharers → M	Put-Ack to req → S	Put-Ack to req → I		Put-Ack to req → S	
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	
S ^D	stall	stall	Put-Ack to req → S ^D	Put-Ack to req → S ^D		Put-Ack to req → S ^D	Data to mem → S

Directory-Based Coherence

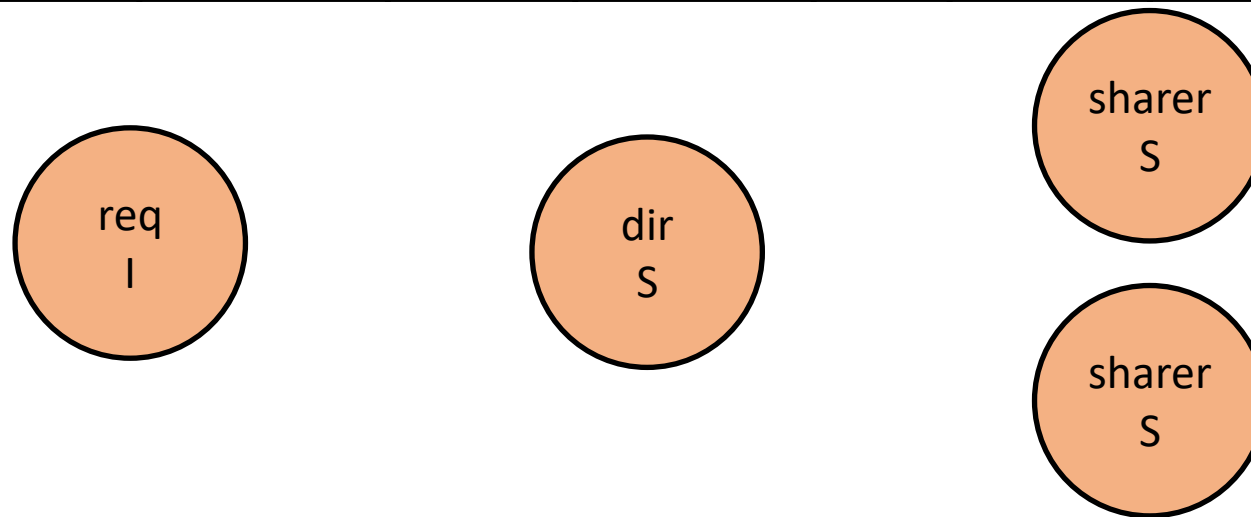
Directory Controller

Current State	Event						
	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
S^D	stall	stall	Put-Ack to req $\rightarrow S^D$	Put-Ack to req $\rightarrow S^D$		Put-Ack to req $\rightarrow S^D$	Data to mem $\rightarrow S$



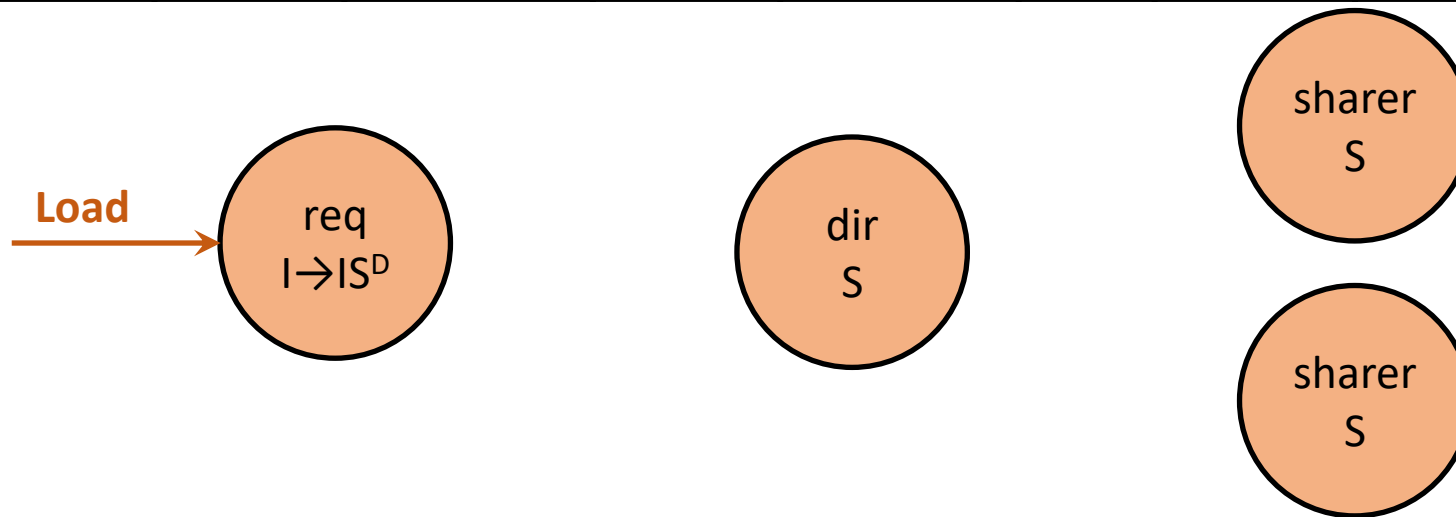
Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		



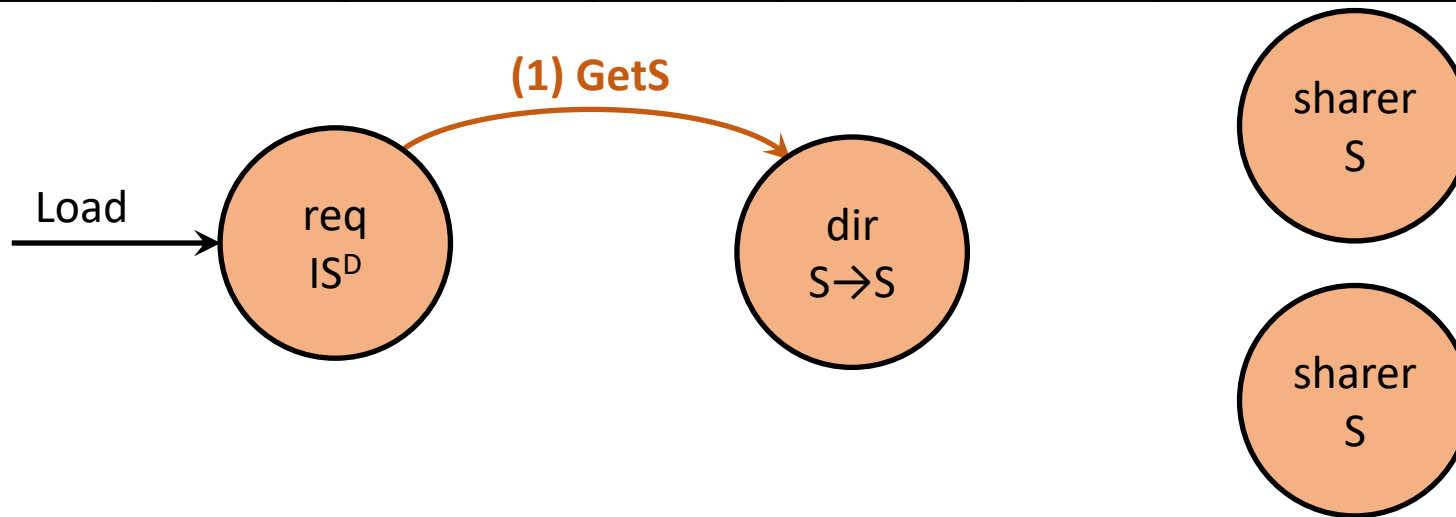
Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		



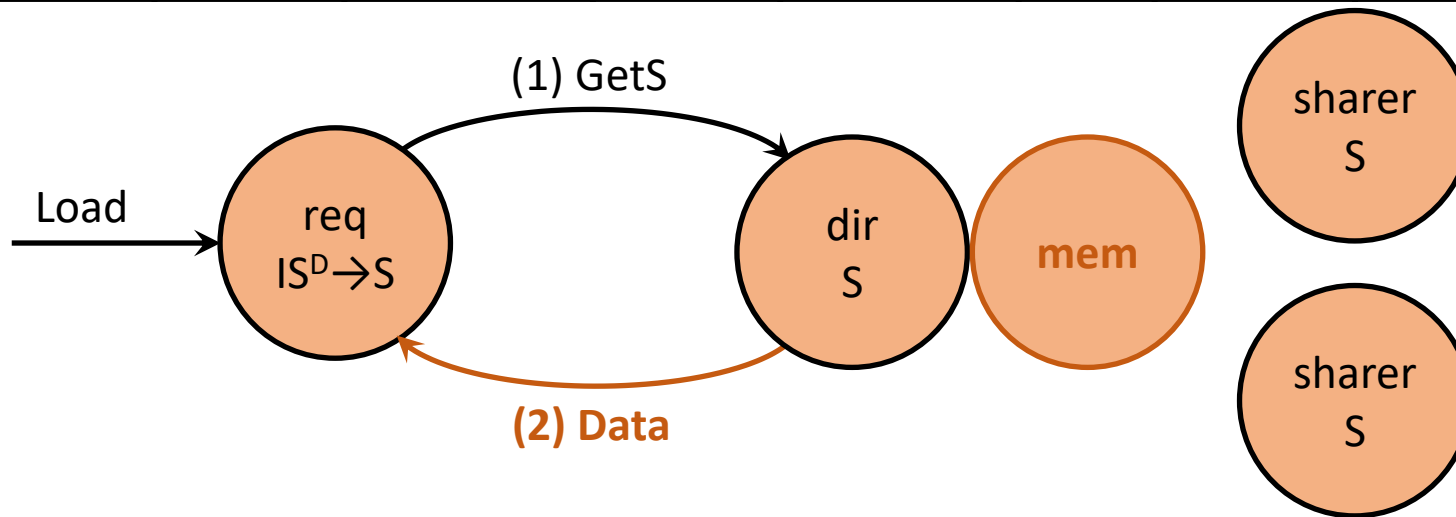
Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		



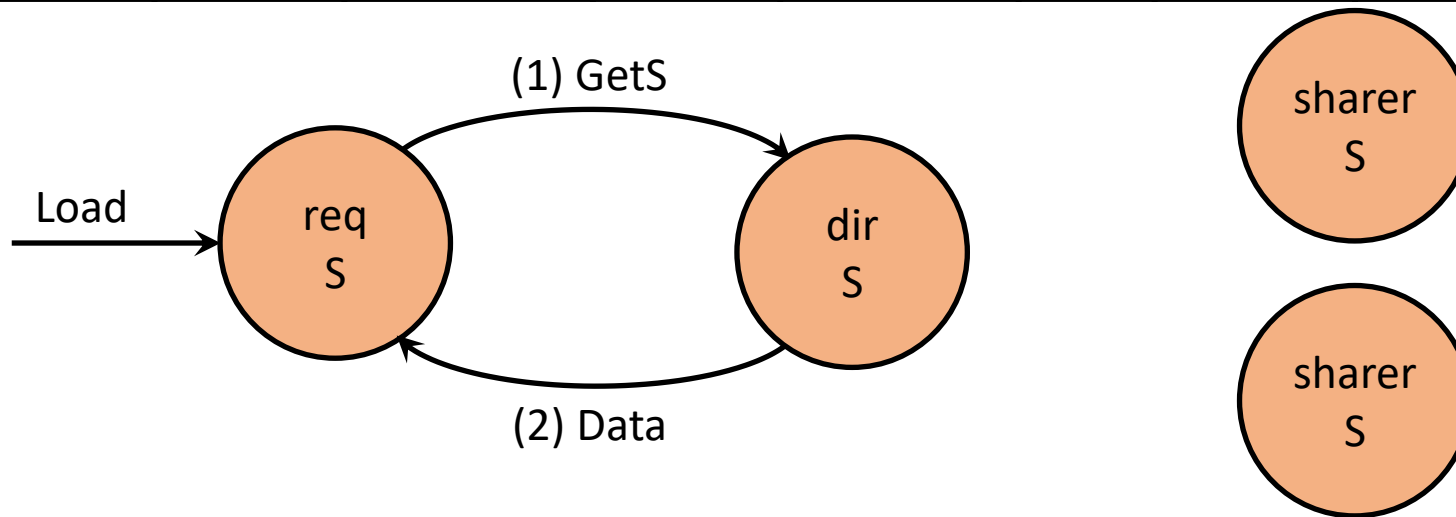
Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		



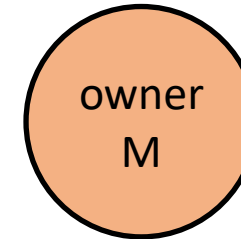
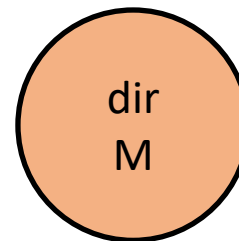
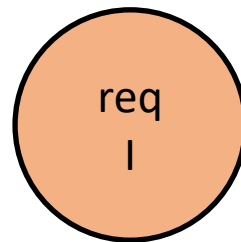
Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		



Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		

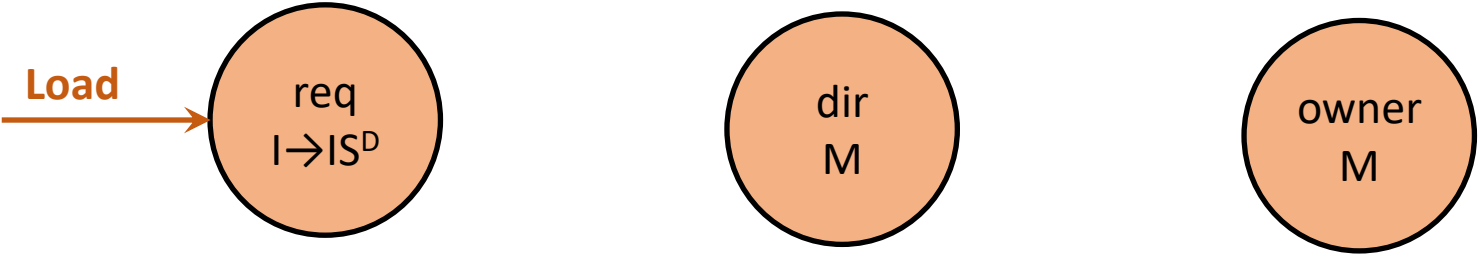


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							



Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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IS ^D	stall	stall	stall			stall		→ S		→ S		

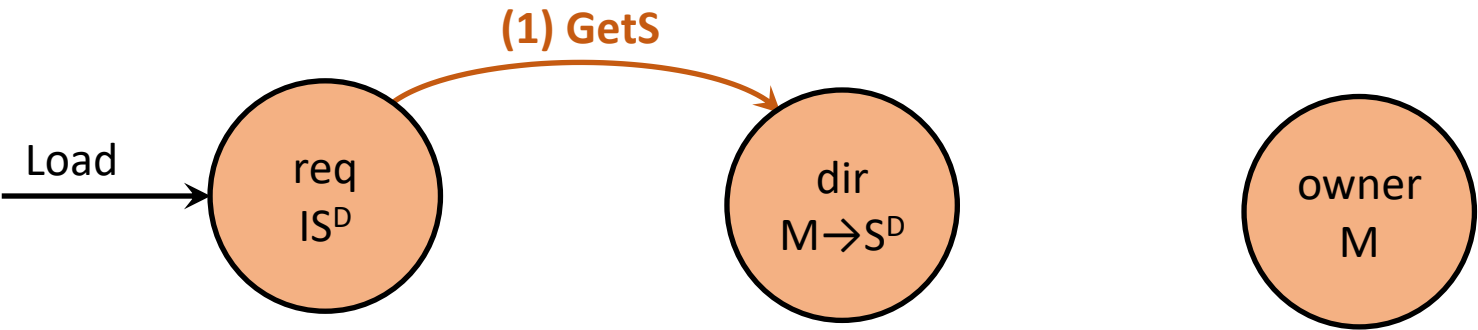


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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IS ^D	stall	stall	stall			stall		→ S		→ S		

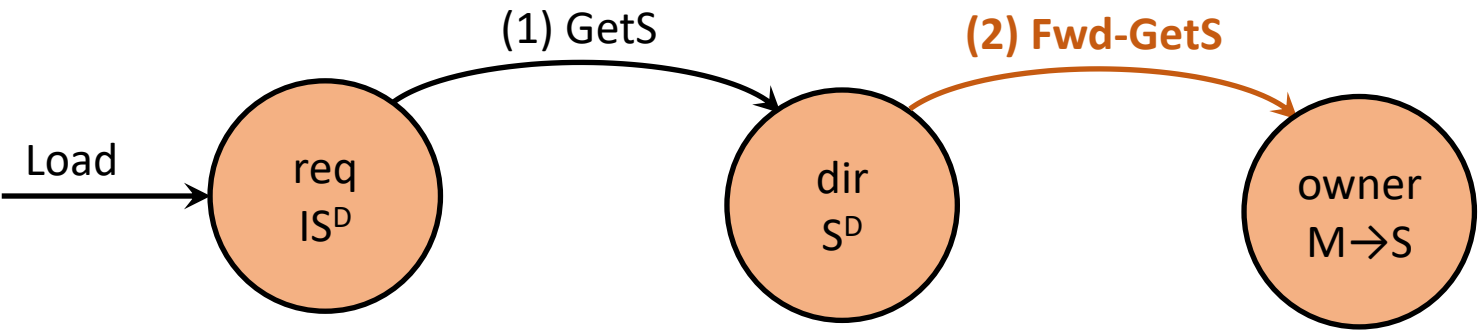


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		

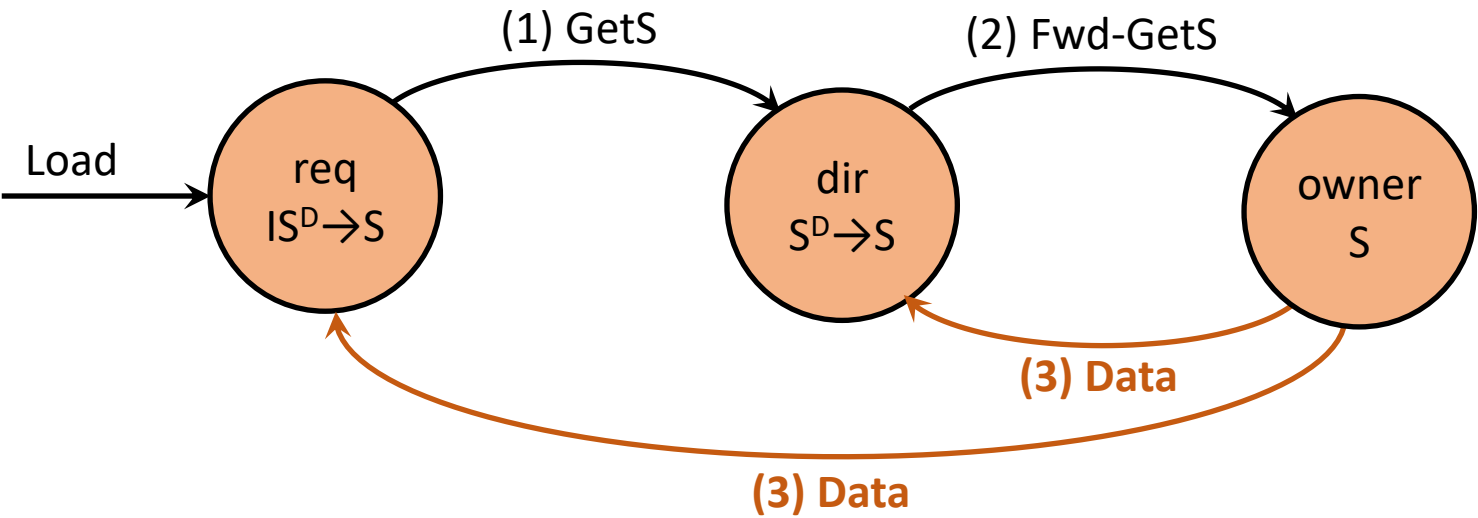


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		

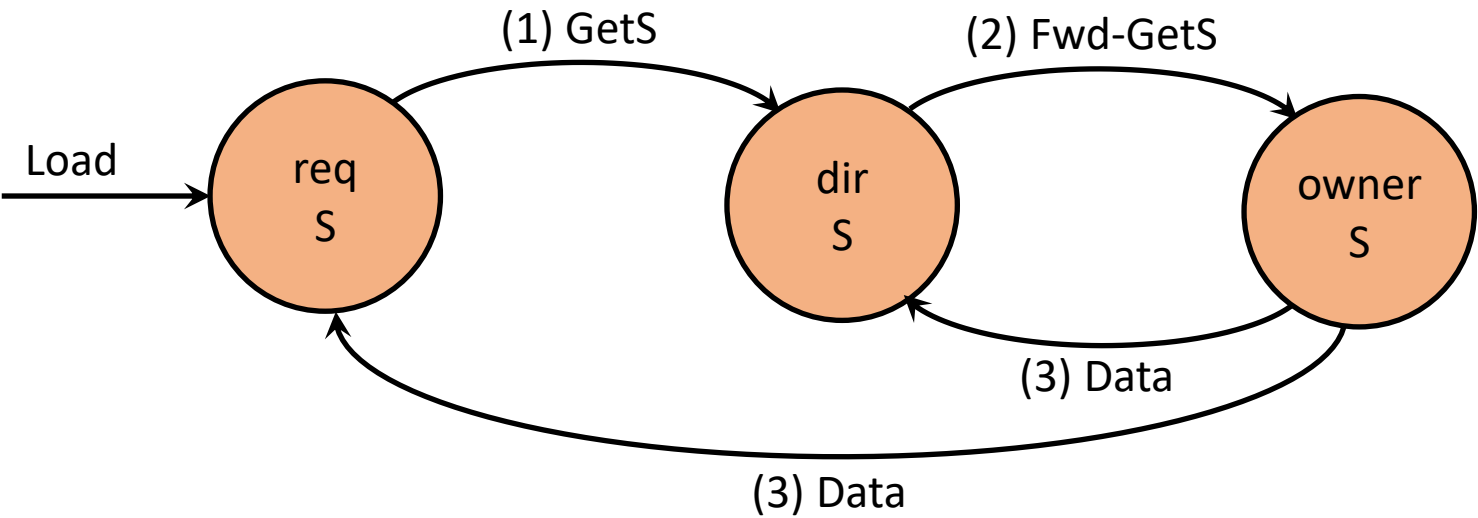


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

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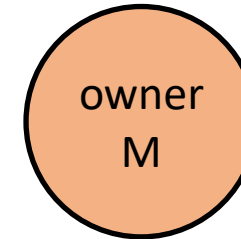
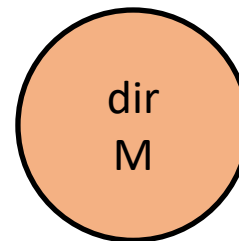
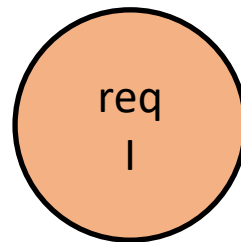


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M

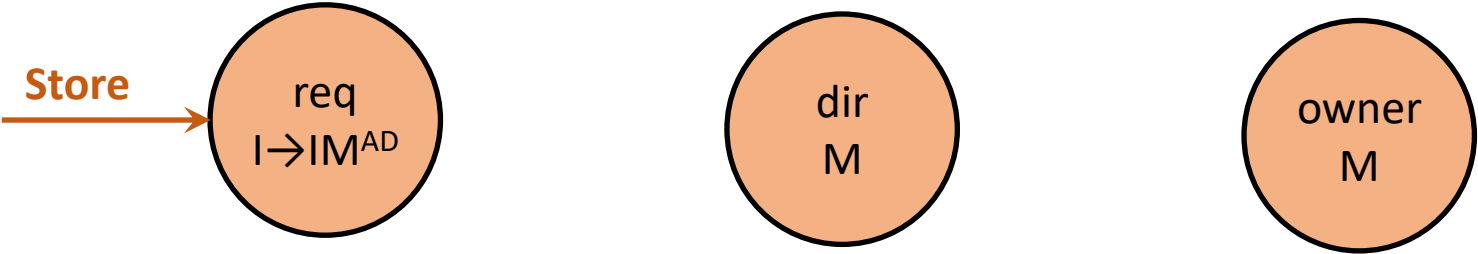


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M

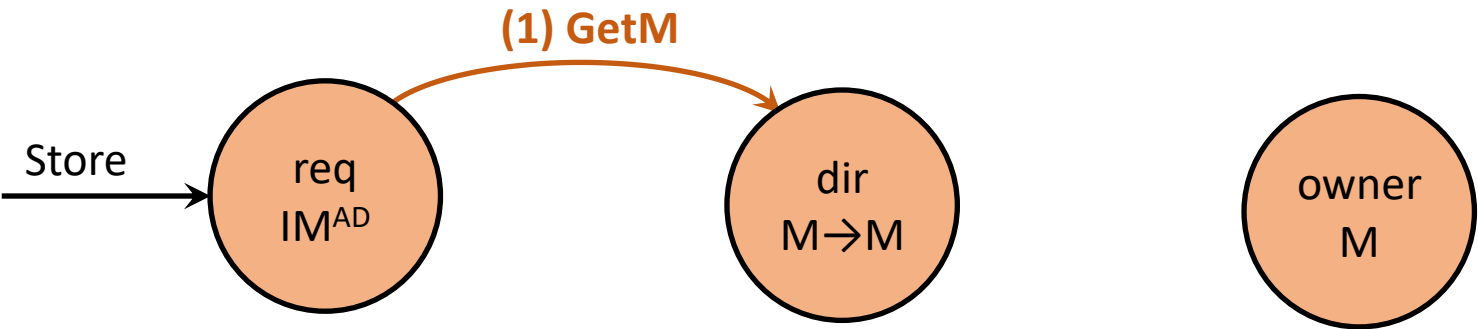


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M

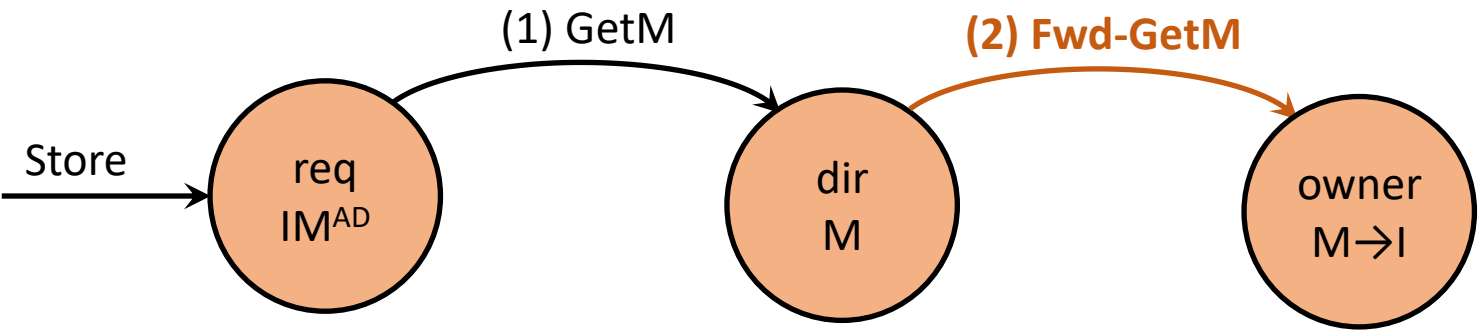


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M

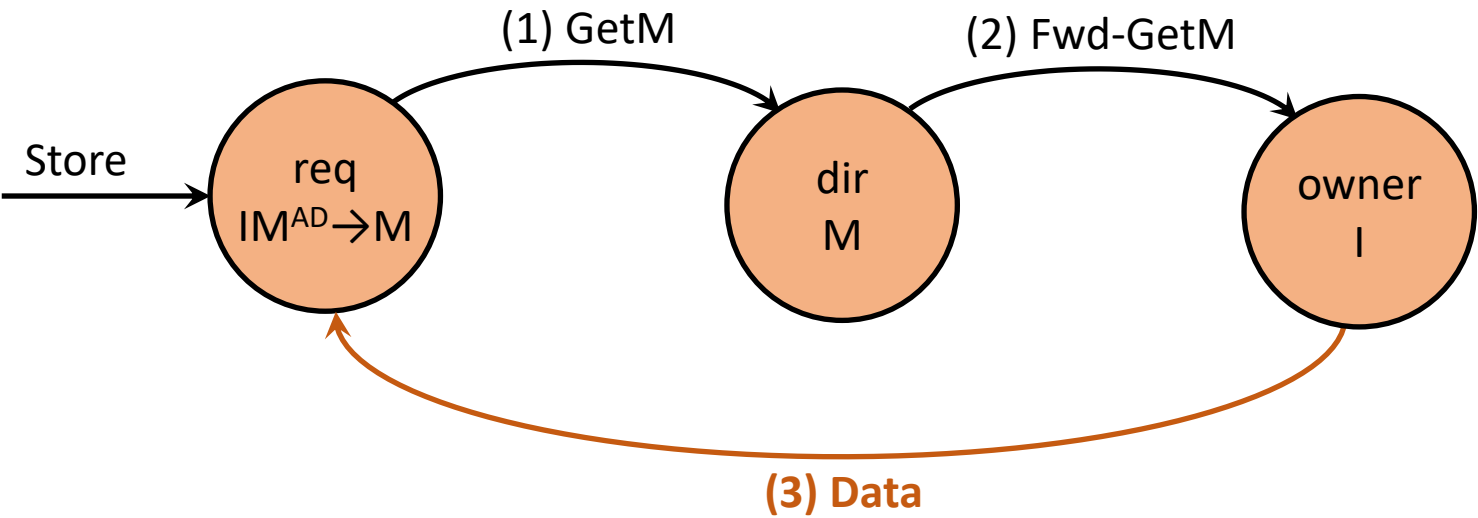


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M

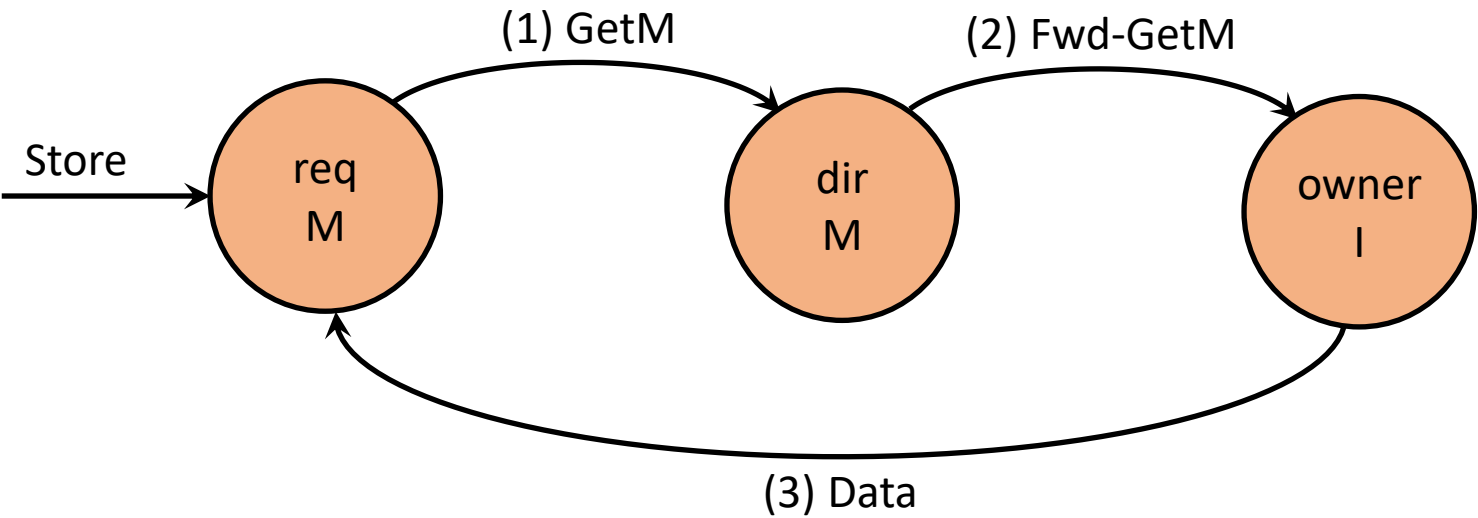


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M

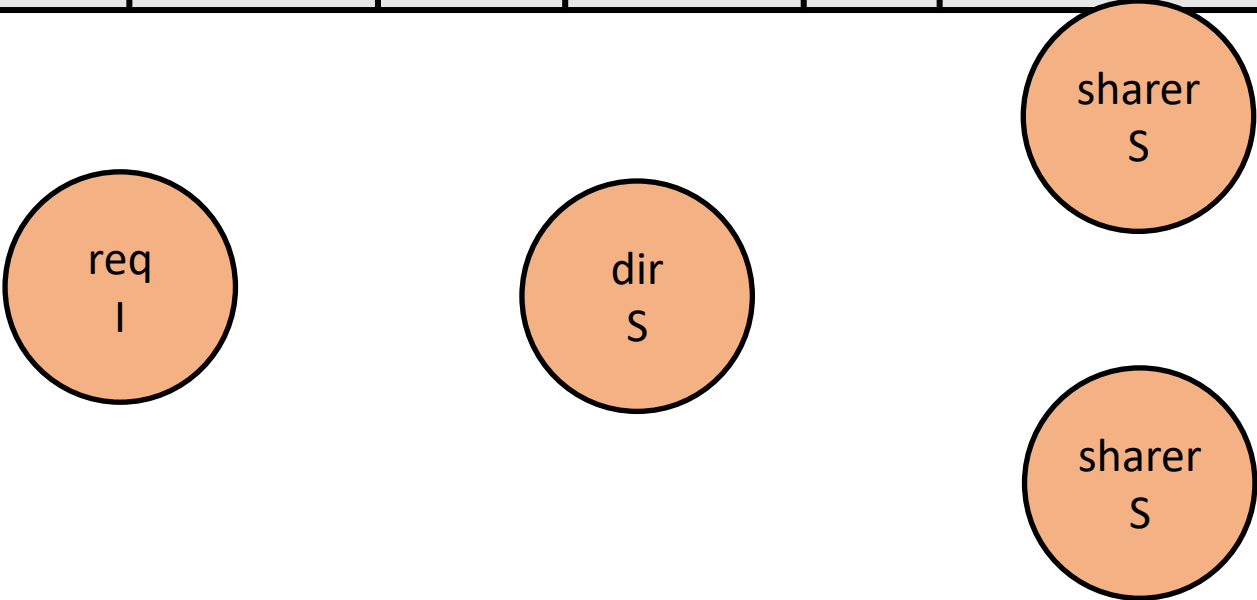


Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

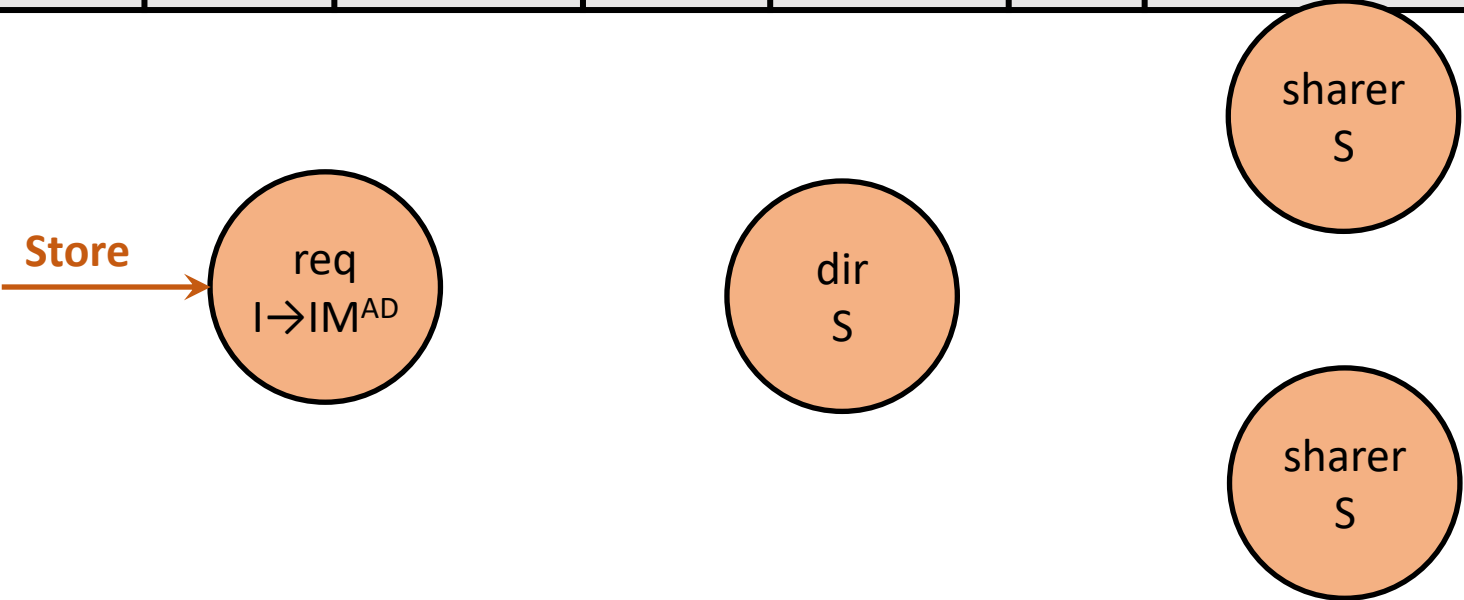
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
S	hit → S	GetM to dir → SM ^{AD}	PutS to dir → SI ^A			Inv-Ack to req → I						
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							

Directory-Based Coherence

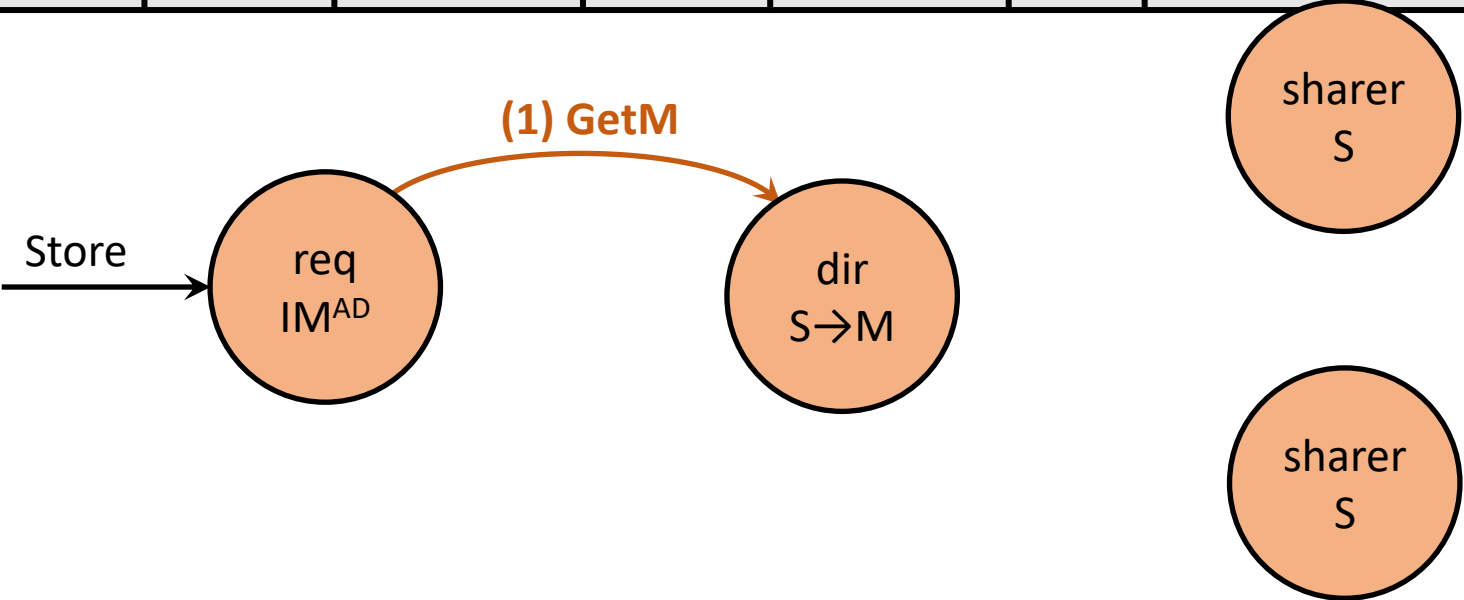
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IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



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Directory-Based Coherence

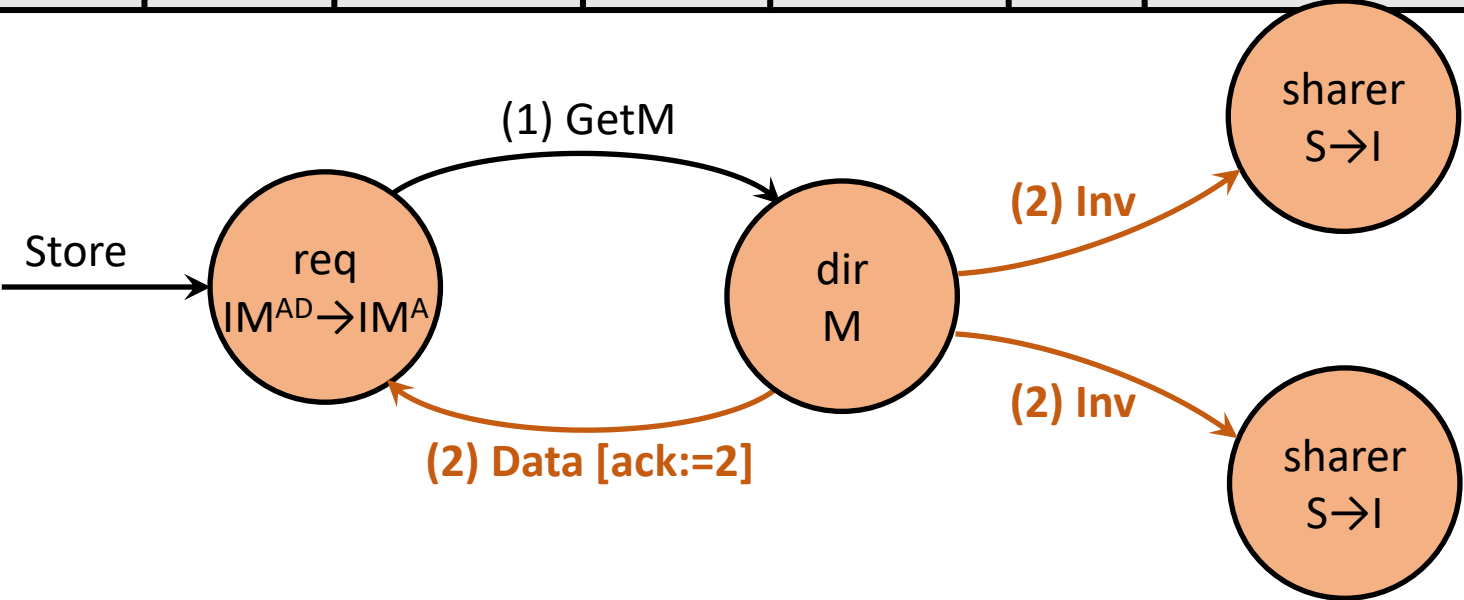
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

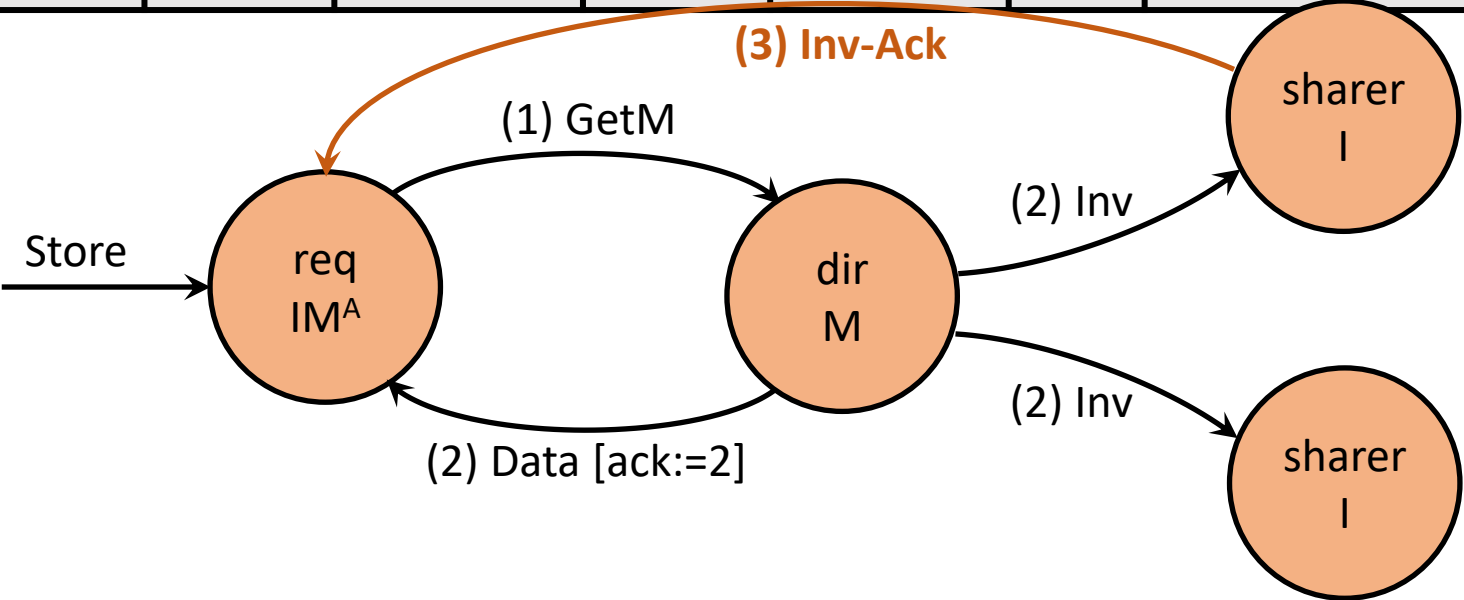
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

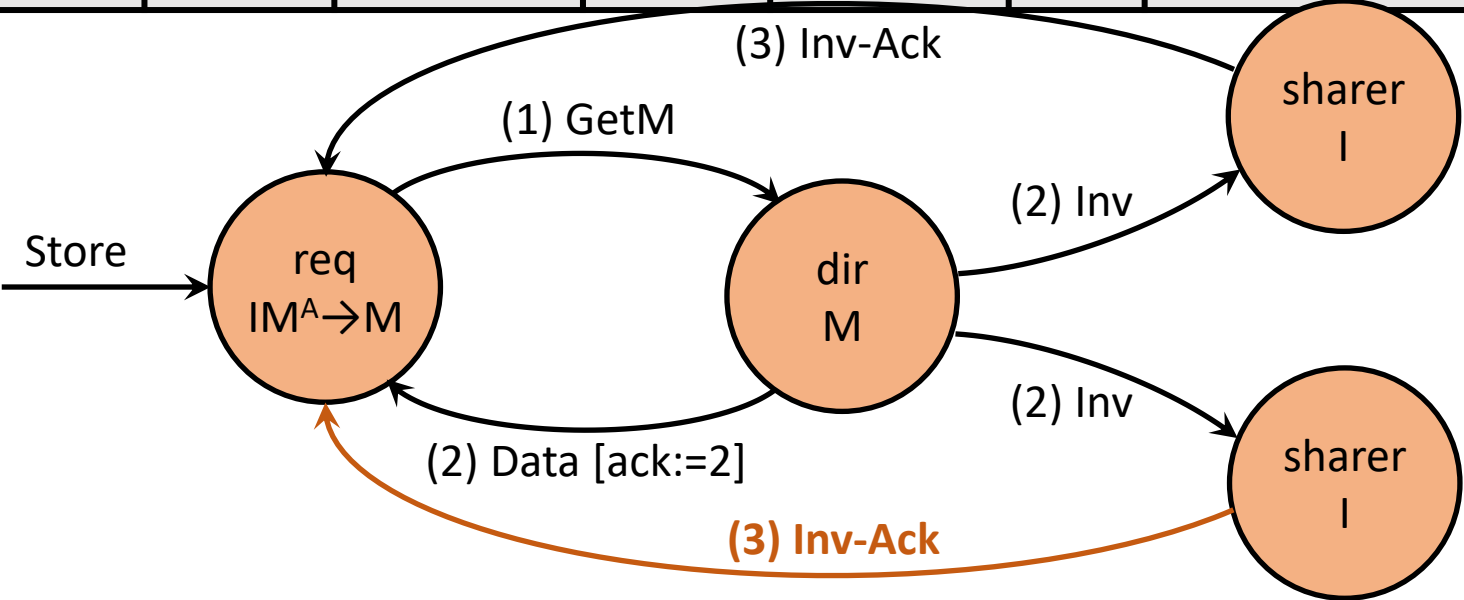
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IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

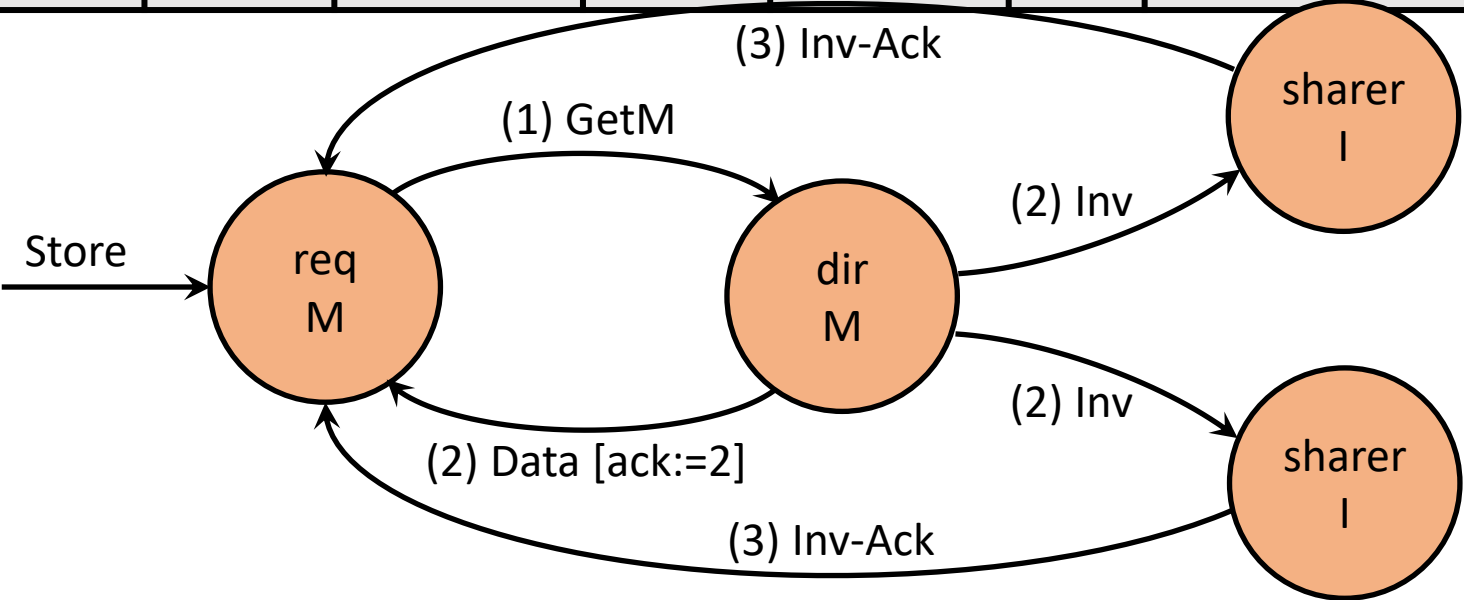
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

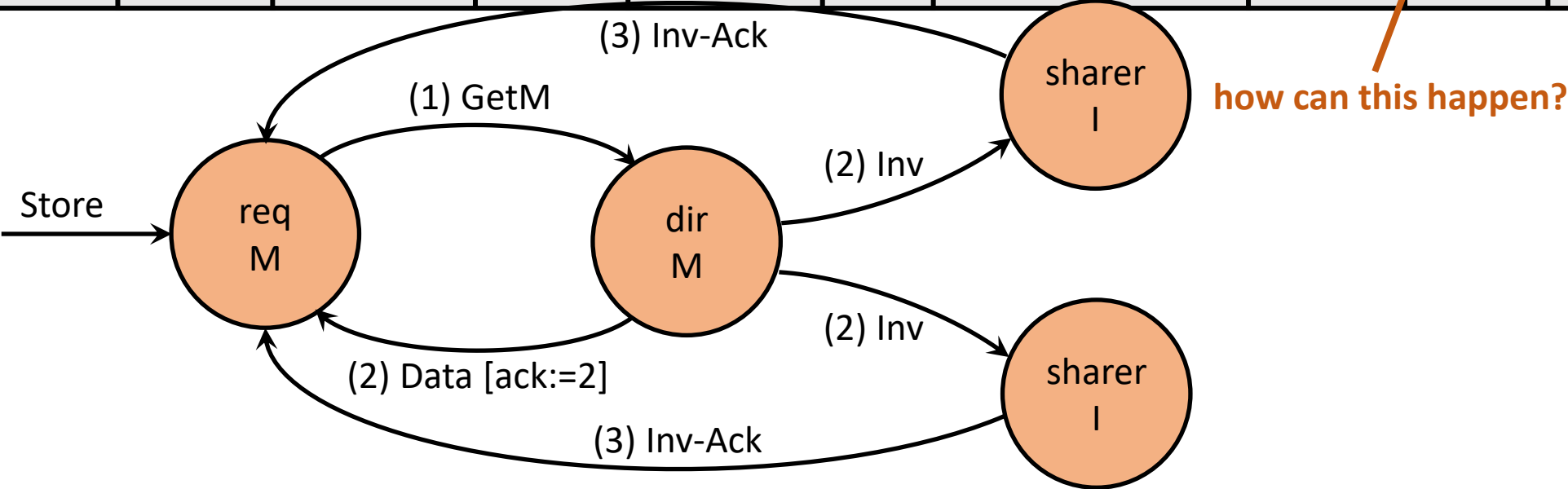
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

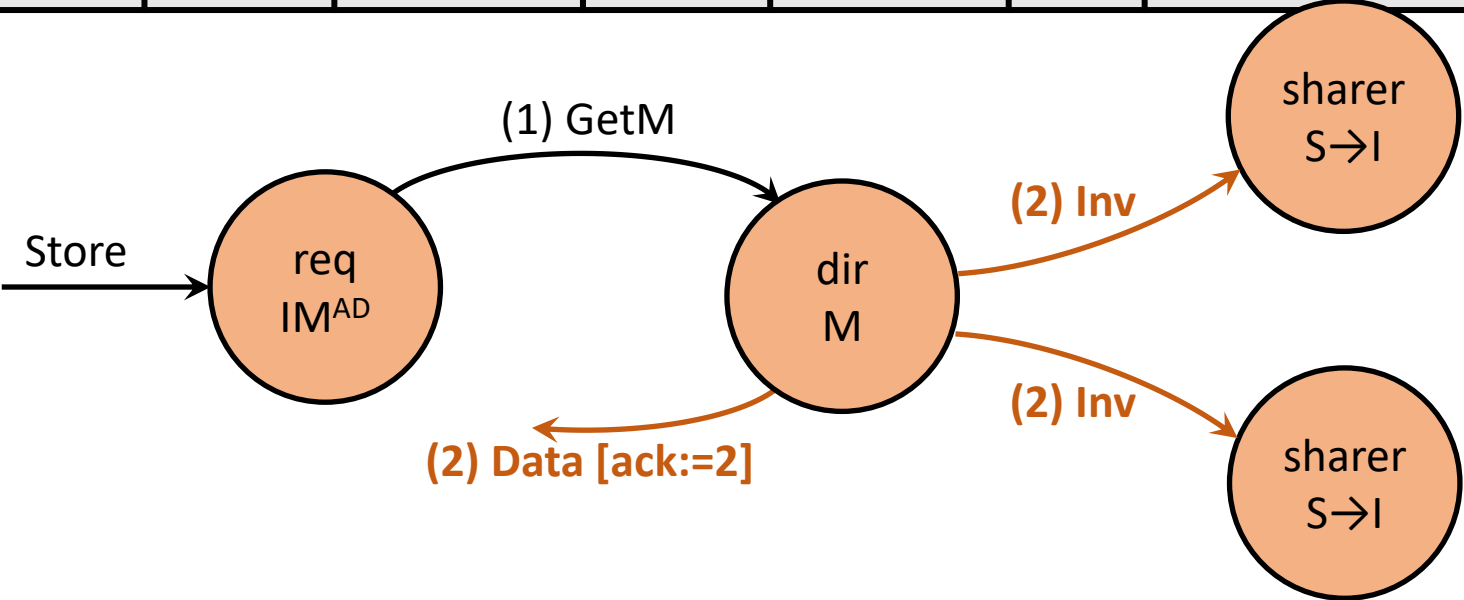
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

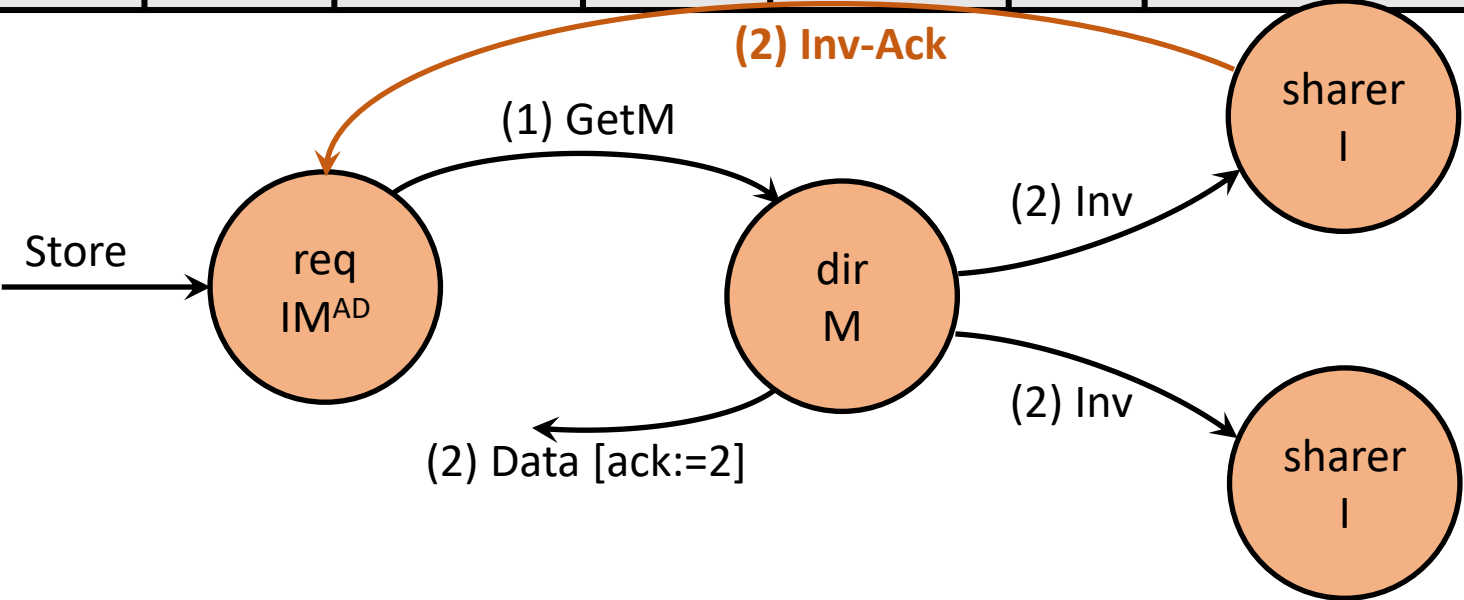
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

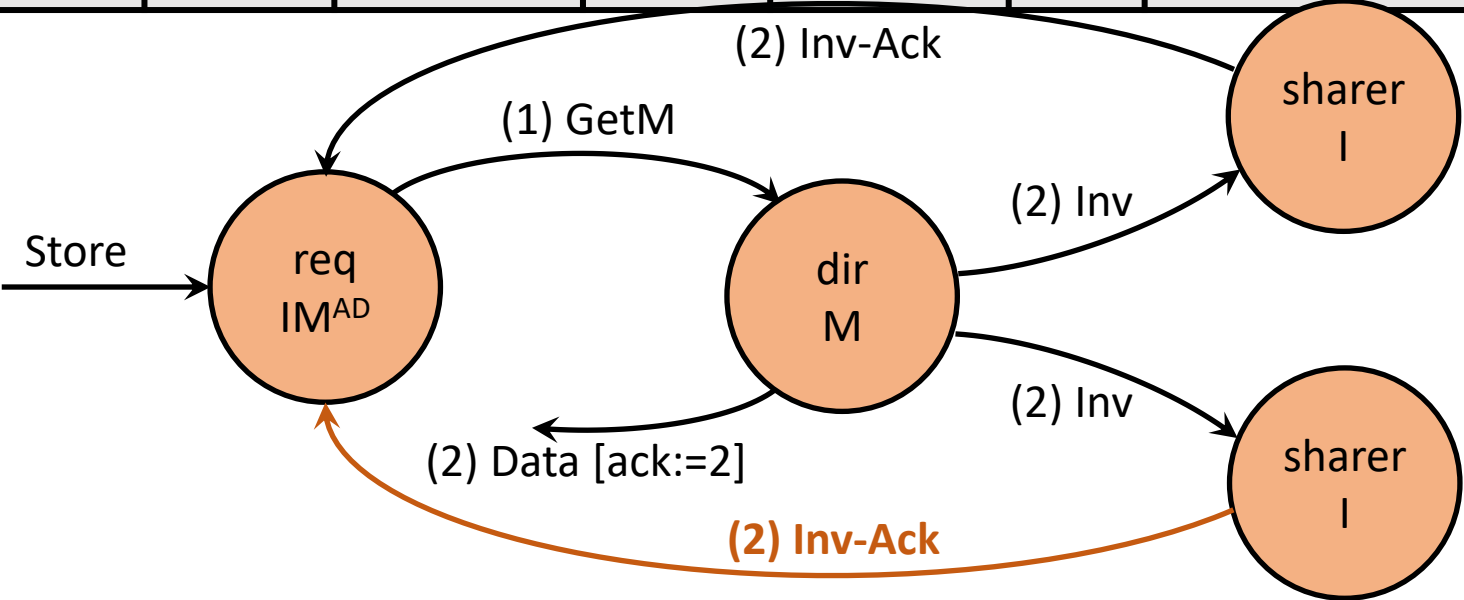
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
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Directory-Based Coherence

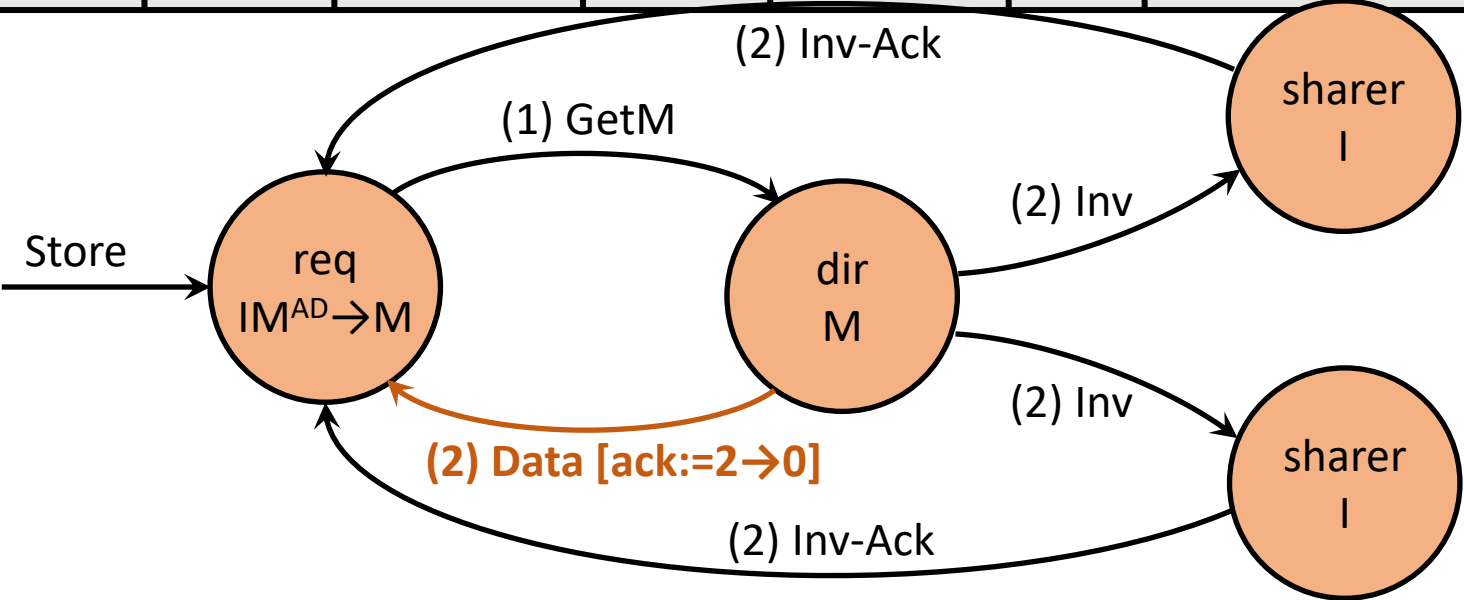
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
S	hit → S	GetM to dir → SM ^{AD}	PutS to dir → SI ^A			Inv-Ack to req → I						
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							

Directory-Based Coherence

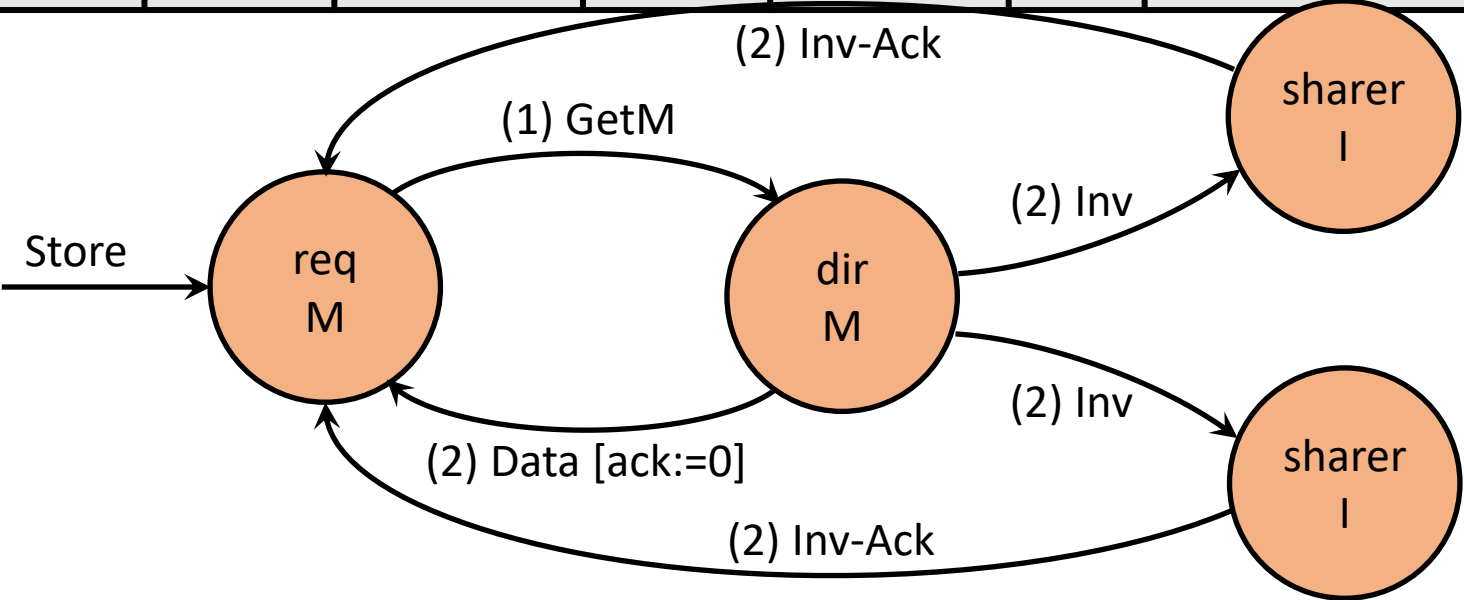
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
S	hit → S	GetM to dir → SM ^{AD}	PutS to dir → SI ^A			Inv-Ack to req → I						
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							

Directory-Based Coherence

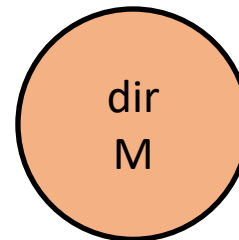
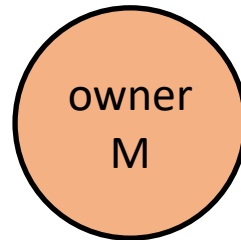
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M



Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
S	hit → S	GetM to dir → SM ^{AD}	PutS to dir → SI ^A			Inv-Ack to req → I						
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

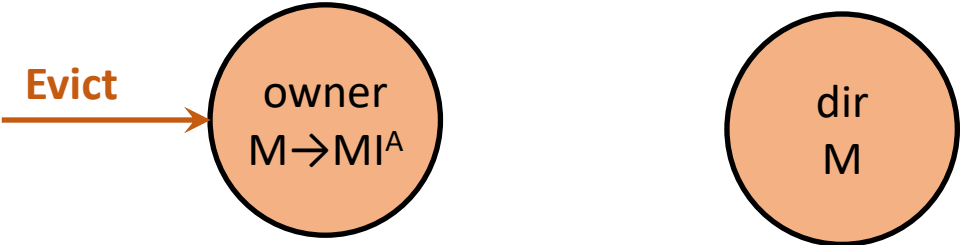


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

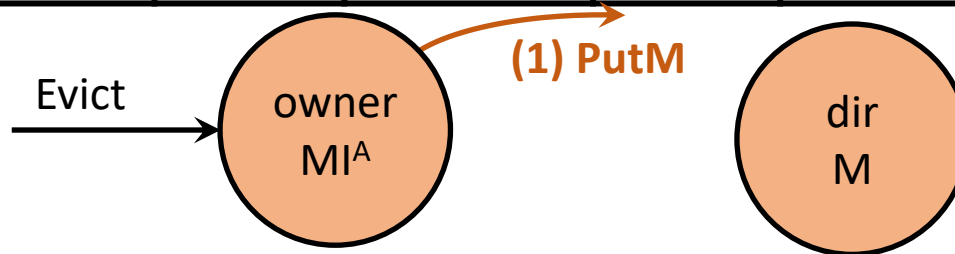


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

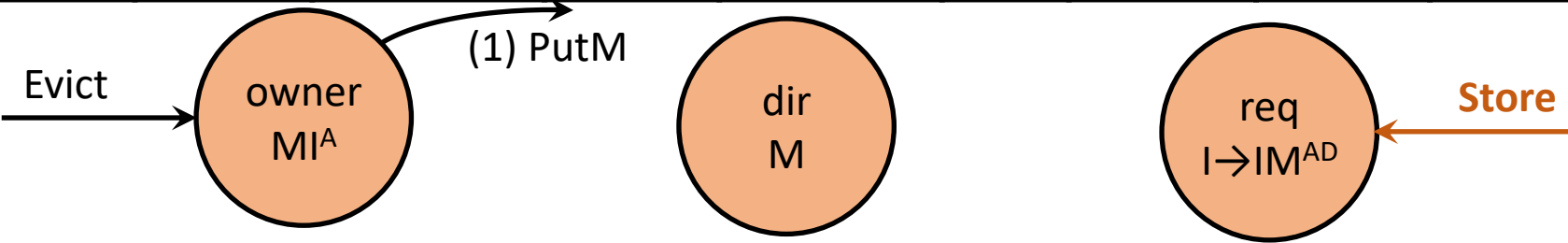


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

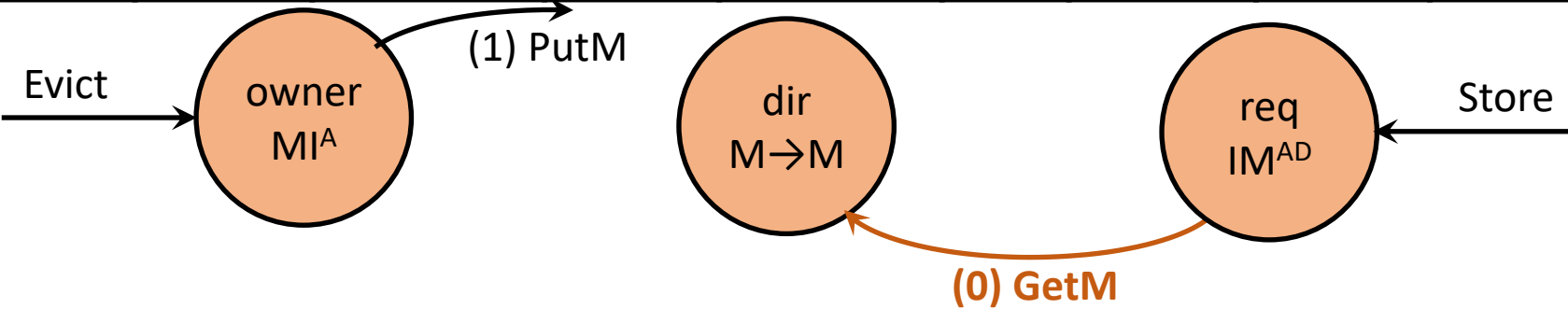


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

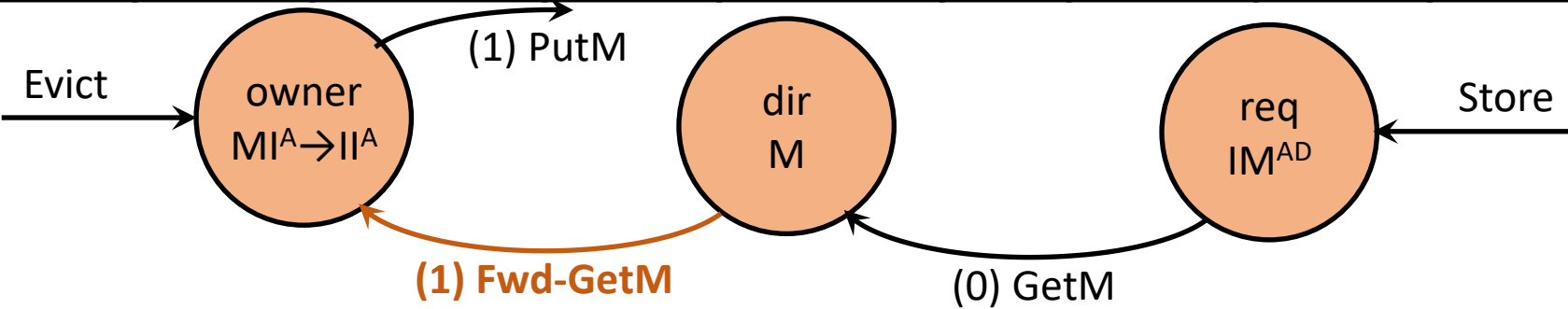


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

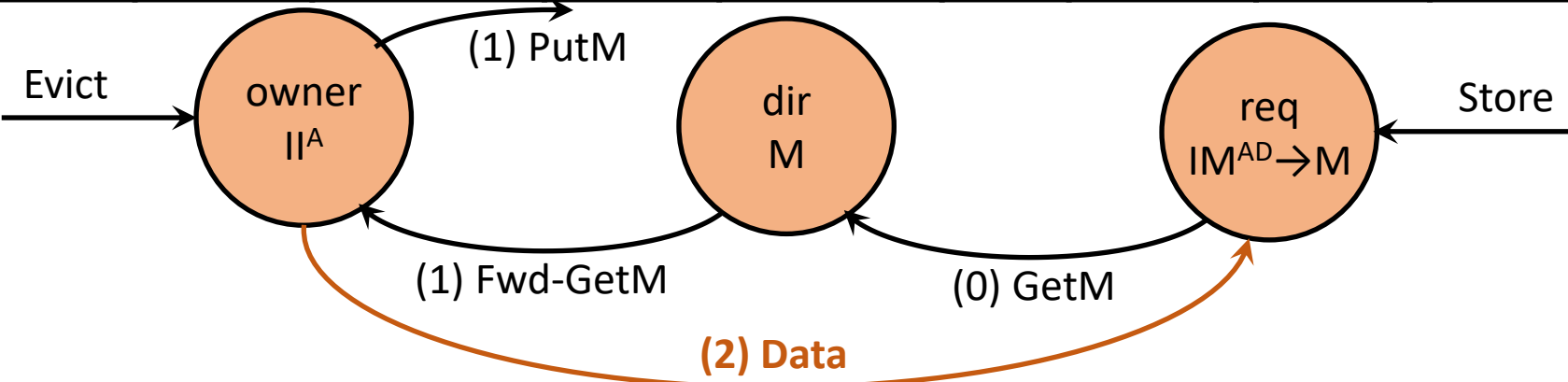


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

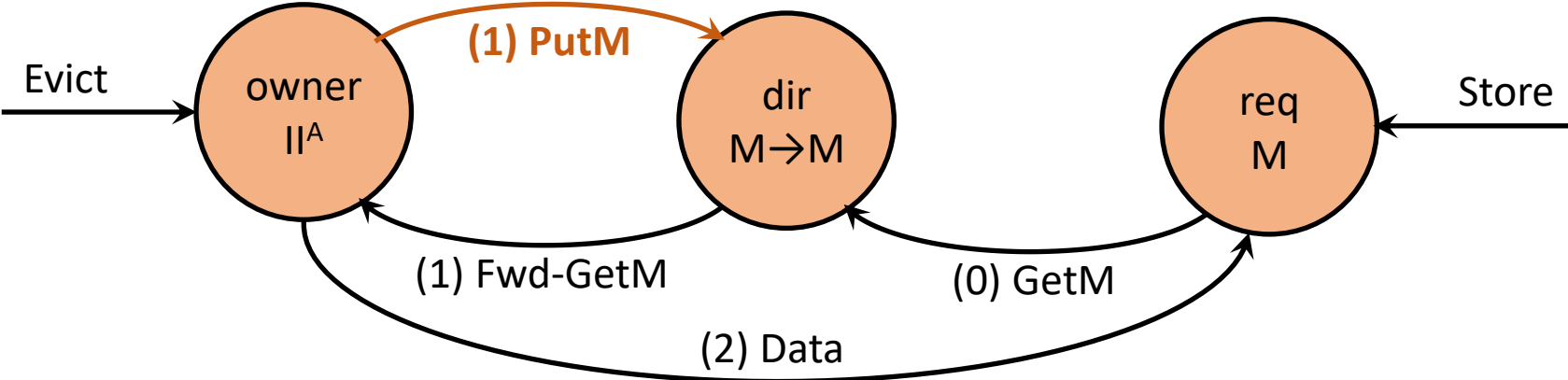


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

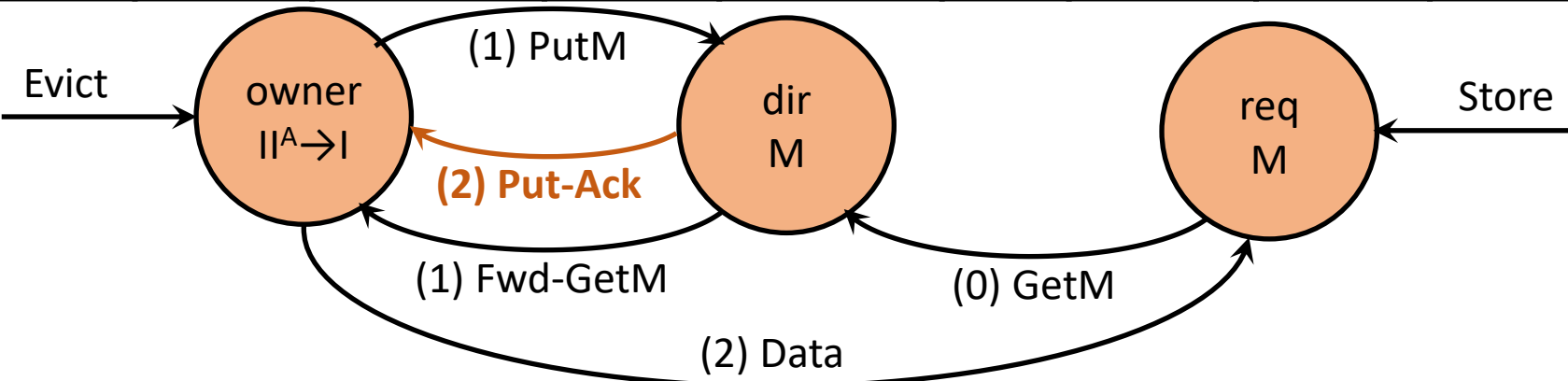


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

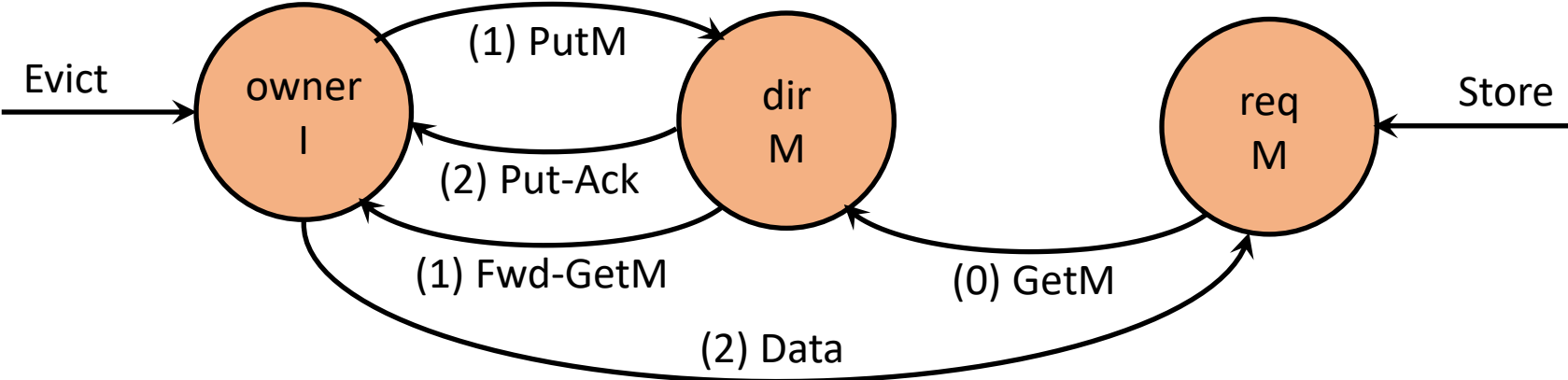


Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					



Directory Controller

Current State	GetS	GetM	PutS (not last)	PutS (last)	PutM (owner)	PutM (not owner)	Data
M	Fwd-GetS to owner → S ^D	Fwd-GetM to owner → M	Put-Ack to req → M	Put-Ack to req → M	Data to mem Put-Ack to req → I	Put-Ack to req → M	

Directory-Based Coherence

	initiating my transaction			someone else's transaction			responses to my transaction					
Current state	Load	Store	Evict	Fwd-GetS	Fwd-GetM	Inv	Put-Ack	Data from dir [ack=0]	Data from dir [ack>0]	Data from owner	Inv-Ack	Inv-Ack (last)
I	GetS to dir → IS ^D	GetM to dir → IM ^{AD}										
IS ^D	stall	stall	stall			stall		→ S		→ S		
IM ^{AD}	stall	stall	stall	stall	stall			→ M	→ IM ^A	→ M	ack-- → IM ^{AD}	
IM ^A	stall	stall	stall	stall	stall						ack-- → IM ^A	→ M
S	hit → S	GetM to dir → SM ^{AD}	PutS to dir → SI ^A			Inv-Ack to req → I						
SM ^{AD}	hit → SM ^{AD}	stall	stall	stall	stall	Inv-Ack to req → IM ^{AD}		→ M	→ SM ^A		ack-- → SM ^{AD}	
SM ^A	hit → SM ^A	stall	stall	stall	stall						ack-- → SM ^A	→ M
M	hit → M	hit → M	PutM to dir → MI ^A	Data to req+dir → S	Data to req → I							
MI ^A	stall	stall	stall	Data to req+dir → SI ^A	Data to req → II ^A		→ I					
SI ^A	stall	stall	stall			Inv-Ack to req → II ^A	→ I					
II ^A	stall	stall	stall				→ I					

Directory-Based Coherence

D. J. Sorin, M. D. Hill and D. A. Wood, *A Primer on Memory Consistency and Cache Coherence*, Synthesis Lectures on Computer Architecture.

Study **Chapter 8**:

- Figure 8.3, Table 8.1 and Table 8.2 for MSI